

13 Empirical Orthogonal Functions

13.0.0 Overview. In this chapter we present a multivariate analysis technique that is to derive the dominant patterns of variability from a statistical field (a random vector, usually indexed by location in space). *Principal Component Analysis*, or *Empirical Orthogonal Function* (EOF) Analysis as it is called in the Earth Sciences, was described by Pearson [309] in 1902 and by Hotelling [186] in 1935. EOF analysis was introduced into meteorology by Lorenz [259] in 1956.

Concepts in linear algebra that are needed to read this chapter (linear bases, matrix properties, eigenanalysis and singular value decomposition) are offered in Appendix B. Empirical Orthogonal Functions are formally defined in Section 13.1. Techniques for estimating EOFs, eigenvalues, and EOF coefficients are explained in Section 13.2. We discuss the quality of estimates in Section 13.3. Several EOF analyses of climate-related problems are given as examples in Section 13.4. *Rotated EOFs*¹ are dealt with in Section 13.5. Finally, a time series analysis technique called *Singular Systems Analysis*, which uses the same mathematics as EOF analysis, is introduced in Section 13.6.

An alternative introduction to EOFs is given by von Storch [387].

13.0.1 Introductory Example:² Daily Profile of Geopotential Height at Berlin. To motivate the concept of Empirical Orthogonal Functions we consider a time series of daily geopotential height profiles as obtained by radiosonde at Berlin (Germany) (Fraedrich and Dümmel [125]). A total of 1080 observations are available in each winter (NDJF) season: 120 days times 9 vertical levels between 950 hPa and 300 hPa. Thus, in a 20-year data set we have 21 600 observations at our disposal to describe the statistics of the geopotential height at Berlin in winter.

The mean state can be estimated by computing the mean value at each level. But how should we describe the variability? One way would be to

compute the standard deviation at each level and to plot it in the vertical. However such a profile does not tell us how the variations are correlated in the vertical. For example, are we likely to observe a positive anomaly (i.e., a positive deviation from the mean profile) at 300 hPa and at 950 hPa at the same time?

EOF analysis is a technique that is used to identify patterns of simultaneous variation. To demonstrate the concept we let $\vec{x}_t$ represent the $m = 9$ level geopotential height profile observed at time t . The mean profile is denoted by $\hat{\mu}$ and to describe the variability we form the *anomalies*

$$\vec{x}'_t = \vec{x}_t - \hat{\mu}.$$

These anomalies are then *expanded* into a finite series

$$\vec{x}'_t = \sum_{i=1}^k \hat{\alpha}_{i,t} \hat{e}^i \quad (13.1)$$

with time coefficients $\hat{\alpha}_{i,t}$ and fixed *patterns* $\hat{e}^i$. Equality is usually only possible when $k = m$, but the variance of the time coefficients $\hat{\alpha}_{i,t}$ usually decreases quickly with increasing index i , so that good approximations are usually possible for k much less than m . The patterns are chosen to be *orthogonal* so that optimal coefficients $\hat{\alpha}_{i,t}$ are obtained by simply projecting the anomalies $\vec{x}'_t$ onto the patterns $\hat{e}^i$. Moreover, the patterns can be specified such that the *error*

$$\sum_t \left(\vec{x}'_t - \sum_{i=1}^k \hat{\alpha}_{i,t} \hat{e}^i \right)^2$$

is minimal. The lag-0 sample cross-correlations of the optimal time coefficients are all zero,

$$\sum_t \hat{\alpha}_{i,t} \hat{\alpha}_{j,t} = 0$$

for $i \neq j$. The patterns $\hat{e}^j$ are estimated *Empirical Orthogonal Functions*.³ The coefficients $\hat{\alpha}_i$ are the *EOF coefficients*.⁴

³Note that the ‘functions’ $\hat{e}^k$ are really *vectors* and not *functions*.

⁴Statisticians refer to the EOF coefficients as *principal components*.

¹A misnomer.

²The mathematics in this subsection are explained in more detail in Sections 13.1 and 13.2.

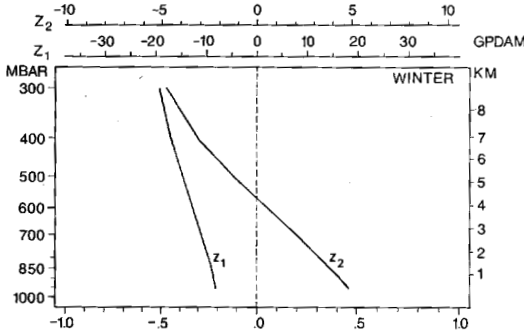


Figure 13.1: The first two EOFs, labelled z_1 and z_2 , of the daily geopotential height over Berlin in winter. From Fraedrich and Dümmler [125].

The analysis of daily Berlin radiosonde data showed that only two patterns are required to describe most of the variability in the observed geopotential height profiles in winter (NDJF) as well as in summer (MJJA). In winter, the first EOF represents 91.2% of the variance (92.6% in summer), and the second EOF represents an additional 8.2% of the variance (7% in summer).⁵ The remaining seven EOFs, which together with the first two EOFs span the full nine-dimensional space, represent only 0.6% of the variance of the height profiles (0.4% in summer). Thus, only two coefficient time series are required to represent the essential information in the time series of geopotential height at the nine levels. Instead of dealing with 1080 numbers per season, only $2 \times 120 = 240$ are needed. This demonstrates one of the advantages of EOFs, namely the ability to often identify a small subspace that contains most of the dynamics of the observed system.⁶

Another advantage is that the patterns can sometimes be seen as *modes of variability*. In the present example the two patterns $\bar{e}^1$ and $\bar{e}^2$ may be identified with the equivalent barotropic mode and the first baroclinic mode of the tropospheric circulation: The first patterns in winter (Figure 13.1) as well as in summer have

⁵When we say that an expansion Y ‘represents’ $p\%$ of the variance of X , we mean that the variance of $Y - X$ is $(100 - p)\%$ of the variance of X . The word ‘explains’ is often used instead of the word ‘represents’ in the literature. This is misleading since nothing is explained causally; only part of the variability of X has been described by Y .

⁶The assumption that the subspace with maximum variance coincides with the dynamically active subspace is arbitrary. In general, it will not be valid and counter examples can easily be constructed. However, in climate research, it is often reasonable to make this assumption. An example demonstrating the dynamical dominance of EOFs is given by Selten [343].

the same sign throughout the troposphere, that is, they exhibit an equivalent barotropic structure. The second EOF, however, changes sign in the middle of the troposphere: it represents the first baroclinic mode.⁷

13.0.2 ‘Complex’ EOFs. EOFs may be derived from real- or complex-valued random vectors. The latter results in complex-valued EOFs. The ‘Complex EOF Analysis’ (CEOF) described in the climate literature (see [181]) is a special case of the EOF analysis of complex random vectors. The time order of the observations is important for these ‘CEOFs,’ or ‘Frequency Domain EOFs,’ since they are the EOFs of a complexified time series. In contrast, the time order of observations is irrelevant in ordinary complex EOF analysis. The original real-valued time series is made complex by adding its Hilbert transform (see Section 16.2) as the imaginary component. The Hilbert transform can be thought of as the time derivative of the original process so that the EOF analysis of the complexified process reveals properties of the variability of the state and its change at the same time. To avoid confusion with the ordinary complex EOF analysis we refer to these EOFs as *Hilbert EOFs* (see Section 16.3).

13.1 Definition of Empirical Orthogonal Functions

13.1.1 Overview. EOFs are introduced formally in this section as parameters of the distribution of an m -dimensional random vector $\vec{X}$.⁸ For the sake of brevity we assume $\bar{\mu} = 0$. We first construct the first EOF, which is the most powerful single pattern in representing the variance of $\vec{X}$. The idea is easily generalized to several patterns and in [13.1.3] the calculations are condensed into a theorem.

13.1.2 The First EOF. The first step is to find one ‘pattern’ $\bar{e}^1$, with $\|\bar{e}^1\| = 1$, such that

$$\epsilon_1 = \mathcal{E}\left(\|\vec{X} - \langle \vec{X}, \bar{e}^1 \rangle \bar{e}^1\|^2\right) \quad (13.2)$$

is minimized.⁹ Equation (13.2) describes the projection of the random vector $\vec{X}$ onto a one-dimensional subspace spanned by the fixed vector

⁷A similar result for the vertical structure of the shelf ocean has been reported by Kundu, Allen, and Smith [234].

⁸Mainly based on [392].

⁹ $\|\cdot\|$ denotes the vector norm, and $\langle \cdot, \cdot \rangle$ denotes the inner product. Note that $\|\vec{X}\| = \langle \vec{X}, \vec{X} \rangle$. See Appendix B.

$\vec{e}^1$. Minimizing ϵ_1 is equivalent to the maximizing of the variance of $\vec{X}$ that is contained in this subspace:

$$\begin{aligned}\epsilon_1 &= \mathcal{E} \left(\|\vec{X}\|^2 - 2\langle \vec{X}, \vec{e}^1 \rangle^* \vec{X}^\dagger \vec{e}^1 \right. \\ &\quad \left. + \langle \vec{X}, \vec{e}^1 \rangle^* \langle \vec{X}, \vec{e}^1 \rangle \right) \\ &= \mathcal{E} (\|\vec{X}\|^2 - \langle \vec{X}, \vec{e}^1 \rangle^* \langle \vec{X}, \vec{e}^1 \rangle) \\ &= \text{Var}(\vec{X}) - \text{Var}(\langle \vec{X}, \vec{e}^1 \rangle),\end{aligned}$$

where the variance of the random vector $\vec{X}$ is defined to be the sum of variances of the elements of $\vec{X}$.¹⁰ Note that

$$\text{Var}(\langle \vec{X}, \vec{e}^1 \rangle) = \vec{e}^{1\dagger} \Sigma \vec{e}^1,$$

where Σ is the covariance matrix of $\vec{X}$. Then minimization of equation (13.2), under the constraint $\|\vec{e}^1\| = 1$, leads to

$$\begin{aligned}\frac{d}{d\vec{e}^1} [-\vec{e}^{1\dagger} \Sigma \vec{e}^1 + \lambda (\vec{e}^{1\dagger} \vec{e}^1 - 1)] \\ = 2\Sigma \vec{e}^1 + 2\lambda \vec{e}^1 = 0\end{aligned}$$

where λ is the Lagrange multiplier associated with the constraint $\|\vec{e}^1\| = 1$.¹¹ Thus, $\vec{e}^1$ is an eigenvector with a corresponding eigenvalue λ , of the covariance matrix Σ . But Σ has m eigenvectors. Therefore, to minimize ϵ_1 , we select the eigenvector that maximizes

$$\begin{aligned}\text{Var}(\langle \vec{X}, \vec{e}^1 \rangle) &= \vec{e}^{1\dagger} \Sigma \vec{e}^1 \\ &= \vec{e}^{1\dagger} \lambda \vec{e}^1 = \lambda.\end{aligned}$$

Thus ϵ_1 is minimized when $\vec{e}^1$ is an eigenvector of Σ associated with its largest eigenvalue λ .¹² This ‘pattern’ is the first EOF.¹³

13.1.3 More EOFs. Having found the first EOF, we now repeat the exercise by finding the ‘pattern’ $\vec{e}^2$ that minimizes

$$\epsilon_2 = \mathcal{E} (\|\vec{X} - \langle \vec{X}, \vec{e}^1 \rangle \vec{e}^1 - \langle \vec{X}, \vec{e}^2 \rangle \vec{e}^2\|^2)$$

subject to the constraint that $\|\vec{e}^2\| = 1$. The result is that $\vec{e}^2$ is the eigenvector of Σ that corresponds

to its second largest eigenvalue λ_2 .¹⁴ This second pattern is orthogonal to the first because the eigenvectors of a Hermitian matrix are orthogonal to one another.

13.1.4 Theorem. The following theorem results from the analysis presented so far.

Let $\vec{X}$ be an m -dimensional random vector with mean $\vec{\mu}$ and covariance matrix Σ . Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$ be the eigenvalues of Σ and let $\vec{e}^1, \dots, \vec{e}^m$ be the corresponding eigenvectors of unit length. Since Σ is Hermitian, the eigenvalues are non-negative and the eigenvectors are orthogonal.

(i) The k eigenvectors that correspond to $\lambda_1, \dots, \lambda_k$ minimize

$$\epsilon_k = \mathcal{E} (\|\vec{X} - \vec{\mu} - \sum_{i=1}^k \langle \vec{X} - \vec{\mu}, \vec{e}^i \rangle \vec{e}^i\|^2). \quad (13.3)$$

$$(ii) \quad \epsilon_k = \text{Var}(\vec{X}) - \sum_{i=1}^k \lambda_i. \quad (13.4)$$

$$(iii) \quad \text{Var}(\vec{X}) = \sum_{i=1}^m \lambda_i. \quad (13.5)$$

The total variance of $\vec{X}$ is broken up into m components. Each of these components is obtained by projecting $\vec{X}$ onto one of the EOFs $\vec{e}^i$. The variance contribution of the k th component to the total variance $\sum_j \lambda_j$ is just λ_k . In relative terms, the proportion of the total variance represented by EOF k is $\lambda_k / \sum_j \lambda_j$. This proportion may be given as a percentage.

If the components are ordered by the size of the eigenvalues then the first component is the most important in representing variance, the second is the second most important and so forth.

Equation (13.3) gives the mean squared error ϵ_k that is incurred when approximating the full m -dimensional random vector $\vec{X}$ in a k -dimensional subspace spanned by the first k EOFs. The construction of the EOFs ensures that the approximation is optimal; the use of any other k -dimensional subspace will lead to mean squared errors at least as large as ϵ_k .

13.1.5 Properties of the EOF Coefficients. The EOF coefficients, or *principal components*,

$$\alpha_i = \langle \vec{X}, \vec{e}^i \rangle = \vec{X}^\dagger \vec{e}^i = \vec{e}^{i\dagger} \vec{X} \quad (13.6)$$

¹⁴Note that $\lambda_1 = \lambda_2$ if $\vec{e}^1$ is degenerate. In fact, if λ_1 has k linearly independent eigenvectors, then k of the m eigenvalues of Σ will be equal to λ .

¹⁰That is, if $\vec{X}$ has covariance matrix Σ , then we define $\text{Var}(\vec{X}) = \text{tr}(\Sigma)$.

¹¹Graybill [148, Section 10.8], describes the differentiation of quadratic forms.

¹²Recall (see Appendix B) that all eigenvalues of the Hermitian matrix $\Sigma = \mathcal{E}(\vec{X}\vec{X}^\dagger)$ are real and non-negative.

¹³The pattern is unique up to sign if Σ has only one eigenvector that corresponds to eigenvalue λ . Otherwise, the pattern can be any vector with unit norm that is spanned by the eigenvectors corresponding to λ . In this case, the EOF is said to be *degenerate*. See Appendix B.

are uncorrelated, and hence independent when $\vec{\mathbf{X}}$ is multivariate normal. In fact, for $i \neq j$,

$$\begin{aligned} \text{Cov}(\alpha_i, \alpha_j) &= \mathcal{E}(\langle (\vec{\mathbf{X}} - \vec{\mu}), \vec{e}^i \rangle \langle (\vec{\mathbf{X}} - \vec{\mu}), \vec{e}^j \rangle^*) \\ &= \vec{e}^{i\dagger} \mathcal{E}((\vec{\mathbf{X}} - \vec{\mu})(\vec{\mathbf{X}} - \vec{\mu})^\dagger) \vec{e}^j \\ &= \vec{e}^{i\dagger} \Sigma \vec{e}^j \\ &= \lambda_j \vec{e}^{i\dagger} \vec{e}^j = 0 \end{aligned}$$

Therefore, the variance of $\mathbf{X}_k$, the k th component of $\vec{\mathbf{X}}$, can also be decomposed into contribution from the individual EOFs as

$$\text{Var}(\mathbf{X}_k) = \sum_{i=1}^m \lambda_i |e_k^i|^2. \quad (13.7)$$

If the elements of $\vec{\mathbf{X}}$ represent locations in space, the spatial distribution of variance can be visualized by plotting $\text{Var}(\mathbf{X}_k)$ as a function of location. Similarly, the variance contribution from the i th EOF can be visualized by plotting $\lambda_i |e_k^i|^2$ or $\lambda_i |e_k^i|^2 / \text{Var}(\mathbf{X}_k)$ as a function of location.

13.1.6 Interpretation. The bulk of the variance of $\vec{\mathbf{X}}$ can often be represented by the first few EOFs. If the original variable has m components the approximation of $\vec{\mathbf{X}}$ by $\vec{\alpha} = (\alpha_1, \dots, \alpha_k)$, with $k \ll m$, leads to a significant reduction of the amount of data while retaining most of the variance. It was shown in the introductory example of Berlin geopotential height [13.0.2] that just two EOFs represent almost all of the information in the data set.

The physical interpretation of EOFs is limited by a fundamental constraint. While it is often possible to clearly associate the first EOF with a known physical process, this is much more difficult with the second (and higher-order) EOF because it is constrained to be orthogonal to the first EOF. However, real-world processes do not need to have orthogonal patterns or uncorrelated indices. In fact, the patterns that most efficiently represent variance do not necessarily have anything to do with the underlying dynamical structure.

13.1.7 Vector Notation. The random vector $\vec{\mathbf{X}}$ may conveniently be written in vector notation by

$$\vec{\mathbf{X}} = \mathcal{P} \vec{\alpha} \quad (13.8)$$

where $\mathcal{P}$ is the $m \times m$ matrix

$$\mathcal{P} = (\vec{e}^1 | \vec{e}^2 | \dots | \vec{e}^m) \quad (13.9)$$

that has EOFs in its columns, and $\vec{\alpha}$ is the m -dimensional (column) vector of EOF coefficients

$\alpha_1, \dots, \alpha_m$. Because the EOFs are orthonormal, the expression (13.8) may be inverted to obtain

$$\vec{\alpha} = \mathcal{P}^\dagger \vec{\mathbf{X}}, \quad (13.10)$$

where $\mathcal{P}^\dagger$ is the conjugate transpose of $\mathcal{P}$. Another consequence of the orthonormality of the EOFs is that

$$\begin{aligned} \Sigma &= \text{Cov}(\vec{\mathbf{X}}, \vec{\mathbf{X}}) \\ &= \mathcal{P} \text{Cov}(\vec{\alpha}, \vec{\alpha}) \mathcal{P}^\dagger \\ &= \mathcal{P} \Lambda \mathcal{P}^\dagger \end{aligned}$$

where Λ is the diagonal $m \times m$ matrix composed of the eigenvalues of Σ ,

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_m).$$

It therefore follows that

$$\begin{aligned} \text{Var}(\vec{\mathbf{X}}) &= \sum_{k=1}^m \text{Var}(\mathbf{X}_k) \\ &= \text{tr}(\Sigma) \\ &= \text{tr}(\mathcal{P} \Lambda \mathcal{P}^\dagger) \\ &= \text{tr}(\Lambda) = \sum_{k=1}^m \lambda_k. \end{aligned}$$

It also follows that the eigenvalues are the m roots of the m th degree *characteristic polynomial* $p_\Sigma(\lambda) = \det(\Sigma - \lambda \mathcal{I})$, where $\mathcal{I}$ is the $m \times m$ identity matrix. In fact

$$\begin{aligned} p_\Sigma(\lambda) &= \det(\mathcal{P} \Lambda \mathcal{P}^\dagger - \lambda \mathcal{P} \mathcal{P}^\dagger) \\ &= \det(\mathcal{P} (\Lambda - \lambda \mathcal{I}) \mathcal{P}^\dagger) \\ &= \det(\Lambda - \lambda \mathcal{I}) = \prod_{i=1}^m (\lambda_i - \lambda). \end{aligned} \quad (13.11)$$

13.1.8 Degeneracy. As noted above, EOFs are not uniquely determined. If $\lambda_\circ$ is a root of multiplicity 1 of $p_\Sigma(\lambda)$ and $\vec{e}$ is a corresponding (normalized) eigenvector, then $\vec{e}$ is unique up to sign, and either $\vec{e}$ or $-\vec{e}$ is chosen as the EOF that corresponds to $\lambda_\circ$. On the other hand, if $\lambda_\circ$ is a root of multiplicity k , the solution space of

$$\Sigma \vec{e} = \lambda_\circ \vec{e}$$

is of dimension k . The solution space is uniquely determined in the sense that it is orthogonal to the space spanned by the $m - k$ eigenvectors of Σ that correspond to eigenvalues $\lambda_i \neq \lambda_\circ$. But any orthonormal basis $\vec{e}^1, \dots, \vec{e}^k$ for the solution space can be used as EOFs. In this case the EOFs are said to be *degenerate*. (An example is discussed in [13.1.9].)

Degeneracy can either be bad or good news. It is bad news if the EOFs are estimated from a sample

of iid realizations of $\vec{X}$. Then degeneracy is mostly a nuisance, because the patterns, which may represent independent processes in the underlying dynamics, can not be disentangled.

However, degeneracy may be good news if the EOFs are estimates from a realization of a stochastic process $\vec{X}_t$. Suppose, for example, that $p_\Sigma(\lambda)$ has a root of multiplicity 2. By construction, the cross-correlation of the two corresponding EOF coefficient time series will be zero at lag-0. But this does not imply that the lagged cross-correlations will be zero, and, in fact, they are often nonzero. This means that a pair of EOFs and their coefficient series could represent a signal that is propagating in space.

The representation of such a spatially propagating signal requires two patterns whose coefficients vary coherently and are 90° out-of-phase. The two patterns representing a propagating signal are not uniquely determined; indeed if any two patterns represent the signal, then any linear combination of the two do so as well. Therefore, degeneracy is a necessary condition for the description of such signals.

13.1.9 Examples. To demonstrate the mathematics of EOF analysis and the phenomenon of degeneracy we now consider the case of a random vector

$$\vec{X} = \sum_{k=1}^m \alpha_k \vec{p}^k \quad (13.12)$$

where coefficients α_k are uncorrelated real univariate random variables and $\vec{p}^1, \dots, \vec{p}^m$ are fixed orthonormal vectors. For simplicity we assume that the α s have mean zero. Then the covariance matrix of $\vec{X}$ is

$$\begin{aligned} \Sigma &= \mathcal{E}\left(\left(\sum_k \alpha_k \vec{p}^k\right)\left(\sum_l \alpha_l \vec{p}^l\right)^T\right) \\ &= \sum_j \text{Var}(\alpha_j) \vec{p}^j \vec{p}^{jT}. \end{aligned} \quad (13.13)$$

It is easily verified that $\text{Var}(\alpha_k)$ is an eigenvalue of this covariance matrix with eigenvector $\vec{p}^k$:

$$\left(\sum_j \text{Var}(\alpha_j) \vec{p}^j \vec{p}^{jT}\right) \vec{p}^k = \text{Var}(\alpha_k) \vec{p}^k.$$

Thus, the chosen orthonormal vectors are the EOFs of the random vector (13.12). The ordering is determined by the variance of the uncorrelated univariate random variables α_k .

The example has two merits. First it may be used as a recipe for constructing random vectors with a given EOF structure. To do so one has to select a

set of orthonormal vectors and associate them with a set of uncorrelated random variables.

The example may also be used to demonstrate the phenomenon of degeneracy. To do so, we assume that all α s have variance 1. Then, the EOFs are degenerate and may be replaced by any other set of orthonormal vectors. One such set of orthonormal vectors are the unit vectors $\vec{u}^k$ with a 1 in the k th row and zeros elsewhere. Then, the representation (13.8), with $\mathcal{P} = (\vec{p}^1 | \dots | \vec{p}^m)$, is transformed as

$$\vec{X} = \mathcal{P} \vec{\alpha} = (\mathcal{P} \mathcal{P}^T) (\mathcal{P} \vec{\alpha}) = \mathcal{U} \vec{\beta}, \quad (13.14)$$

where the new EOFs are the columns of

$$\mathcal{U} = \mathcal{P} \mathcal{P}^T = \mathcal{I} = (\vec{u}^1 | \dots | \vec{u}^m)$$

and the EOF coefficients are given by

$$\vec{\beta} = \mathcal{P} \vec{\alpha}.$$

These coefficients are uncorrelated as well because of $\text{Var}(\alpha_k) = 1$ for all k :

$$\begin{aligned} \text{Cov}(\vec{\beta}, \vec{\beta}) &= \text{Cov}(\mathcal{P} \vec{\alpha}, \mathcal{P} \vec{\alpha}) \\ &= \mathcal{P} \text{Cov}(\alpha, \alpha) \mathcal{P}^T \\ &= \mathcal{P} \mathcal{I} \mathcal{P}^T = \mathcal{I}. \end{aligned}$$

Obviously, the only meaningful information the EOF analysis offers in this case is that there is no preferred direction in the phase space. The only property that matters is the uniformity of the variance in all directions.

13.1.10 Coordinate Transformations. Let us consider two m -variate random vectors $\vec{X}$ and $\vec{Z}$ that are related to each other by

$$\vec{Z} = \mathcal{L} \vec{X} \quad (13.15)$$

where $\mathcal{L}$ is an invertible matrix so that $\vec{X} = \mathcal{L}^{-1} \vec{Z}$. Both vectors represent the same information but the data are given in different coordinates. The covariance matrix of $\vec{Z}$ is

$$\Sigma_{ZZ} = \mathcal{L} \Sigma_{XX} \mathcal{L}^\dagger, \quad (13.16)$$

for $\mathcal{L}^\dagger$ the conjugate transpose of $\mathcal{L}$. Suppose the transformation is orthogonal (i.e., $\mathcal{L}^{-1} = \mathcal{L}^\dagger$), and also let λ be an eigenvalue of Σ_{XX} and let $\vec{e}^X$ be the corresponding eigenvector. Then, since $\Sigma_{XX} \vec{e}^X = \lambda \vec{e}^X$,

$$\begin{aligned} \Sigma_{ZZ} \mathcal{L} \vec{e}^X &= \mathcal{L} \Sigma_{XX} \mathcal{L}^\dagger \mathcal{L} \vec{e}^X \\ &= \mathcal{L} \Sigma_{XX} \vec{e}^X \\ &= \lambda \mathcal{L} \vec{e}^X. \end{aligned}$$

Thus λ is also an eigenvalue of Σ_{ZZ} and the EOFs of $\vec{Z}$ are related to those of $\vec{X}$ through

$$\vec{e}^Z = \mathcal{L}\vec{e}^X. \quad (13.17)$$

Thus eigenvectors are transformed (13.17) just as a random vector is transformed (13.15).

Another consequence of using an orthogonal transformation is that the EOF coefficients are invariant. To see this, let $\mathcal{P}_X$ be the matrix composed of the $\vec{X}$ -EOFs $\vec{e}^X$ and let $\mathcal{P}_Z$ be the corresponding matrix of $\vec{Z}$ -EOFs. We see from equation (13.17) that

$$\mathcal{P}_Z = \mathcal{L}\mathcal{P}_X. \quad (13.18)$$

Using equation (13.10) and transformation (13.15), the vector of $\vec{X}$ -EOF coefficients

$$\begin{aligned} \vec{\alpha}_X &= \mathcal{P}_X^\dagger \vec{X} = \mathcal{P}_X^\dagger \mathcal{L}^\dagger \vec{Z} = (\mathcal{L}\mathcal{P})^\dagger \vec{Z} = \mathcal{P}_Y^\dagger \vec{Z} \\ &= \vec{\alpha}_Z \end{aligned}$$

is seen to be equal to the vector of $\vec{Z}$ -EOF coefficients. Thus the EOF coefficients are invariant under *orthogonal transformations*. They are generally not invariant under non-orthogonal transformations. This becomes important when different variables such as precipitation and temperature are combined in a data vector. The EOFs of such a random vector depend on the units in which the variables are expressed.

A special case of transformation (13.15) occurs when $\vec{X}$ has already been transformed into EOF coordinates using (13.10),

$$\vec{Z} = \vec{\alpha} = \mathcal{P}^\dagger \vec{X}.$$

That is, $\mathcal{L} = \mathcal{P}^\dagger$. Using transformation (13.18), we see that the $\vec{\alpha}$ -EOFs are

$$\mathcal{P}_\alpha = \mathcal{L}\mathcal{P} = \mathcal{P}^\dagger \mathcal{P} = \mathcal{I}.$$

Thus, in the new coordinates the EOFs are unit vectors. This fact may be used to test EOF programs.

13.1.11 Further Aspects. Some other aspects of EOFs are worth mentioning.

- Empirical Orthogonal Functions may be generalized to continuous functions, in which case they are known as *Karhunen-Loève* functions. The standard inner product $\langle \cdot, \cdot \rangle$ is replaced by an integral, and the eigenvalue problem is no longer a matrix problem but an operator problem. (See, e.g., North et al. [296].)

- The EOFs of some random vectors or random functions are given by sets of analytic orthogonal functions. For instance, if the covariance structure of a spatial process is independent of the location, then the EOFs on a continuous or regularly discretized sphere (circle) are the spherical harmonics (trigonometric functions). See North et al. [296].
- The analysed vector $\vec{X}$ may be a combination of small vectors that are expressed on different scales, such as temperature and precipitation or geopotential height at 700 and 200 hPa. Then the technique is sometimes called *Combined Principal Component Analysis* (see, e.g., Bretherton, Smith, and Wallace [64]). Vector $\vec{X}$ might also consist of smaller vectors representing a single field observed at different times, in which case the technique is called *Extended EOF Analysis* (EEOF; see Weare and Nasstrom [417]) or *Multichannel Singular Spectrum Analysis* (MSSA; see Section 13.6).
- Any m -dimensional vector $\vec{y}$ can be projected onto an EOF $\vec{e}$ of $\vec{X}$ by computing the inner product $\langle \vec{y}, \vec{e} \rangle$. Vector $\vec{y}$ can then by approximated by $\vec{y} \approx \sum_{i=1}^k \langle \vec{y}, \vec{e}^i \rangle \vec{e}^i$.
- Where are the units? When we expand the random vector $\vec{X}$ into EOFs

$$\vec{X} \approx \sum_{i=1}^k \alpha_i \vec{e}^i \quad (13.19)$$

with $\alpha_i = \langle \vec{X}, \vec{e}^i \rangle$, where do we place the units of $\vec{X}$ on the right side of approximation (13.19)? Formally the answer is that the coefficients carry the units while the patterns are dimensionless. However, in practice approximation (13.19) is often replaced by

$$\vec{X} \approx \sum_{i=1}^k \alpha_i^+ \vec{e}^i \quad (13.20)$$

with re-normalized coefficients

$$\alpha_i^+ = \frac{1}{\sqrt{\lambda_i}} \alpha_i \quad (13.21)$$

and patterns

$$\vec{e}^{i+} = \sqrt{\lambda_i} \vec{e}^i \quad (13.22)$$

so that $\text{Var}(\alpha_i^+) = 1$. The re-normalized pattern then carries the units of $\vec{\mathbf{X}}$, and represents a ‘typical’ anomaly pattern if we regard $\alpha_i^+ = \pm 1$ as a ‘typical event’.

The decomposition of the local variance, as given by equation (13.7), takes a particularly simple form with this normalization, namely

$$\text{Var}(\mathbf{X}_k) = \sum_{i=1}^m |e_k^{i+}|^2. \quad (13.23)$$

Note that the coefficient α_i^+ can be expressed as

$$\alpha_i^+ = \frac{1}{\lambda_i} \langle \vec{\mathbf{X}}, \vec{e}^i \rangle. \quad (13.24)$$

13.2 Estimation of Empirical Orthogonal Functions

13.2.1 Outline. After having defined the eigenvalues and EOFs of random vector $\vec{\mathbf{X}}$ as parameters that characterize its covariance matrix, the question naturally arises as to how to *estimate* these parameters from sample $\{\vec{\mathbf{x}}_1, \dots, \vec{\mathbf{x}}_n\}$ of realizations of $\vec{\mathbf{X}}$. It turns out that useful estimators may be defined by replacing the covariance matrix Σ with the sample covariance matrix $\hat{\Sigma}$ and by replacing the expectation operator $\mathcal{E}(\cdot)$ with averaging over the sample. An important little trick for reducing the amount of calculation when the sample size n is less than the dimension of $\vec{\mathbf{X}}$ (as is often true) is presented in [13.2.5]. A computational alternative to solving the eigenproblem is to perform a singular value decomposition [13.2.8].

13.2.2 Strategies for Estimating EOFs. The eigenvalues and EOFs are parameters that characterize the covariance matrix of a random vector $\vec{\mathbf{X}}$. In practice, the distribution of $\vec{\mathbf{X}}$, and thus the covariance matrix Σ and its eigenvalues and eigenvectors, is unknown. They must therefore be estimated from a finite sample $\{\vec{\mathbf{x}}_1, \dots, \vec{\mathbf{x}}_n\}$.

There are two reasonable approaches for estimation.

- Since the eigenvalues and EOFs characterize the covariance matrix Σ of $\vec{\mathbf{X}}$, one reasonable approach is to estimate the covariance matrix and then estimate the eigenvalues λ_i and eigenvectors $\vec{e}^i$ with the eigenvalues λ_j and the eigenvectors $\vec{e}^j$ of the estimated covariance matrix $\hat{\Sigma}$.

- After [13.1.3] the EOFs form an orthonormal set of vectors that is most efficient in representing the variance of $\vec{\mathbf{X}}$ (13.3). Thus another reasonable approach is to use a set of orthonormal vectors that represent as much of the sample variance of the finite sample as possible.

The two approaches are equivalent and lead to the following.

13.2.3 Theorem. Let $\hat{\Sigma} = \frac{1}{n} \sum_{j=1}^n (\vec{\mathbf{x}}_j - \hat{\mu})(\vec{\mathbf{x}}_j - \hat{\mu})^\dagger$, where $\dagger$ indicates the conjugate transpose and $\hat{\mu} = \frac{1}{n} \sum_{j=1}^n \vec{\mathbf{x}}_j$, derived from a sample $\{\vec{\mathbf{x}}_1, \dots, \vec{\mathbf{x}}_n\}$ be the estimated covariance matrix of n realizations of $\vec{\mathbf{X}}$. Let $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_m$ be the eigenvalues of $\hat{\Sigma}$ and let $\hat{e}^1, \dots, \hat{e}^m$ be corresponding eigenvectors of unit length. Since $\hat{\Sigma}$ is Hermitian, the eigenvalues are non-negative and the eigenvectors are orthogonal.

(i) The k eigenvectors that correspond to $\hat{\lambda}_1, \dots, \hat{\lambda}_k$ minimize

$$\hat{\epsilon}_k = \sum_{j=1}^n \left| \vec{\mathbf{x}}_j - \sum_{i=1}^k (\vec{\mathbf{x}}_j, \hat{e}^i) \hat{e}^i \right|^2. \quad (13.25)$$

$$(ii) \quad \hat{\epsilon}_k = \widehat{\text{Var}}(\vec{\mathbf{X}}) - \sum_{j=1}^k \hat{\lambda}_j. \quad (13.26)$$

$$(iii) \quad \widehat{\text{Var}}(\vec{\mathbf{X}}) = \sum_{j=1}^m \hat{\lambda}_j, \quad (13.27)$$

where $\widehat{\text{Var}}(\vec{\mathbf{X}}) = \text{tr}(\hat{\Sigma})$.

13.2.4 The Estimated Covariance Matrix $\hat{\Sigma}$. The covariance between the j th and k th elements of $\vec{\mathbf{X}}$ is estimated by

$$\hat{\sigma}_{jk} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_{ji} - \bar{\mathbf{x}}_j)(\mathbf{x}_{ki} - \bar{\mathbf{x}}_k),$$

where $\mathbf{x}_{ji}$ and $\mathbf{x}_{ki}$ are the j th and k th elements of $\vec{\mathbf{x}}_i$. This sum of products can be expressed as a *quadratic form*:¹⁵

$$\hat{\Sigma} = \frac{1}{n} \mathcal{X}(\mathcal{I} - \frac{1}{n} \mathcal{J})(\mathcal{I} - \frac{1}{n} \mathcal{J}) \mathcal{X}^\dagger \quad (13.28)$$

where $\mathcal{X}$ is the *data matrix*¹⁶

$$\mathcal{X} = \begin{pmatrix} \mathbf{x}_{11} & \mathbf{x}_{12} & \dots & \mathbf{x}_{1n} \\ \mathbf{x}_{21} & \mathbf{x}_{22} & \dots & \mathbf{x}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}_{m1} & \mathbf{x}_{m2} & \dots & \mathbf{x}_{mn} \end{pmatrix}, \quad (13.29)$$

¹⁵A quadratic form is a matrix product of the form $\mathcal{A} \mathcal{A}^\dagger$ or $\mathcal{A}^\dagger \mathcal{A}$.

¹⁶Sometimes also called the design matrix.

and $\mathcal{X}^\dagger$ is the conjugate transpose of $\mathcal{X}$, $\mathcal{I}$ is the $n \times n$ identity matrix, and $\mathcal{J}$ is the $n \times n$ matrix composed entirely of units. The n columns of the $m \times n$ data matrix $\mathcal{X}$ are the sample vectors $\vec{\mathbf{x}}_1, \dots, \vec{\mathbf{x}}_n$; the rows mark the m coordinates in the original space. The matrix product $\mathcal{X}\mathcal{X}^\dagger$ is a square matrix even if $\mathcal{X}$ is not.

13.2.5 Theorem. The following theorem is often useful when computing eigenvalues and eigenvectors [391].

Let $\mathcal{A}$ be any $m \times n$ matrix. If λ is a nonzero eigenvalue of multiplicity s of $\mathcal{A}^\dagger \mathcal{A}$ with s linearly independent eigenvectors $\vec{e}^1, \dots, \vec{e}^s$, then λ is also an s -fold eigenvalue of $\mathcal{A}\mathcal{A}^\dagger$ with s linearly independent eigenvectors $\mathcal{A}\vec{e}^1, \dots, \mathcal{A}\vec{e}^s$.

A proof is given in Appendix M.

13.2.6 Recipe. The message of Theorem [13.2.5] is that the nonzero eigenvalues of $\mathcal{A}\mathcal{A}^\dagger$ are identical to those of $\mathcal{A}^\dagger \mathcal{A}$ and that the eigenvectors of the two matrices associated with nonzero eigenvalues are related through a simple linear relationship. Thus the following recipe may be used to estimate EOFs.

- If the sample size, n , is larger than the dimension of the problem, m , then the EOFs are calculated directly as the normalized eigenvectors of the $m \times m$ matrix $\frac{1}{n}\mathcal{X}(\mathcal{I} - \frac{1}{n}\mathcal{U})(\mathcal{I} - \frac{1}{n}\mathcal{U})\mathcal{X}^\dagger$.
- If the sample size, n , is smaller than the dimension of the problem, m , the EOFs may be obtained by first calculating the normalized eigenvectors $\vec{g}$ of the $n \times n$ matrix $\frac{1}{n}(\mathcal{I} - \frac{1}{n}\mathcal{J})\mathcal{X}^\dagger \mathcal{X}(\mathcal{I} - \frac{1}{n}\mathcal{J})$ and then computing the EOFs as

$$\vec{e} = \frac{\mathcal{X}(\mathcal{I} - \frac{1}{n}\mathcal{J})\vec{g}}{\|\mathcal{X}(\mathcal{I} - \frac{1}{n}\mathcal{J})\vec{g}\|}.$$

13.2.7 Properties of the Coefficients of the Estimated EOFs. There are several properties worth noting.

- As with the true EOFs, the estimated EOFs span the full m -dimensional vector space. Random vector $\vec{\mathbf{X}}$ can therefore be expanded in terms of the estimated EOFs as $\vec{\mathbf{X}} = \sum_{j=1}^m \hat{\alpha}_j \vec{e}^j$, where

$$\hat{\alpha}_j = \langle \vec{\mathbf{X}}, \vec{e}^j \rangle. \quad (13.30)$$

- When $\vec{\mathbf{X}}$ is multivariate normal, the distribution of $\hat{\alpha}$, where $\hat{\alpha}$ is the m -dimensional vector of EOF coefficients $\hat{\alpha}_j$, conditional upon the samples used to estimate the EOFs is multivariate normal with mean $\mathcal{E}(\hat{\alpha}|\vec{\mathbf{x}}_1, \dots, \vec{\mathbf{x}}_n) = \hat{\mathcal{P}}^\dagger \vec{\mu}$ and covariance matrix $\text{Cov}(\hat{\alpha}|\vec{\mathbf{x}}_1, \dots, \vec{\mathbf{x}}_n) = \hat{\mathcal{P}}^\dagger \Sigma \hat{\mathcal{P}}$. Matrix $\hat{\mathcal{P}}$, which has $\vec{e}^j$ in column j , is a complicated function of $\vec{\mathbf{x}}_1, \dots, \vec{\mathbf{x}}_n$.

- $\hat{\lambda}_j$ is the variance of the EOF coefficients computed from the sample used to estimate the EOFs. That is, if $\hat{\alpha}_{ji} = \langle \vec{\mathbf{X}}_i, \vec{e}^j \rangle$, then $\frac{1}{n} \sum_{i=1}^n |\hat{\alpha}_{ji} - \bar{\alpha}_j|^2 = \hat{\lambda}_j$.

Note that $\hat{\lambda}_j$ has at least two interpretations as a variance estimate. We could regard $\hat{\lambda}_j$ as an estimate of the variance of the true EOF coefficient $\alpha_j = \langle \vec{\mathbf{X}}, \vec{e}^j \rangle$ (see [13.3.3]). Alternatively, we could view the estimated EOFs $\vec{e}^j$ as fixed, not quite optimal, proxies for $\vec{e}^j$. Then $\hat{\lambda}_j$ could be viewed as an estimator of the variance of $\hat{\alpha}_i = \langle \vec{\mathbf{X}}, \vec{e}^j \rangle$ when $\vec{e}^j$ is fixed (see [13.3.2]). These two variances are not equal, although they become asymptotically equivalent as $n \rightarrow \infty$. Thus, at least one of the interpretations makes $\hat{\lambda}_j$ a biased estimator. In fact, they are both poor estimators when the sample is small. In the former case there is uncertainty because the EOFs must be estimated. In the latter case the EOFs are regarded as fixed, but there is a bias because independent data are not used to estimate $\text{Var}(\hat{\alpha}_i)$. See also [13.3.2,3].

- The sample covariance of a pair of EOF coefficients computed from the sample used to estimate the EOFs is zero. That is, $\frac{1}{n} \sum_{i=1}^n (\hat{\alpha}_{ji} - \bar{\alpha}_j)(\hat{\alpha}_{ki} - \bar{\alpha}_k)^* = 0$ if $j \neq k$.

As with $\hat{\lambda}_j$, the covariance has two interpretations. It correctly estimates the covariance of the true EOF coefficients $\alpha_j = \langle \vec{\mathbf{X}}, \vec{e}^j \rangle$ and $\alpha_k = \langle \vec{\mathbf{X}}, \vec{e}^k \rangle$. Alternatively, if we view the estimated EOFs $\vec{e}^j$ as being fixed, then it incorrectly estimates $\text{Cov}(\hat{\alpha}_j, \hat{\alpha}_k)$. The latter, the (j, k) element of $\hat{\mathcal{P}}^\dagger \Sigma \hat{\mathcal{P}}$, can be substantially different from zero if $\vec{e}^j$ and $\vec{e}^k$ are computed from a small sample.

13.2.8 Gappy Data. Data are often incomplete, that is, there are irregularly distributed gaps in the data vectors caused by missing observations. Estimated EOFs and EOF coefficients can be derived in this case, but the procedure is slightly

different. Each element of Σ is estimated by forming sums of all available products

$$\hat{\sigma}_{ij} = \frac{1}{|K_i \cap K_j|} \sum_{k \in K_i \cap K_j} (\mathbf{x}_{ki} - \hat{\mu}_i)(\mathbf{x}_{kj} - \hat{\mu}_j)^* \quad (13.31)$$

where $K_i = \{k: \text{component } i \text{ of } \bar{\mathbf{x}}_k \text{ is not missing}\}$, and where $\hat{\mu}_i = \frac{1}{|K_i|} \sum_{k \in K_i} \mathbf{x}_{ki}$. The estimated EOFs are then the eigenvectors $\hat{\mathbf{e}}^i$ of this covariance matrix estimate. The set $K_i \cap K_j$ is the set of all indices such that $\mathbf{x}_{ki}$ and $\mathbf{x}_{kj}$ are not missing. The $|\cdot|$ notation is used to indicate the size of the enclosed set.

The EOF coefficient $\hat{\alpha}_i$ of a gappy data vector $\bar{\mathbf{x}}$ can not be obtained as a simple dot product of the gappy data vector $\bar{\mathbf{x}}$ and the estimated EOF $\hat{\mathbf{e}}^i$, as in equation (13.30), but a least squares estimate can be obtained by choosing $\hat{\alpha}_i$ to minimize $\|\bar{\mathbf{x}} - \hat{\alpha}_i \hat{\mathbf{e}}^i\|$. The least square estimate is given by

$$\hat{\alpha}_i = \frac{\sum_{j \in K} \mathbf{x}_j \hat{\mathbf{e}}_j^{i*}}{\sum_{j \in K} |\hat{\mathbf{e}}_j^i|^2} \quad (13.32)$$

where $\mathbf{x}_j$ and $\hat{\mathbf{e}}_j^i$ are the j th components of $\mathbf{x}$ and $\hat{\mathbf{e}}^i$, respectively, and where $K = \{j: \mathbf{x}_j \text{ is not missing}\}$. Note that equation (13.32) reduces to $\hat{\alpha}_i = \langle \bar{\mathbf{x}}, \hat{\mathbf{e}}^i \rangle$ when there are no gaps in $\mathbf{x}$.

13.2.9 Computing Eigenvalues and Eigenvectors. One approach to computing eigenvalues and eigenvectors is to use a ‘canned’ eigenanalysis routine such as those that are contained in EISPACK [352], IMSL [193], or NAG [298]. Press et al. [322, p. 454] discuss the origins of these routines and give further references.

An alternative approach uses Singular Value Decomposition (Appendix B, and see also Press et al. [322, pp. 51–63] and Kelly [218, 219]). The SVD of the conjugate transpose of the $m \times n$ centred data matrix

$$\mathcal{X}' = \mathcal{X} \left(\mathcal{I} - \frac{1}{n} \mathcal{J} \right), \quad (13.33)$$

where $\mathcal{X}$ is given by equation (13.29), $\mathcal{I}$ is the $n \times n$ identity matrix, and $\mathcal{J}$ is the $n \times n$ matrix of units, is

$$\mathcal{X}'^\dagger = \mathcal{U} \mathcal{S} \mathcal{V}^\dagger, \quad (13.34)$$

where $\mathcal{U}$ is $n \times m$, $\mathcal{S}$ and $\mathcal{V}$ are each $m \times m$, n is the sample size, and m is the dimension of

$\bar{\mathbf{X}}$.¹⁷ Since $n\hat{\Sigma} = \mathcal{X}'\mathcal{X}'^\dagger$, we infer from equation (B.6) that the right singular vectors $\bar{\mathbf{v}}^i$ are equal to the estimated EOFs $\hat{\mathbf{e}}^i$. The singular values s_i are related to the estimated eigenvalues by $\hat{\lambda}_i = \frac{1}{n}s_i^2$. The left singular vectors $\bar{\mathbf{u}}^i$ are given by (B.5)

$$\bar{\mathbf{u}}^i = \frac{1}{s_i} \mathcal{X}'^\dagger \bar{\mathbf{v}}^i. \quad (13.35)$$

The k th column of $\mathcal{X}'$ represents the vector of deviations $\bar{\mathbf{x}}_k - \hat{\mu}$ so that

$$u_k^i = \frac{1}{s_i} (\bar{\mathbf{x}}_k - \hat{\mu})^\dagger \bar{\mathbf{v}}^i. \quad (13.36)$$

Thus, u_k^i is the i th normalized EOF coefficient (13.21) of the anomalies $\bar{\mathbf{x}}_k - \hat{\mu}$. Note that the sample variance of the i th normalized EOF coefficient is

$$\begin{aligned} \frac{1}{n} \sum_{k=1}^n \left(\frac{1}{s_i} (\bar{\mathbf{x}}_k - \hat{\mu})^\dagger \bar{\mathbf{v}}^i \right)^2 &= \frac{1}{s_i^2} \widehat{\text{Var}}(\hat{\alpha}_i) \\ &= \frac{1}{s_i^2} \hat{\lambda}_i = 1. \end{aligned} \quad (13.37)$$

Note also that equations (13.35)–(13.37) are only valid for those EOFs that correspond to nonzero eigenvalues. The number of nonzero eigenvalues, which is determined by the rank¹⁸ of the centred data matrix, is no greater than $\min(m, n - 1)$.

Thus SVD extracts the same information from the sample as a conventional EOF analysis.

13.3 Inference

13.3.1 General. We consider the reliability of eigenvalues and EOF estimates in this section. This is a somewhat different question from that which users generally have in mind when they enquire about the ‘significance’ of an EOF. The null hypothesis that is usually implicit in the latter is that the EOF in question describes only an aspect of the covariance structure of the ‘noise’ in the observed system, and the alternative hypothesis is that the EOF also describes at least part of the dynamics of the observed system. Unfortunately, discrimination between ‘noise’ and ‘signal’ in

¹⁷We have implicitly assumed here that $m \leq n$. The problem is approached similarly when $m > n$, except we begin by obtaining the SVD of $\mathcal{X}'$. Note also that in some texts $\mathcal{U}$ and $\mathcal{V}$ are $n \times n$ and $m \times m$ orthogonal matrices respectively and $\mathcal{S}$ is $n \times m$. The singular values are placed in the diagonal part of $\mathcal{S}$ and the rest of the matrix is zero. We use the decomposition given in (13.34) because it is commonly used in SVD subroutines (see, e.g., Press et al. [322]).

¹⁸The rank of a matrix is the dimension of the sub-space spanned by the columns of that matrix.

this way is fraught with difficulty. We discuss this further in [13.3.4]. However, we first briefly consider the variance of EOF coefficients in [13.3.2] and the bias of eigenvalue estimates in [13.3.3]. We consider the sampling error of the EOFs themselves in [13.3.5,6].

13.3.2 The Variance of EOF Coefficients of a Given Set of Estimated EOFs. Assume we are given a set of eigenvalues $\hat{\lambda}_i$ and EOFs $\hat{e}^i$ that are derived from a finite sample $\{\vec{x}_1 \dots \vec{x}_n\}$. Then any random vector $\vec{X}$ can be represented in the space spanned by these estimated EOFs by using the transformation $\hat{\alpha} = \mathcal{P}^+ \vec{X}$. The transformed random variables $\hat{\alpha}_i = \langle \vec{X}, \hat{e}^i \rangle$ have their own moments, such as variances $\text{Var}(\hat{\alpha}_i)$ and covariances $\text{Cov}(\hat{\alpha}_i, \hat{\alpha}_j)$. In the following we view the estimated EOFs $\hat{e}^i$ as being ‘fixed’ (or frozen) rather than random.

Intuitively one would hope that the variance of $\hat{\alpha}_i$ is equal to that of the real EOF coefficient α_i . Unfortunately, this is not the case (see [13.2.6]). Consider, for example, the first EOF $\hat{e}^1$ and the corresponding EOF coefficient $\alpha_1 = \langle \vec{X}, \hat{e}^1 \rangle$. The first EOF minimizes

$$\epsilon_1 = \mathcal{E}(\|\vec{X} - \langle \vec{X}, \hat{e}^1 \rangle \hat{e}^1\|^2).$$

Replacing $\hat{e}^1$ with any other vector, such as $\hat{e}^1$, increases ϵ_1 . Thus

$$\begin{aligned} \text{Var}(\vec{X}) - \text{Var}(\alpha_1) &= \mathcal{E}(\|\vec{X} - \langle \vec{X}, \hat{e}^1 \rangle \hat{e}^1\|^2) \\ &< \mathcal{E}(\|\vec{X} - \langle \vec{X}, \hat{e}^1 \rangle \hat{e}^1\|^2) \\ &= \text{Var}(\vec{X}) - \text{Var}(\hat{\alpha}_1), \end{aligned}$$

that is, $\text{Var}(\alpha_1) > \text{Var}(\hat{\alpha}_1)$. Similar arguments lead to

- $\text{Var}(\hat{\alpha}_i) < \text{Var}(\alpha_i)$ for the first few EOFs (those corresponding to the largest eigenvalues).

Since the total variance $\text{Var}(\vec{X}) = \sum_{j=1}^m \text{Var}(\vec{x}_j)$ is estimated with nearly zero bias by $\widehat{\text{Var}}(\vec{X}) = \sum_{j=1}^m \hat{\lambda}_j$, it follows that

- $\text{Var}(\hat{\alpha}_i) > \text{Var}(\alpha_i)$ for the last few EOFs.

Examples show that these deviations may be considerable, in particular for small eigenvalues [392].

13.3.3 The Bias in Estimating Eigenvalues. It is natural to ask questions about the reliability of eigenvalues and EOF estimates, such as the extent to which the estimated patterns resemble the true patterns and how close the estimated and true eigenvalues are. These questions do not have completely satisfactory answers, but there are a number of potentially useful facts. One of these facts is the following set of asymptotic formulae that apply to eigenvalue estimates computed from samples that can be represented by n independent and identically distributed normal random vectors (Lawley [245]):

$$\mathcal{E}(\hat{\lambda}_i) = \lambda_i \left(1 + \frac{1}{n} \sum_{\substack{j=1 \\ j \neq i}}^m \frac{\lambda_j}{\lambda_i - \lambda_j} \right) + \mathcal{O}(n^{-2}) \quad (13.38)$$

$$\begin{aligned} \text{Var}(\hat{\lambda}_i) &= \frac{2\lambda_i^2}{n} \left(1 - \frac{1}{n} \sum_{\substack{j=1 \\ j \neq i}}^m \left(\frac{\lambda_j}{\lambda_i - \lambda_j} \right)^2 \right) \\ &+ \mathcal{O}(n^{-3}). \end{aligned} \quad (13.39)$$

As usual, m is the dimension of the random vectors. The symbol $\mathcal{O}(n^{-s})$ represents a term that converges to zero as $n \rightarrow \infty$ at least as quickly as n^{-s} does.

By equations (13.38) and (13.39), the eigenvalue estimators are consistent:

$$\lim_{n \rightarrow \infty} \mathcal{E}((\hat{\lambda}_i - \lambda_i)^2) = 0. \quad (13.40)$$

However, something unwanted is hidden in equation (13.38), namely that the estimators of the largest and of the smallest eigenvalues are biased. For the largest eigenvalues, almost all of the denominators in equation (13.38) are positive so that the entire sum is positive, that is, $\mathcal{E}(\hat{\lambda}_i) > \lambda_i$ for large eigenvalues. Similarly, $\mathcal{E}(\hat{\lambda}_i) < \lambda_i$ for the smallest eigenvalues.

Together with the results from [13.3.2] this finding shows that

- for the largest eigenvalues λ_i ,

$$\mathcal{E}(\hat{\lambda}_i) > \lambda_i = \text{Var}(\alpha_i) > \text{Var}(\hat{\alpha}_i), \quad (13.41)$$

- for the smallest eigenvalues λ_i ,

$$\mathcal{E}(\hat{\lambda}_i) < \lambda_i = \text{Var}(\alpha_i) < \text{Var}(\hat{\alpha}_i). \quad (13.42)$$

Relations (13.41) and (13.42) illustrate that we must be cautious when using estimated eigenvalues. First, the estimates are biased: the

large eigenvalues are overestimated and the small ones are underestimated. More important, though, is the inequality $\mathcal{E}(\hat{\lambda}_i) \neq \text{Var}(\hat{\alpha}_i)$: The sample eigenvalue $\hat{\lambda}$ is a biased estimator of the variance of $\hat{\alpha}_i = (\bar{\mathbf{X}}, \hat{\mathbf{e}}^i)$, for any frozen set of estimated EOFs $\hat{\mathbf{e}}^i$. Similarly, $\text{Cov}(\hat{\alpha}_i, \hat{\alpha}_j) \neq \widehat{\text{Cov}}(\hat{\alpha}_i, \hat{\alpha}_j) = 0$.

13.3.4 ‘Selection Rules.’ Many so-called *selection rules* have been proposed that supposedly separate the physically relevant EOFs from those that are not.¹⁹ One popular procedure of this type is ‘Rule N’ [321]. The basic supposition is that the full phase space can be partitioned into one subspace that contains only noise and another that contains dynamical variations (or ‘signals’). It is assumed that the signal-subspace is spanned by well-defined EOFs while those in the noise-subspace are degenerate. Thus, the idea is to attempt to identify the signal-subspace as the space spanned by the EOFs that are associated with large, well-separated eigenvalues.

The selection rules compare the eigenspectrum²⁰ estimated from the sample with distributions of sample eigenspectra that are obtained under the assumption that all or the smallest $\tilde{m}$ true eigenvalues are equal. The number $\tilde{m}$ is either specified *a priori* or determined recursively. All estimated eigenvalues that are larger than, say, the 95% percentile of the (marginal) distribution of the reference ‘noise’ spectra, are identified as being ‘significant’ at the 5% level.

One problem with this approach is that this selection rule is mistakenly understood to be a *statistical test* of the null hypothesis that EOFs $\bar{\mathbf{e}}^1, \dots, \bar{\mathbf{e}}^{\tilde{m}}$, for $\tilde{m} < m$, span noise against the alternative hypothesis that they span the signal-subspace. The connection between this alternative and the determination of a ‘signal-subspace’ is vague. Also, the approach sketched above does not consider the reliability of the estimated patterns since the selection rules are focused only on the eigenvalues.

The other problem with the ‘selection rule’ approach is that there need not be any connection between the shape of the eigenspectrum on the one hand and the presence or absence of ‘dynamical structure’ on the other. To illustrate, suppose that a process $\bar{\mathbf{X}}_t = \bar{\mathbf{D}}_t + \bar{\mathbf{N}}_t$, containing both dynamical

and noise components (recall [10.1.1]), has eigenvalues $\lambda_1, \dots, \lambda_m$ and EOFs $\bar{\mathbf{e}}^1, \dots, \bar{\mathbf{e}}^m$. Now construct a multivariate white noise $\bar{\mathbf{Z}}_t$ from iid $\mathcal{N}(0, \mathcal{I})$ random vectors. Then the multivariate white noise process $\bar{\mathbf{Y}}_t = \mathcal{P}\Lambda^{1/2}\bar{\mathbf{Z}}_t$, which is completely devoid of ‘dynamics,’ has the same eigenvalues and eigenvectors as $\bar{\mathbf{X}}_t$. Thus we cannot always diagnose dynamical structure from the zero lag covariance structure of a process.²¹

Our recommendation is to avoid using selection rules. We outline a better approach, based on *North’s Rule-of-Thumb*, in the next subsection.

13.3.5 North’s Rule-of-Thumb. Using a scale argument, North et al. [296] obtained an approximation for the ‘typical’ error of the estimated EOFs

$$\Delta \hat{\mathbf{e}}^i \approx \sqrt{\frac{2}{n}} \sum_{\substack{j=1 \\ j \neq i}}^m \frac{c}{\lambda_j - \lambda_i} \bar{\mathbf{e}}^j \quad (13.43)$$

where c is a constant and n is the number of *independent* samples. There are three things to notice about this equation.

- The first-order error $\Delta \hat{\mathbf{e}}^i$ is of the order of $\sqrt{\frac{1}{n}}$. Thus convergence to zero is slow.
- The first-order error $\Delta \hat{\mathbf{e}}^i$ is orthogonal to the true EOF $\bar{\mathbf{e}}^i$.
- The estimate of the i th EOF $\bar{\mathbf{e}}^i$ is most strongly contaminated by the patterns of those other EOFs $\bar{\mathbf{e}}^j$ that correspond to the eigenvalues λ_j closest to λ_i . The smaller the difference between λ_j and λ_i , the more severe the contamination.

Lawley’s formulae (13.38, 13.39) yield a first-order approximation of the ‘typical error’ in $\hat{\lambda}_i$:

$$\Delta \lambda_i \approx \sqrt{\frac{2}{n}} \lambda_i. \quad (13.44)$$

Combining this with a simplified version of approximation (13.44), North et al. [296] finally obtain

$$\Delta \hat{\mathbf{e}}^i \approx \frac{c' \Delta \lambda_i}{\lambda_j - \lambda_i} \bar{\mathbf{e}}^j \quad (13.45)$$

where c' is a constant and λ_j is the the closest eigenvalue to λ_i . North’s ‘Rule-of-Thumb’ follows from approximation (13.45): ‘If the sampling error

¹⁹See, for example, Preisendorfer, Zwiers, and Barnett [321].

²⁰An *eigenspectrum* is the distribution of variance (i.e., eigenvalues), with EOF index. The eigenspectrum is an analogue of the power spectrum (see Section 11.2) since both describe the distribution of variance across the coefficients of orthonormal basis functions.

²¹We would need to also analyse at least part of the lagged covariance structure of $\bar{\mathbf{X}}_t$ to reveal the ‘dynamics’ in this example.

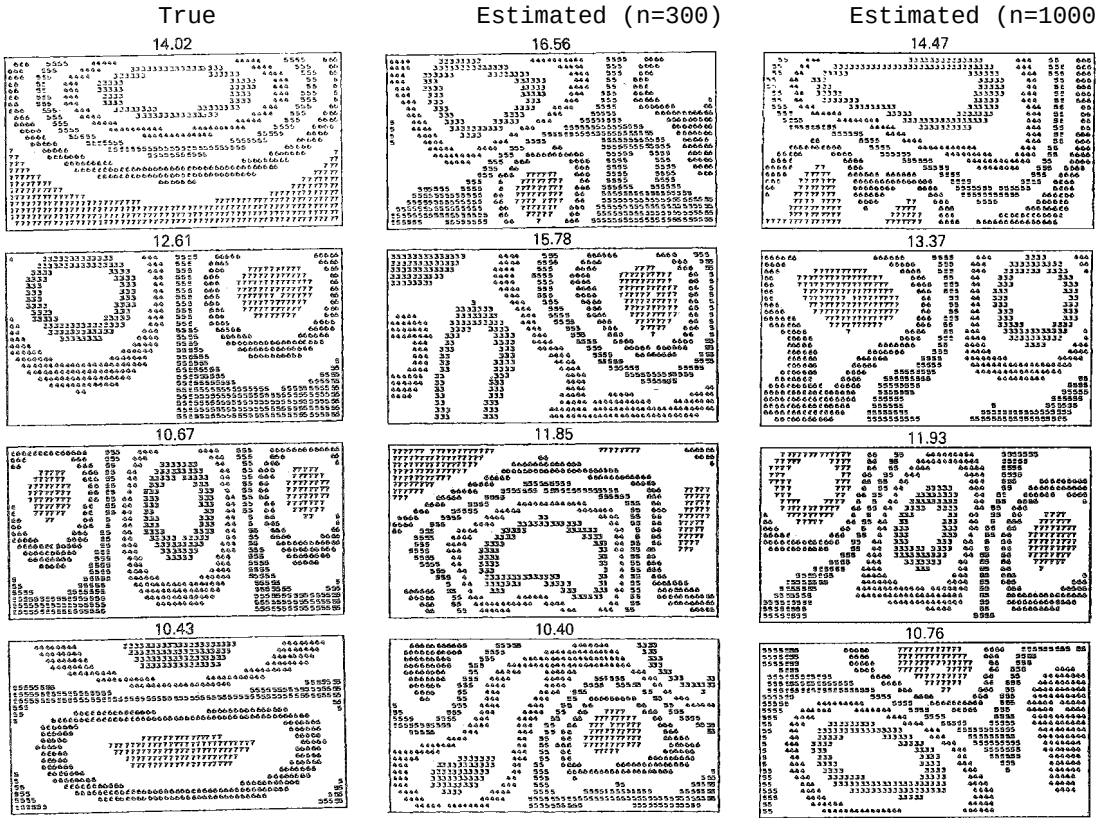


Figure 13.2: North et al.'s [296] illustration of North's Rule-of-Thumb [13.3.5]. From [296].

Left: The first four true eigenvalues and EOFs.
Middle: Corresponding estimates obtained from a random sample of size $n = 300$.
Right: As middle column, except $n = 1000$.

of a particular eigenvalue $\Delta\lambda$ is comparable to or larger than the spacing between λ and a neighbouring eigenvalue, then the sampling error $\Delta\tilde{e}$ of the EOF will be comparable to the size of the neighbouring EOF²².

13.3.6 North et al.'s Example. North et al. [296] constructed a synthetic example in which the first four eigenvalues are 14.0, 12.6, 10.7 and 10.4 to illustrate North's Rule-of-Thumb [13.3.4]. The first four (true) EOFs are shown in the left hand column of Figure 13.2. According to approximation (13.44) the typical error for the first four estimated eigenvalues is $\Delta\lambda_i \approx \pm 1$ for $n = 300$ and $\Delta\lambda_i \approx \pm 0.6$ for $n = 1000$. Since $\lambda_1 - \lambda_2 = 1.4$, $\lambda_2 - \lambda_3 = 2$ and $\lambda_3 - \lambda_4 = 0.3$, one would expect the first two EOFs to be mixed²² when $n = 300$ and the third and fourth EOF to be mixed for both $n = 300$ and $n = 1000$. That this is

a reasonable guess is demonstrated in the middle and right hand columns of Figure 13.2, which displays EOFs estimated from random samples of size $n = 300$ and $n = 1000$, respectively.

13.4 Examples

13.4.1 Overview. We will present two examples of conventional EOF analysis in this section. This first case, on the globally distributed SST, is most straight forward. The second example involves a data vector that is constructed by combining the same variable at several levels in the vertical.

13.4.2 Monthly Mean Global Sea-surface Temperature. The first two EOFs of monthly mean sea-surface temperature (SST) of the global ocean between 40°S and 60°N are shown in Figure 13.3. They represent 27.1% and 7.9% of the total variance, respectively.

²²That is, we expect the first two EOFs to be a combination of the EOFs that correspond to nearby eigenvalues.

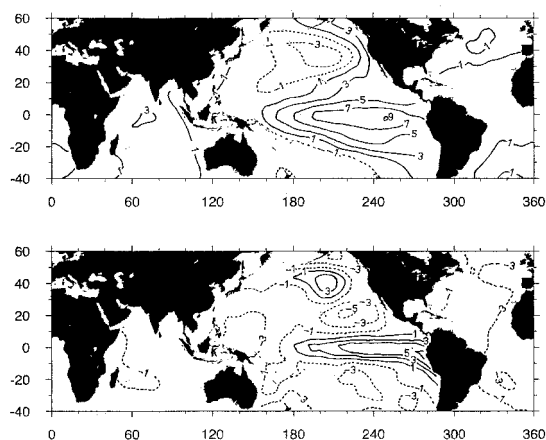


Figure 13.3: EOFs 1 (top) and 2 (bottom) of monthly mean sea-surface temperature (SST). Units: 10^{-2} . Courtesy Xu.

The first EOF, which is concentrated on the Pacific Ocean, represents ENSO. Its time coefficient, shown as curve ‘D’ in Figure 13.4, is highly correlated with the two Southern Oscillation indices (Darwin minus Papeete SLP, curve ‘E’, and SST area average, curve ‘F’) introduced previously. The large ‘centre of action’ in the North Pacific represents the oceanic response to anomalous extratropical winds which, in turn, were excited by the anomalous tropical state.

The second EOF of the SST field also involves the tropical Pacific Ocean. The most prominent feature is the narrow tongue of water in the eastern and central equatorial Pacific with temperatures that vary coherently. While the coefficient time series (Figure 13.4, curve ‘A’) reflects ENSO events (e.g., 1982/83) in part, the connection with the SOI is not as clear as with the first EOF. The coefficient appears to have a downward trend from about 1976 onwards, which would correspond to cooling in the eastern and central tropical Pacific or warming elsewhere. It remains to determine whether the trend is real, part of the global ocean’s natural low-frequency variability, or just an artifact of the way in which these data have been collected and analysed.

13.4.3 Monthly Mean of Zonal Wind at Various Levels. The next example is on the monthly mean zonal wind in the troposphere (Xu, personal communication). A joint analysis of the wind field at the 850, 700, 500, 300 and 200 hPa levels was performed. The size of the problem was kept

manageable by performing the analysis in two steps. Separate EOF analyses were first performed at each level. In each analysis, the coefficients representing 90% of the variance were retained. A combined vector, composed of EOF coefficients selected for the five levels, is used as input for the eventual EOF analysis of the three-dimensional zonal wind field.

The first two EOFs are shown in Figure 13.5, and their coefficient time series are shown as traces ‘B’ and ‘C’ in Figure 13.4. The first EOF, representing 11% of the total monthly variance, is mostly barotropic, not only in the extratropics but also in the tropics. Its coefficient time series exhibits a trend parallel to that found in the coefficient of the second SST EOF. The mean westerly winds in the Southern Hemisphere were analysed as being weaker in the 1970s than in the mid 1980s (negative sign indicates easterly wind anomalies). At the same time the mean low-level easterlies along the equatorial Pacific were weaker in the early 1970s and stronger in the mid 1980s (positive anomalies represent anomalous westerly winds). The results of the EOF analysis of the SST in [13.4.2] are consistent with this representation: The second SST EOF described an equatorial Pacific that was warmer in the early 1970s and cooler in the 1980s, a phenomenon that should be accompanied by strengthening easterly trades during this period.

Because independent analysis techniques are used to derive the SST and zonal wind fields, we can conclude that the trend found in both EOF analyses is not due to data problems. However, it is still not possible to determine whether the trend originates from a natural low-frequency variation or from some other cause.

13.5 Rotation of EOFs

13.5.1 Introduction. This section describes a class of basis vector ‘rotation’ procedures that is widely used in climate research. The procedures are usually applied to EOFs in the hope that the resulting ‘rotated EOFs’ can be more easily interpreted than the EOFs themselves. The term ‘rotated EOFs’ is a mild misnomer that may lead to confusion; ‘rotation’ transforms the Empirical Orthogonal Function into a *non-orthogonal* linear basis. Also, ‘rotation’ can be performed on any linear basis, not just EOFs.

We will first explain the general idea and will then describe ‘varimax’-rotation in some detail. We use three examples to describe the merits

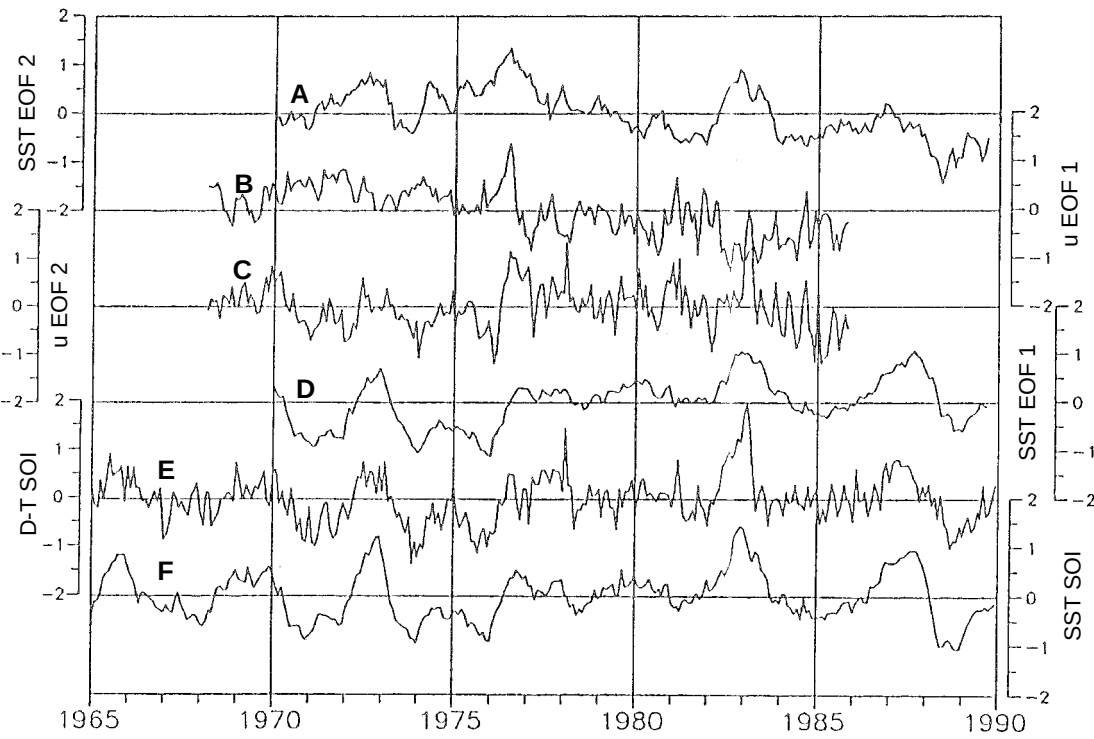


Figure 13.4: EOF coefficients of monthly mean sea-surface temperature [13.4.2] (curves D and A), monthly mean zonal wind [13.4.3] (curves B and C) and two Southern Oscillation indices [1.2.2] (curves E and F). All data are normalized.

A: 2nd SST EOF coefficient. B: 1st zonal wind EOF coefficient. C: 2nd zonal wind EOF coefficient. D: 1st SST EOF coefficient. E: Darwin minus Papeete SLP index of the Southern Oscillation. F: SST index of the Southern Oscillation.

Courtesy Xu.

of this procedure. The first of these examples is on the successful and reproducible identification of teleconnection patterns (cf. Section 17.4). The second example deals with a case in which the effect of the rotation is negligible. The third case illustrates pathological behaviour by showing that rotation sometimes splits features into different patterns even though they are part of the *same physical pattern*.

13.5.2 The Concept of ‘Rotation.’ Having used EOF analysis, or some other technique, to identify a low-dimensional subspace that contains a substantial fraction of the total variance, it is sometimes of interest to look for a linear basis of this subspace with specified properties, such as the following.

- Basis vectors that contain simple geometrical patterns. Simplicity could mean that the patterns are confined regionally, or that the patterns are composed of two regions, one

with large positive values and another with large negative values.

- Basis vectors that have time coefficients with specific types of behaviour, such as having nonzero values only during some compact time episodes.

Richman [331] lists five vague criteria for simple structure and there are many proposals of ‘simplicity’ functionals. The minimization of these functionals is generally non-trivial since the functionals are nonlinear. Numerical algorithms used to obtain approximate solutions can only be applied to bases of moderate size.

The results of a rotation exercise depend on the number and length of the ‘input vectors’, and on the measure of simplicity. Successful application of the rotation technique requires some experience and the novice may find Richman’s [331] review paper on rotation useful. Interesting examples are

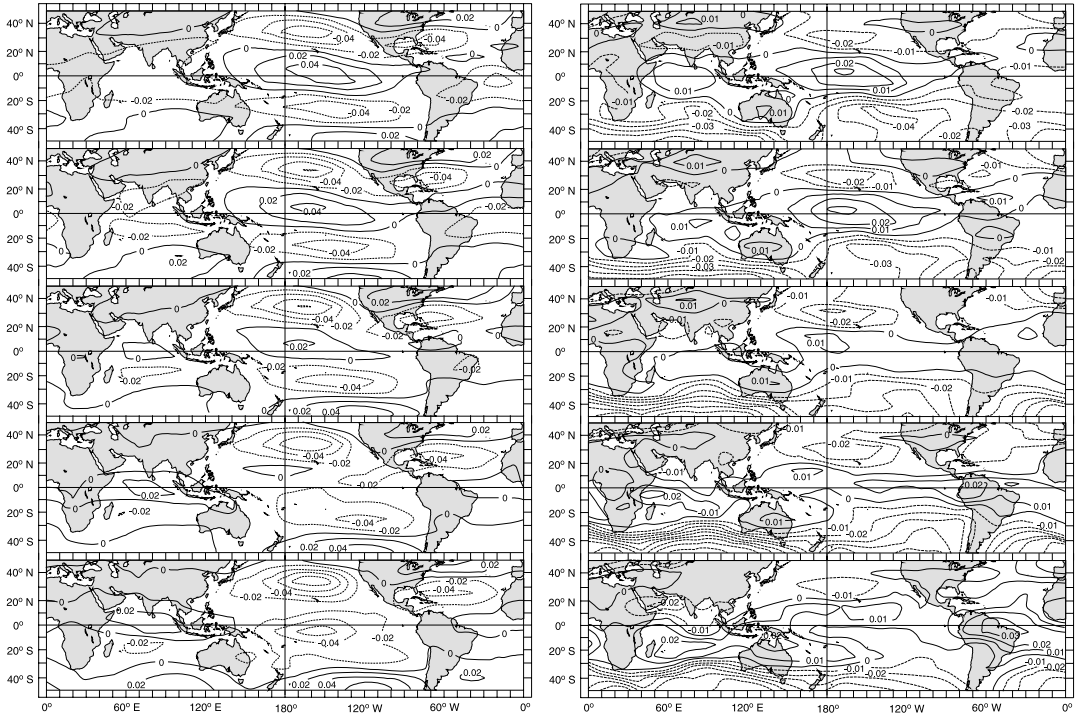


Figure 13.5: First two EOFs of the tropospheric zonal wind between 45°S and 45°N , at 850, 700, 500, 300 and 200 hPa (from bottom to top). First EOF on the left, second on the right.

Courtesy Xu.

offered by Barnston and Livezey [27] and Chelliah and Arkin [80], among many others.

The opinion in the community is divided on the subject of rotation. Part of the community advocates the use of rotation fervently, arguing that it is a means with which to diagnose physically meaningful, statistically stable patterns from data. Several arguments are raised in favour of the rotated EOFs.

- The technique produces compact patterns that can be used for ‘regionalization,’ that is, to divide an area in a limited number of homogeneous sub-areas.
- Rotated EOFs are less sensitive to the distribution of observing locations than conventional EOFs.
- Rotated EOFs are often statistically more stable than conventional EOFs (see, e.g., Cheng, Nitsche, and Wallace [82]). That is, the sampling variance of rotated EOFs is often less than that of the input vectors.

Others in the scientific community are less convinced because of the heuristic arguments that

motivate the simplicity functionals, and thus the heuristic basis for the interpretation of the result. Jolliffe [198] lists four drawbacks of the routine use of rotation, namely i) the arbitrary choice of the rotation criterion, ii) the sensitivity of the result to the normalization of the EOFs (see [13.5.3]), iii) the need to redo the entire calculation if the number of EOFs is changed (see [13.5.4]), and iv) the loss of information about the dominant sources of variation in the data.

13.5.3 The Mathematics of ‘Rotation.’ ‘Rotation’ consists of the transformation of a set of ‘input vectors’ $\mathcal{P} = (\vec{p}^1 | \dots | \vec{p}^K)$ into another set of vectors $\mathcal{Q} = (\vec{q}^1 | \dots | \vec{q}^K)$ by means of an invertible $K \times K$ matrix $\mathcal{R} = (r_{ij})$:

$$\mathcal{Q} = \mathcal{P}\mathcal{R} \quad (13.46)$$

or, for each vector $\vec{q}^i$:

$$\vec{q}^i = \sum_{j=1}^K r_{ij} \vec{p}^j. \quad (13.47)$$

The matrix $\mathcal{R}$ is chosen from a class of matrices, such as orthonormal matrices, subject to the

constraint that a functional $V(\mathcal{Q})$ is minimized. An example of such a functional is described in the next subsection.

Under some conditions, operation (13.46) can be viewed as a rotation of the ‘input vectors.’ Since these are often the first K EOFs, the resulting vectors $\tilde{q}^i$ are called ‘rotated EOFs.’

When matrix $\mathcal{R}$ is orthonormal, the operation is said to be an ‘orthonormal rotation’; otherwise it is said to be ‘oblique.’

Now let $\tilde{\mathbf{X}}$ be a random vector that takes values in the space spanned by the input vectors. That is

$$\tilde{\mathbf{X}} = \mathcal{P}\tilde{\alpha} \quad (13.48)$$

where $\tilde{\alpha}$ is a k -dimensional vector of random expansion coefficients. Then, because of operation (13.46)

$$\tilde{\mathbf{X}} = (\mathcal{P}\mathcal{R})(\mathcal{R}^{-1}\tilde{\alpha}) = \mathcal{Q}\tilde{\beta} \quad (13.49)$$

where $\tilde{\beta} = \mathcal{R}^{-1}\tilde{\alpha}$ is the k -dimensional vector of random expansion coefficients for the rotated patterns.

Let us assume for the following that the matrix $\mathcal{R}$ is orthonormal so that $\tilde{\beta} = \mathcal{R}^T\tilde{\alpha}$.²³

- When the input vectors are orthogonal, the scalar products between all possible pairs of rotated vectors are given by the matrix

$$\mathcal{Q}^T\mathcal{Q} = \mathcal{R}^T\mathcal{P}^T\mathcal{P}\mathcal{R} = \mathcal{R}^T\mathcal{D}\mathcal{R}, \quad (13.50)$$

where $\mathcal{D} = (\tilde{p}^1\tilde{p}^1, \dots, \tilde{p}^K\tilde{p}^K)$. Thus the rotated vectors are orthogonal only if $\mathcal{D} = \mathcal{I}$, or, in other words, if the input vectors are normalized to unit length.

- Similarly, if the expansion coefficients of the input vectors are pairwise uncorrelated, so that $\Sigma_{\alpha\alpha} = \text{diag}(\sigma_1^2, \dots, \sigma_K^2)$, then the coefficients of the rotated patterns are also pairwise uncorrelated only if coefficients α_j have unit variance. Then

$$\begin{aligned} \Sigma_{\beta\beta} &= \text{Cov}(\mathcal{R}^T\tilde{\alpha}, \mathcal{R}^T\tilde{\alpha}) \\ &= \mathcal{R}^T\Sigma_{\alpha\alpha}\mathcal{R}. \end{aligned} \quad (13.51)$$

Equations (13.50) and (13.51) imply that rotated patterns derived from normalized EOFs, as defined in [13.1.2,3] so that $\tilde{p}^j = \tilde{e}^j$, are also orthonormal, but their time coefficients are not uncorrelated. If, on the other hand, the EOFs are

re-normalized as in equations (13.21) and (13.22) so that $\tilde{p}^j = \tilde{e}^j$ and $\text{Var}(\alpha_j^+) = 1$, then the rotated patterns are no longer orthogonal but the coefficients remain pairwise uncorrelated.

Thus two important conclusions may be drawn.

- The result of the rotation exercise depends on the lengths of the input vectors. Differently scaled but directionally identical sets of input vectors lead to sets of rotated patterns that are directionally different from one another. Jolliffe [199] demonstrates that the differences can be large.

The rotated vectors are a function of the input vectors rather than the space spanned by the input vectors.

- After rotating EOF patterns, the new patterns and coefficients are not orthogonal and uncorrelated at the same time. When the coefficients are uncorrelated, the patterns are not orthogonal, and vice versa. Thus, the percentage of variance represented by the individual patterns is no longer additive.

13.5.4 The ‘Varimax’ Method. ‘Varimax’ is a widely used orthonormal rotation that minimizes the ‘simplicity’ functional

$$V(\tilde{q}^1, \dots, \tilde{q}^K) = \sum_{i=1}^K f_V(\tilde{q}^i) \quad (13.52)$$

where $\tilde{q}^i$ is given by equation (13.47) and f_V is defined by

$$f_V(\tilde{q}) = \frac{1}{m} \sum_{i=1}^m \left(\frac{q_i}{s_i} \right)^4 - \frac{1}{m^2} \left(\sum_{i=1}^m \left(\frac{q_i}{s_i} \right)^2 \right). \quad (13.53)$$

The constants s_i are chosen by the user. The *raw varimax* rotation is obtained when $s_i = 1$, $i = 1, \dots, K$, and the *normal varimax* rotation is obtained by setting $s_i = \sum_{j=1}^K (p_i^j)^2$. Another option is to define s_i as the standard deviation of the i th component of

$$\tilde{\mathbf{X}}^{(K)} = \sum_{j=1}^K \alpha_j \tilde{p}^j,$$

which is the projection of the original full random vector $\tilde{\mathbf{X}}$ onto the subspace spanned by the K vectors $\{\tilde{p}^1 \dots \tilde{p}^K\}$.

Note that $f_V(\tilde{q})$ (13.53) can be viewed as the spatial variance of the normalized squares $(q_i/s_i)^2$. That is, $f_V(\tilde{q})$ measures the ‘weighted

²³All matrices and vectors in this section are assumed to be real valued. Thus orthonormal matrices satisfy $\mathcal{R}\mathcal{R}^T = \mathcal{R}^T\mathcal{R} = \mathcal{I}$.

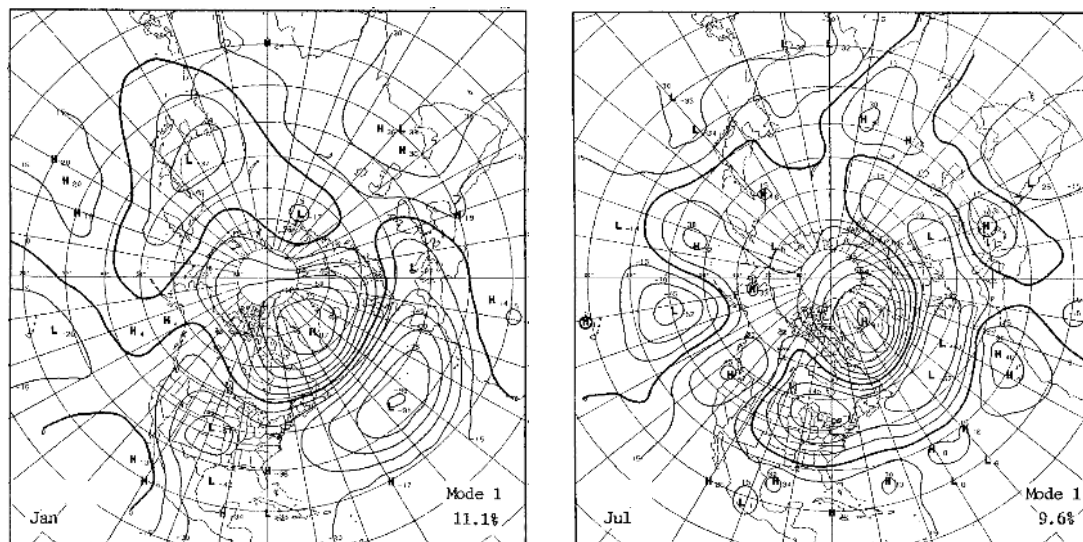


Figure 13.6: January (left) and July (right) versions of the North Atlantic Oscillation pattern derived by Barnston and Livezey [27] by applying varimax rotation to the first 10 normalized EOFs of January and July mean 700 hPa height, respectively. Courtesy R. Livezey.

square amplitude' variance of $\bar{q}$. Therefore, minimizing function (13.52) is equivalent to finding a matrix $\mathcal{R}$ such that the sum of the total weighted square amplitude variance of the K patterns $(\bar{q}^1 | \dots | \bar{q}^k) = \mathcal{P}\mathcal{R}$ is minimized. See Richman [331] for further details.

13.5.5 Example: Low-frequency Atmospheric Circulation Patterns. Barnston and Livezey [27] argued extensively that rotated EOF analysis is a more effective tool for the analysis of atmospheric circulation patterns than the 'teleconnection' analysis (Wallace and Gutzler [409]; see also Section 17.4). They used a varimax rotation of re-normalized EOFs (13.21, 13.22) to isolate the dominant circulation patterns in the Northern Hemisphere (NH) on the monthly time scale. The EOFs used in the study were computed separately for each month of the year from correlation matrices derived from a 35-year data set of monthly mean 700 hPa heights analysed on a 358-point grid. The data set itself was carefully screened to remove known analysis biases. Rotation was performed on the first 10 EOFs in each month. They represent about 80% of the total variance in winter and 70% in summer.

The result of the exercise is an extensive collection of NH circulation patterns. Barnston and Livezey identified 13 patterns: nine cold season patterns, two warm season patterns, and two transition season patterns. Only one pattern,

the North Atlantic Oscillation (NAO, Figure 13.6) is evident in every month of the year. Barnston and Livezey estimate that it represents between 15.4% (March) and 7.4% (October) of the total variance. The NAO is the dominant circulation pattern in the solstitial seasons (DJFM and MJJAS). The NAO is characterized by a 'high' (this adjective is arbitrary since the sign of the pattern is arbitrary) that is centred, roughly, over Greenland and a low pressure band to the south. Figure 13.6 displays 'typical' configurations in winter and summer. The Greenland centre is located at about 70°N and $40\text{--}60^\circ\text{W}$ in winter, and has a zero line at about 50°N . This centre retreats northward in summer and a second zero line appears at about $30\text{--}35^\circ\text{N}$.

Another pattern extracted by Barnston and Livezey that has been studied by many others is the Pacific/North American (PNA) pattern (Figure 13.7). The PNA is characterized by two centres of the same sign over the Aleutian Islands and the southeastern United States that flank a centre of opposite sign located over western North America. The PNA is evident in winter (December to April) and again in September and October. It is strongest in February when Barnston and Livezey estimate that it represents 13.2% of the total variance.

Even though the rotated EOFs appear to be less prone to 'mixing' than ordinary EOFs, a great deal of sampling variability still clouds the patterns that are produced, and a considerable

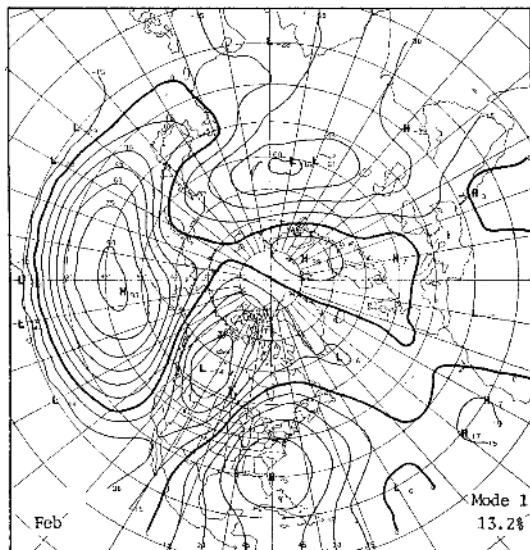


Figure 13.7: As Figure 13.6, except the February Pacific/North American pattern is displayed. Courtesy R. Livezey.

amount of skill and subjective judgement are needed to classify and name the patterns. This is amply illustrated by Barnston and Livezey [27], who discuss the types of latitude they permitted themselves in developing their classification. Their illustration of six renditions of the NAO obtained for different times of the year (we show two of these in Figure 13.6; see Barnston and Livezey [27, Figure 2]) demonstrates the kind of variability that the analyst must be able to penetrate when classifying estimated patterns.

13.5.6 Example: Atlantic Sea-Level Air Pressure. In this subsection we consider the EOFs and varimax-rotated EOFs of North Atlantic monthly mean SLP in DJF.²⁴

The first three EOFs of SLP (Figure 13.8, left hand column) represent 41%, 26% and 9% of the total variance, respectively. The first EOF has almost uniform sign and exhibits one large feature. The second and third EOFs have dipole structures that reflect the constraint that the higher-order EOFs must be orthogonal to the first EOF.

The EOFs of North Atlantic SLP have simple structure, even without rotation. It is therefore not surprising that the application of the varimax rotation technique to these EOFs (Figure 13.8,

middle and right hand columns) results in little change.

- Figure 13.8 (middle column) displays the result of the rotation using $K = 5$ normalized EOFs as input vectors. The rotated EOFs represent 38%, 24% and 10% of the total variance. Similar results are obtained when $K = 5$ non-normalized EOFs are used (not shown).
- The result of the rotation using the first $K = 10$ non-normalized EOFs is shown in the right hand column of Figure 13.8. The patterns represent 26%, 15%, and 13% of the total variance, respectively. These patterns deviate somewhat from those in the left hand and middle columns of the diagram. They are noisier than the other sets of patterns, including the rotated patterns derived from $K = 10$ normalized EOFs (not shown).

Intuitively this is what we expect since the non-normalized patterns enter the minimization functional with equal weights. Thus the poorly estimated EOFs are as influential as the well-estimated EOFs in the determination of matrix $\mathcal{R}$. In contrast, normalization gives the well-estimated EOFs relatively more influence on the form of $\mathcal{R}$.

The first rotated pattern represents less variance than the first EOF, simply because the first EOF was constructed to maximize the variance. Higher-order rotated EOFs typically represent more variance than the respective EOFs (see, e.g., Table 1 of Barnston and Livezey [27]).

In this example, little is gained by processing the original EOF patterns with the varimax machinery. The rotated EOFs become noisy when too many non-normalized EOFs were used as input.

13.5.7 Example: North Atlantic Sea-surface Temperature. The first three EOFs of the monthly mean SST in DJF represent 26%, 17% and 10% of the total variance, respectively (Figure 13.9, left hand column). These EOFs do not have simple structure. The first contains three well-separated centres of location located in the West Atlantic off the North American coast, south of Greenland, and in the upwelling region off the west coast of Africa.

Varimax rotation leads to a substantially different distribution of variance between patterns (Figure 13.9, middle and right hand columns).

²⁴The analysis presented here and in [13.5.7] were provided by V. Kharin (personal communication). Note that all eigenvalues, EOFs, and rotated EOFs presented here are estimates.

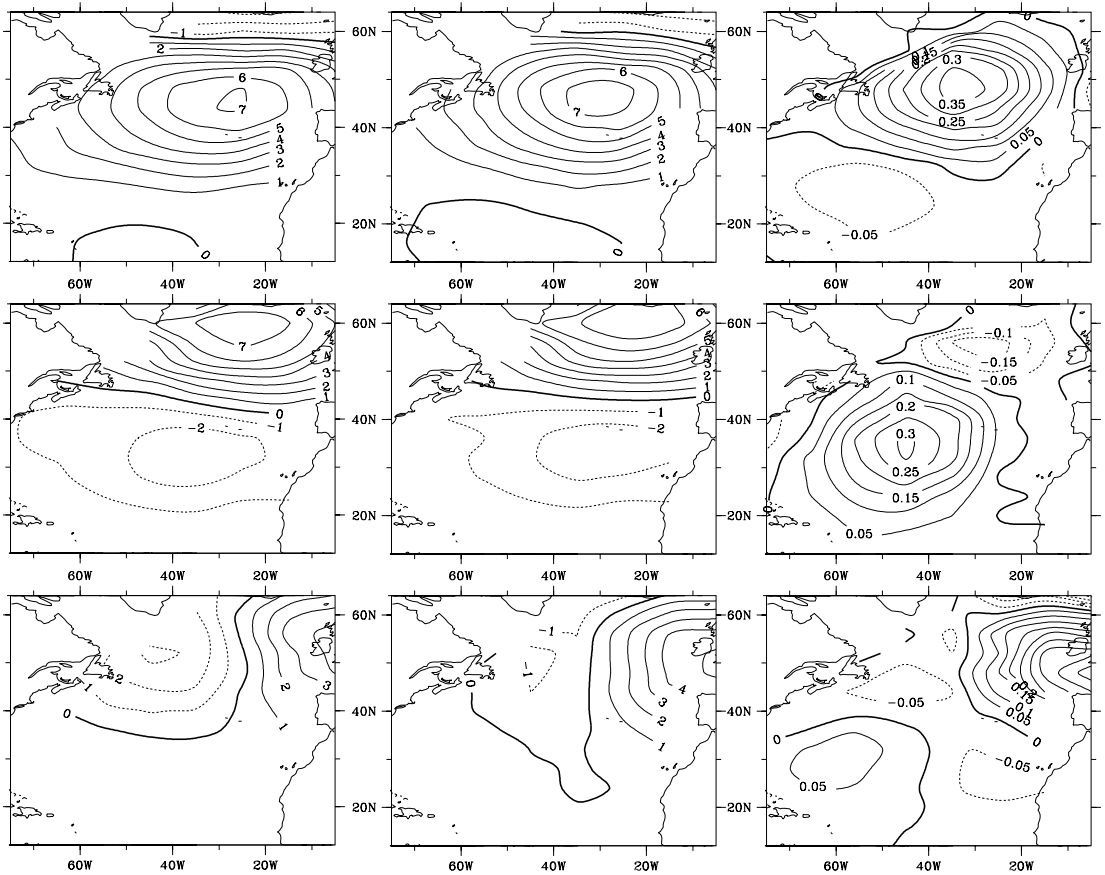


Figure 13.8: First three rotated and unrotated EOFs of North Atlantic SLP in winter. From top to bottom $j = 1$, $j = 2$, $j = 3$. Courtesy V. Kharin.

Left column: Normalized EOFs $\tilde{e}^j$.

Middle column: Rotated EOFs derived from $K = 5$ normalized EOFs.

Right column: Rotated EOFs derived from $K = 10$ non-normalized EOFs.

- When the input is $K = 5$ normalized EOFs (Figure 13.9, middle column) the three centres of action in the first EOF are separated and distributed to the first three rotated patterns (which represent 21%, 16% and 15% of the variance, respectively).
- When the input is $K = 5$ non-normalized EOFs (i.e., all EOFs have unit length; Figure 13.9, right hand column) the three rotated EOFs represent about the same percentage of variance, namely 15%, 15% and 13%, respectively.²⁵ Note that the sequence of patterns is changed from that

obtained with the normalized EOFs (which have unequal lengths). When more input vectors are used, the rotated patterns become noisier and represent less variance (not shown).

We will revisit the analysis of North Atlantic SST and SLP in [14.3.1]. There we will see that the first two conventional EOFs of the North Atlantic SST reflect two forcing mechanisms, two characteristic variations in the large-scale atmospheric state that are encoded in the first two SLP EOFs shown in Figure 13.8. Thus, in this case, the rotation makes interpretation more difficult by masking the underlying physics (see [14.3.2]).

²⁵Note that the concept of degeneracy is irrelevant for rotated EOFs, since degeneracy is immaterial for the minimization of the functional V .

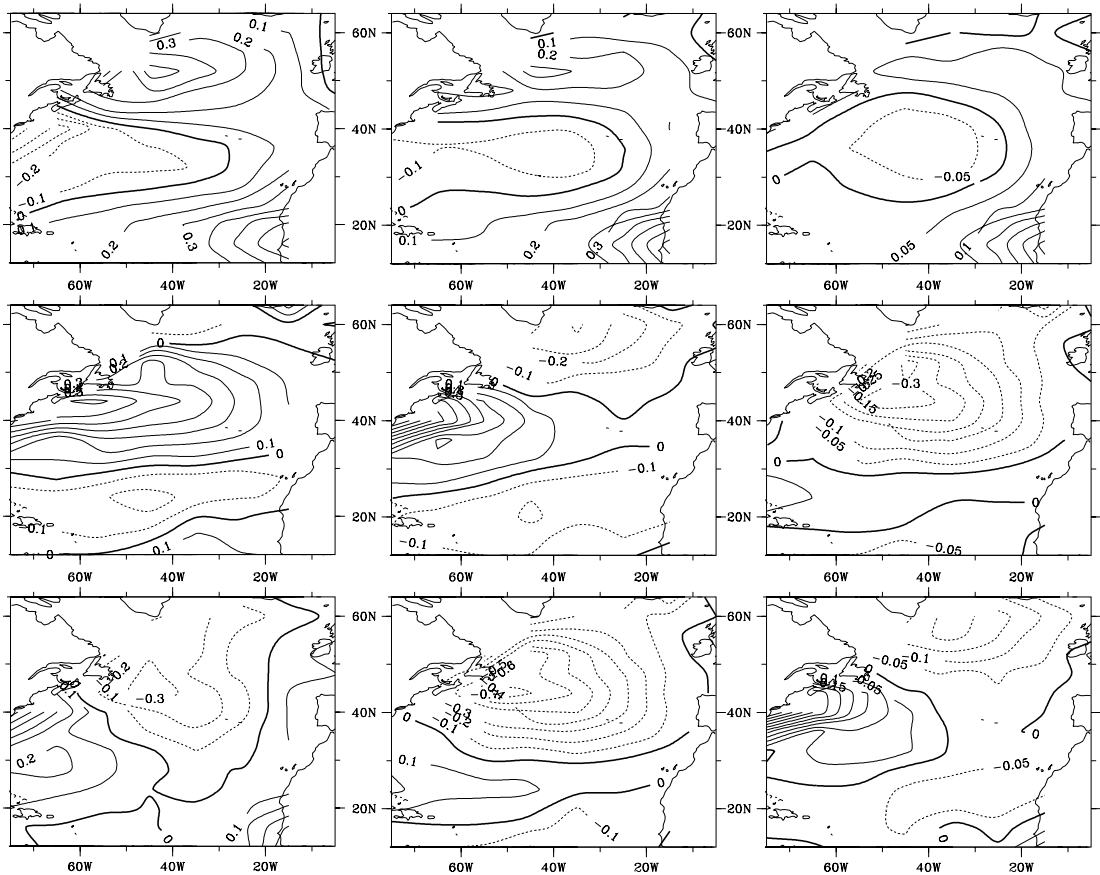


Figure 13.9: First three unrotated (left hand column) and rotated (middle and right hand columns) EOFs of North Atlantic monthly mean SST in DJF. From top to bottom: $j = 1$, $j = 2$, $j = 3$. Courtesy V. Kharin.

Left column: Normalized EOFs $\tilde{e}^{j+}$.
Middle column: Rotated EOFs derived from $K = 5$ normalized EOFs.
Right column: Rotated EOFs derived from $K = 5$ non-normalized EOFs.

13.5.8 Rotation: a Postscript. EOF rotation is often useful, but it is not meant to be a default operation after every EOF analysis. Instead its use should be guided by the problem under consideration.

Jolliffe [198] points out that rotation should be used routinely for subsets of EOFs that have equal, or near-equal, eigenvalues. The corresponding EOFs are not well defined because of their degeneracy (cf. [13.1.8]), and thus the patterns contained by the degenerate EOFs may be arbitrarily rotated within the space that they span. The sensitivity of the rotation to the normalization of the EOFs becomes less relevant since all eigenvalues are similar.

13.6 Singular Systems Analysis and Multichannel SSA

13.6.1 General. The *Singular Systems Analysis* (SSA; see Vautard, Yiou, and Ghil [381] or Vautard [380]) and the *Multichannel Singular Spectrum Analysis* (MSSA, see Plaut and Vautard [317]) are time series analysis techniques used to identify recurrent patterns in univariate time series (SSA) and multivariate time series (MSSA). Mathematically, SSA and MSSA are variants of conventional EOF analysis, but the application of the mathematics is markedly different. Vautard [380] reviews recent applications of SSA and MSSA. Allen and colleagues [8, 9,

10] have investigated various aspects of these methods.

13.6.2 Singular Systems Analysis. Univariate time series $\mathbf{X}_t$ are considered in SSA. An m -dimensional vector time series $\vec{\mathbf{Y}}_t$ is derived from $\mathbf{X}_t$ by setting:

$$\vec{\mathbf{Y}}_t = (\mathbf{X}_t, \mathbf{X}_{t+1}, \dots, \mathbf{X}_{t+m-1})^T. \quad (13.54)$$

A Singular Systems Analysis is an EOF analysis of $\vec{\mathbf{Y}}_t$.

The vector space occupied by $\vec{\mathbf{Y}}_t$ is called the *delay-coordinate space*.

The (zero lag) covariance matrix of $\vec{\mathbf{Y}}$, $\Sigma_{YY} = \text{Cov}(\vec{\mathbf{Y}}_t, \vec{\mathbf{Y}}_t)$, is a *Töplitz matrix*.²⁶ Element (j, k) of Σ_{YY} , say σ_{jk} , is the covariance between the j th element of $\vec{\mathbf{Y}}_t$ ($\mathbf{X}_{t+j-1}$) and its k th element $\mathbf{X}_{t+k-1}$. Thus

$$\sigma_{jk} = \gamma_{xx}(|j - k|),$$

where $\gamma_{xx}(\cdot)$ is the auto-correlation function of $\mathbf{X}_t$. All off-diagonal elements of Σ_{YY} are identified by $|j - k| = \tau$ and have the same value $\gamma_{xx}(\tau)$. Thus, matrix Σ_{YY} is band-structured and contains all auto-covariances of $\mathbf{X}_t$ up to lag $m - 1$. The covariance and correlation matrices of $\vec{\mathbf{Y}}_t$ differ by only a constant factor ($1/\sigma_x^2$). They therefore have the same eigenvectors. The eigenvalues of the two matrices differ by the same constant factor.

The eigenvectors $\vec{e}^i$ of Σ_{YY} , sometimes called *time EOFs*, are interpreted as a sequence in time. Each eigenvector $\vec{e}^i$ is a normalized sequence of m time-ordered numbers,

$$\vec{e}^i = (e_0^i, \dots, e_{m-1}^i)^T, \quad (13.55)$$

that may be understood as a ‘typical’ sequence of events. The orthogonality of the eigenvectors in the delay-coordinate space, $\vec{e}^j{}^T \vec{e}^k = \delta_{jk}$, is equivalent to the *temporal orthogonality* of any two typical sequences $(e_0^j, \dots, e_{m-1}^j)$ and $(e_0^k, \dots, e_{m-1}^k)$:

$$\sum_{i=0}^{m-1} e_i^j e_i^k = \delta_{jk}. \quad (13.56)$$

The EOF coefficients

$$\alpha_k(t) = \langle \vec{\mathbf{y}}_t, \vec{e}^k \rangle = \sum_{i=0}^{m-1} \mathbf{X}_{t+i} e_i^k \quad (13.57)$$

²⁶The elements on each diagonal of a Töplitz matrix are equal. That is, if $\mathcal{A}$ is an $m \times m$ matrix and if there are constants $c_{-(n-1)}, \dots, c_{(n-1)}$ such that $\mathcal{A}_{i,j} = c_{j-i}$, then $\mathcal{A}$ is Töplitz. Graybill [148] describes some of their properties (see Section 8.15).

are empirically determined averages (recall Section 10.5) of length m . That is, $\alpha_k(t)$ is a *filtered*²⁷ version of the original time series $\mathbf{X}_t$, with filter weights that are given by the k th eigenvector. When $\mathbf{X}_t$ is dominated by high-frequency variations, the dominant eigenvectors will be high-pass filters, and when most of the variance of $\mathbf{X}_t$ is concentrated at low frequencies the dominant eigenvectors will act as low-pass filters. The eigenvectors will generally *not* form symmetric filters. Thus we need to be aware that operation (13.57) causes a frequency-dependent phase shift.

As with ordinary EOF analysis, SSA distributes the total variance of $\vec{\mathbf{Y}}_t$ to the m eigenvalues λ_i . The total variance of $\vec{\mathbf{Y}}_t$ is equal to m times the variance of $\mathbf{X}_t$. Thus

$$\sum_{i=1}^m \lambda_i = m \text{Var}(\mathbf{X}_t). \quad (13.58)$$

The vector-matrix version of (13.57) is

$$\vec{\alpha}(t) = \mathcal{P} \vec{\mathbf{Y}}_t$$

where $\vec{\alpha}(t)$ and $\mathcal{P}$ are defined in the usual way. Thus the auto-correlation function of the multivariate coefficient process $\vec{\alpha}(t)$ is related to the auto-correlation function of $\mathbf{X}_t$ by

$$\Sigma_{\alpha\alpha}(\tau) = \mathcal{P} \Sigma_{YY}(\tau) \mathcal{P}^T$$

where $\Sigma_{YY}(\tau)$ is the matrix whose (i, j) th entry is given by

$$[\Sigma_{YY}(\tau)]_{i,j} = \gamma_{xx}(\tau + j - i). \quad (13.59)$$

Note that

$$\Sigma_{\alpha\alpha}(0) = \Lambda = \text{diag}(\lambda_1, \dots, \lambda_m).$$

13.6.3 Reconstruction in the Time Domain. Also, as with ordinary EOFs,

$$\vec{\mathbf{Y}}_t = \sum_{i=1}^m \alpha_i(t) \vec{e}^i. \quad (13.60)$$

Thus, using equation (13.60) to expand $\vec{\mathbf{Y}}_t, \vec{\mathbf{Y}}_{t-1}, \dots, \vec{\mathbf{Y}}_{t-m+1}$, we find that $\mathbf{X}_t$ has m equivalent time expansions in the m ‘SSA-signals’:

$$\begin{aligned} \mathbf{X}_t &= \sum_{i=1}^m \alpha_i(t) e_1^i \\ &= \sum_{i=1}^m \alpha_i(t-1) e_2^i \\ &\vdots \\ &= \sum_{i=1}^m \alpha_i(t-m+1) e_m^i. \end{aligned} \quad (13.61)$$

²⁷See Section 17.5.

Each of these expansions distributes the variance of the SSA-signals differently. In fact, using the orthogonality of the EOF coefficients, it is easily shown that

$$\text{Var}(\mathbf{X}_t) = \sum_{i=1}^m \lambda_i (e_k^i)^2 \quad (13.62)$$

for all k .²⁸ If we consider the normalized representation (13.22) we find that the SSA patterns add to the same numbers:

$$\sum_i (e_k^i)^2 = \text{constant}, \quad (13.63)$$

for all lags k .

13.6.4 Paired Eigenvectors and Oscillatory Components. We now consider, briefly, time series that contain an oscillatory component. For simplicity, we suppose that $\mathbf{X}_t$ is pure cosine so that

$$\begin{aligned} \bar{\mathbf{Y}}_t &= (\mathbf{X}_t, \dots, \mathbf{X}_{t+m-1})^T \\ &= \left(\cos\left(2\pi \frac{t}{m}\right), \dots, \cos\left(2\pi \frac{t+m-1}{m}\right) \right)^T. \end{aligned} \quad (13.64)$$

By equation (13.60), the time EOFs must be able to represent this structure. Suppose one of the time EOFs contains the cosine pattern, that is,

$$\bar{e}^i = \left(1, \cos\left(\frac{2\pi}{m}\right), \dots, \cos\left(\frac{2\pi(m-1)}{m}\right) \right)^T.$$

Then $\bar{\mathbf{Y}}_0 = \bar{e}^i$. However, one time step later, we have

$$\begin{aligned} \bar{\mathbf{Y}}_1 &= \left(\cos\left(\frac{2\pi}{m}\right), \dots, \cos\left(\frac{2\pi(m-1)}{m}\right), 1 \right)^T \\ &= \cos\left(\frac{2\pi}{m}\right) \left(1, \cos\left(\frac{2\pi}{m}\right), \dots, \cos\left(\frac{2\pi(m-1)}{m}\right) \right)^T \\ &\quad - \sin\left(\frac{2\pi}{m}\right) \left(0, \sin\left(\frac{2\pi}{m}\right), \dots, \sin\left(\frac{2\pi(m-1)}{m}\right) \right)^T \\ &= \cos\left(\frac{2\pi}{m}\right) \bar{e}^i - \sin\left(\frac{2\pi}{m}\right) \bar{e}^j, \end{aligned}$$

where $\bar{e}^j = \left(0, \sin\left(\frac{2\pi}{m}\right), \dots, \sin\left(\frac{2\pi(m-1)}{m}\right) \right)^T$ is another eigenvector of $\bar{\mathbf{Y}}_t$. At time t ,

$$\begin{aligned} \bar{\mathbf{Y}}_t &= \cos\left(\frac{2\pi t}{m}\right) \bar{e}^i - \sin\left(\frac{2\pi t}{m}\right) \bar{e}^j \\ &= \alpha_i(t) \bar{e}^i + \alpha_j(t) \bar{e}^j \end{aligned}$$

where $\alpha_i(t) = \cos\left(\frac{2\pi t}{m}\right)$ and $\alpha_j(t) = \sin\left(\frac{2\pi t}{m}\right)$. Note that both coefficients have the same ‘variance’ (i.e., $\lambda_i = \lambda_j$), and that the coefficients are 90° out-of-phase.

While the example is artificial, the properties of the eigenvectors and coefficients above characterize what happens when $\mathbf{X}_t$ contains an oscillatory

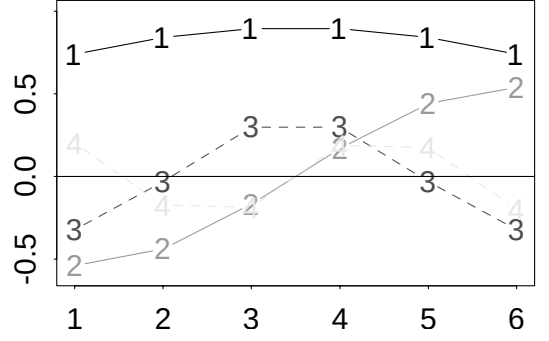


Figure 13.10: The first four time EOFs of an AR(1) process with $a = 0.8$ obtained using window length $m = 6$. The patterns are normalized with the square root of the eigenvalue.

signal. That is, we expect to find a pair of degenerate EOFs with coefficients that vary coherently and are 90° out-of-phase with each other.²⁹ The pair of patterns and their coefficients may be written as one complex pattern and one complex coefficient.

13.6.5 SSA of White Noise. A white noise process $\{\mathbf{X}_t\}$ (see [10.2.3]) consists of a sequence of independent, identically distributed random variables. It has auto-covariance function $\gamma_{xx}(\tau)$ such that $\gamma_{xx}(0) = \text{Var}(\mathbf{X}_t)$ and $\gamma_{xx}(\tau) = 0$ for nonzero τ . Thus

$$\Sigma_{YY} = \text{Var}(\mathbf{X}_t) \mathcal{I},$$

where $\bar{\mathbf{Y}}_t$ is the delay-coordinate space version of $\mathbf{X}_t$ (13.58) and $\mathcal{I}$ is the $m \times m$ identity matrix. Hence $\bar{\mathbf{Y}}_t$ has m eigenvalues $\lambda_i = \text{Var}(\mathbf{X}_t)$ and m degenerate eigenvectors. One possible set of eigenvectors are the unit vectors, $\bar{e}^i = (0, \dots, 1, \dots, 0)$ with the 1 in the i th column.

13.6.6 SSA of Red Noise. Red noise processes³⁰ have exponentially decaying auto-covariance functions $\gamma_{xx}(\tau) = \sigma_X^2 a^{|\tau|}$, where $\sigma_X^2 = \text{Var}(\mathbf{X}_t)$. Thus

$$\Sigma_{YY} = \sigma_X^2 \begin{pmatrix} 1 & a & \dots & a^{m-1} \\ a & 1 & \dots & a^{m-2} \\ \vdots & \vdots & \ddots & \vdots \\ a^{m-1} & a^{m-2} & \dots & 1 \end{pmatrix}.$$

²⁹Compare with the discussion of complex POP coefficients in Chapter 15.

³⁰AR(1), or ‘red noise,’ processes were introduced in [10.3.2]. They can be represented by a stochastic difference equation $\mathbf{X}_t = a\mathbf{X}_{t-1} + \mathbf{Z}_t$, where $\mathbf{Z}_t$ is white noise. The auto-covariance function was derived in [11.1.6]. We represent the lag-1 correlation coefficient by ‘ a ’ instead of ‘ α ’ to avoid confusion with our notation for the EOF coefficients.

²⁸Note that (13.62) is just a special case of (13.7).

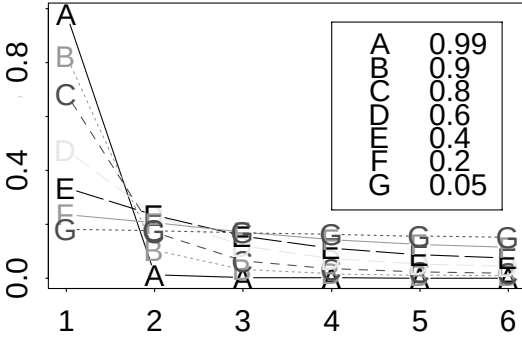


Figure 13.11: Eigenspectra obtained with window length $m = 6$ for AR(1) processes with $a = 0.99, 0.90, 0.80, 0.60, 0.40, 0.20$, and 0.05 . The spectra are normalized by the variance of the process.

When $m = 2$, Σ_{YY} has eigenvalues $(1 - a)$ and $(1 + a)$ and corresponding time EOFs $(1/\sqrt{2}, -1/\sqrt{2})^T$ and $(1/\sqrt{2}, 1/\sqrt{2})^T$. The order of the eigenvalues and EOFs depends upon the sign of a . Given a specific window length m , the same EOFs are obtained for all AR(1) processes $\mathbf{X}_t$.

The first four AR(1) time EOFs for window length $m = 6$ are shown in Figure 13.10. The patterns are multiplied by the square root of the eigenvalue, as in equation (13.22). The k th pattern crosses the zero line $k - 1$ times. Thus, the time EOFs are ordered by time scale, with most variance contributed by the variability with longest time scales.

One characteristic of the time EOFs is that no two patterns have the same number of zeros. Thus oscillatory behaviour, such as that described in [13.6.4], is not possible. This is consistent with the discussion in [11.1.2], when we also found no indication of oscillatory behaviour in AR(1) processes.

Figure 13.11 shows the eigenspectra of several AR(1) processes for the same window length m . The larger the ‘memory’ a , the steeper the spectrum. In the extreme case with $a = 0.99$, almost all variance is contributed by the ‘almost constant’ first time EOF. At the other end of the memory scale ($a = 0.05$) all time EOFs contribute about the same amount of variance.

13.6.7 SSA of an AR(2) process. An AR(2) process has the form

$$\mathbf{X}_t = a_1 \mathbf{X}_{t-1} + a_2 \mathbf{X}_{t-2} + \mathbf{Z}_t$$

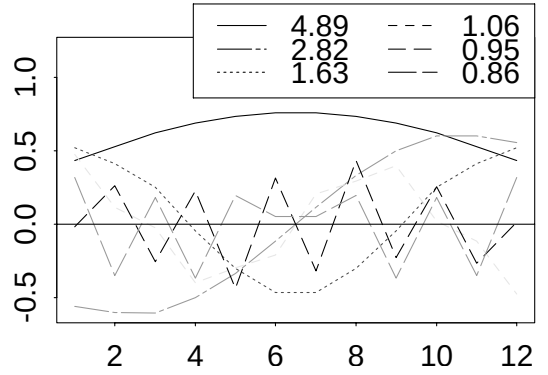


Figure 13.12: First six time EOFs of an AR(2) process with $\vec{a} = (0.3, 0.3)$ obtained using window length $m = 12$. The patterns are normalized with the square root of the eigenvalue. The eigenvalues are given at the bottom.

[10.3.3]. The auto-covariance function is either the sum of the auto-covariance functions of two red noise processes (11.4), or it is a damped oscillatory function (11.9) (for details, see [11.1.9]).

The process with coefficients $a_1 = a_2 = 0.3$ was found to belong to the former ‘non-oscillatory’ group. The first six time EOFs obtained using window length $m = 12$ are shown in Figure 13.12. Similar to the AR(1) process, all eigenvectors have different patterns, with the k th eigenvector having $(k - 1)$ zeros. All eigenvalues are well separated. Consistent with the discussion in [11.1.7] and [11.2.6], an oscillatory mode is not identified. Note the similarity between the patterns in Figure 13.12 and the AR(1) patterns shown in Figure 13.10. (The patterns in Figure 13.12 are not sensitive to the choice of m .)

The other AR(2) process considered previously has $\alpha_1 = 0.9, \alpha_2 = -0.8$. This process has oscillatory behaviour with a ‘period’ of 6 time steps (see [10.3.3], [11.1.7] and [11.2.6]). The time EOFs of this process, obtained using window length $m = 12$, are shown in Figure 13.13. The first two time EOFs are sinusoidal, with a period of 6 time steps, and phase-shifted by 1 to 2 time steps (a quarter of a period, 1.5 time steps, can not be represented in time steps of 1). The two time EOFs share similar eigenvalues (4.2 and 4.1) and obviously represent an oscillatory mode as described in [13.6.4]. The higher index time EOFs are reminiscent of the time EOFs obtained for AR(1) processes.

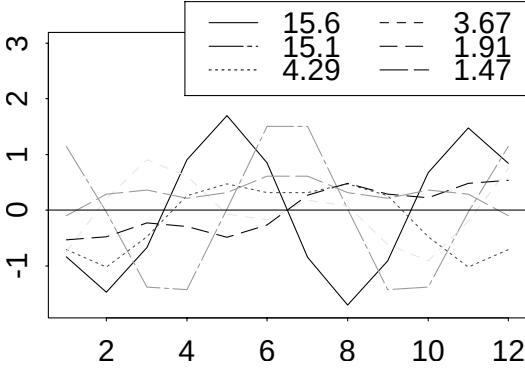


Figure 13.13: First six time EOFs of an AR(2) process with $\vec{a} = (0.9, -0.8)$ obtained using window length $m = 12$. The patterns are normalized with the square root of the eigenvalue. The eigenvalues are given at the bottom.

13.6.8 Multichannel Singular Spectrum Analysis.

MSSA (see Vautard [380]) differs from SSA only in the dimension of the basic time series, which is now m' -dimensional rather than one-dimensional. The derived random vector $\vec{Y}_t$ is therefore mm' -dimensional. Thus MSSA is *Extended EOF analysis* [13.1.8] in which m consecutively observed fields are concatenated together. The number of fields m is usually small compared with the field dimension m' in EEOF analysis. The opposite, $m > m'$, is often true in MSSA.

13.6.9 Estimation. We conclude with a brief comment on the estimation of eigenvalues and time EOFs in SSA. The same applies, by extension, to the eigenvalues and space-time EOFs in MSSA.

SSA is applied to a finite sample of observations $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ with $n \gg m$ by first forming $\vec{Y}$ -vectors,

$$\vec{y}_1 = (\mathbf{x}_1, \dots, \mathbf{x}_m)^T$$

$$\vec{y}_2 = (\mathbf{x}_2, \dots, \mathbf{x}_{m+1})^T$$

$$\vdots$$

$$\vec{y}_{n-m+1} = (\mathbf{x}_{n-m+1}, \dots, \mathbf{x}_n)^T.$$

Conventional EOF analysis is applied to the resulting sample of $n - m + 1$ $\vec{Y}$ -vectors. The estimated eigenvalues and EOFs can be computed from either the estimated covariance matrix of $\vec{Y}$ (see [13.2.4]) or by means of SVD (see [13.2.8]).

Note that neither *North's Rule-of-Thumb* [13.3.5] nor Lawley's formulae (13.38, 13.39) can be used to assess the reliability of the estimate directly because consecutive realizations of $\vec{Y}_t$ are auto-correlated (see (13.33)). The effects of temporal dependence must be accounted for (see Section 17.1) when using these tools.

Allen and co-workers [8, 9, 10] discuss the problem of discriminating between noisy components and truly oscillatory modes in detail.

14 Canonical Correlation Analysis

14.0.0 Overview. Just as EOF analysis (Chapter 13) is used to study the variability of a random vector $\vec{X}$, *Canonical Correlation Analysis* (CCA) is used to study the correlation structure of a pair of random vectors $\vec{X}$ and $\vec{Y}$.

CCA and EOF analyses share similar objectives and similar mathematics. One interpretation of the first EOF $\vec{e}^1$ of $\vec{X}$ is that $\vec{X}^T \vec{e}^1$ is the linear combination of elements of $\vec{X}$ with the greatest variance (subject to $\|\vec{e}^1\| = 1$). The second EOF $\vec{e}^2$ provides the linear combination $\vec{X}^T \vec{e}^2$ with greatest variance that is uncorrelated with $\vec{X}^T \vec{e}^1$, and so on. The objective of CCA is to find a pair of patterns $\vec{f}_X^1$ and $\vec{f}_Y^1$ (subject to $\|\vec{f}_X^1\| = \|\vec{f}_Y^1\| = 1$) so that the *correlation* between linear combinations $\vec{X}^T \vec{f}_X^1$ and $\vec{Y}^T \vec{f}_Y^1$ is maximized.¹ A second pair of patterns $\vec{f}_X^2$ and $\vec{f}_Y^2$ is found so that $\vec{X}^T \vec{f}_X^2$ and $\vec{Y}^T \vec{f}_Y^2$ are the most strongly correlated linear combinations of $\vec{X}$ and $\vec{Y}$ that are not correlated with $\vec{X}^T \vec{f}_X^1$ and $\vec{Y}^T \vec{f}_Y^1$, and so on.

Canonical Correlation Analysis was first described by Hotelling [187].

The ‘Canonical Correlation Patterns’ of a paired random vector $(\vec{X}, \vec{Y})$ are defined in Section 14.1, and their estimation is described in Section 14.2. Examples of some applications are given in Section 14.3. A closely related technique, called *Redundancy Analysis*, is described in Section 14.4.

14.0.1 Introductory Example: Large-scale Temperature and SLP over Europe and Local Weather Elements in Bern. Gyalistras et al. [152] analysed the simultaneous variations of the local climate in Bern (Switzerland) and the troposphere over the North Atlantic in DJF. The state of the local climate in a given season was represented by a 17-dimensional random vector $\vec{X}$ consisting of the number of days in the season with at least 1 mm of precipitation, and the

seasonal means and daily standard deviations of the daily mean, minimum, maximum, and range of temperature, precipitation, wind speed, relative humidity, and relative sunshine duration. The large-scale state of the atmosphere was represented by a vector $\vec{Y}$ consisting of the near-surface temperature and sea-level pressure (SLP) fields over Europe and the Northeast Atlantic Ocean.

CCA was used to analyse the joint variability of $\vec{X}$ and $\vec{Y}$. As noted above, this technique finds pairs of patterns such that the correlation between two corresponding pattern coefficients is maximized.

The pair of patterns with the largest correlation is shown in Figure 14.1. The two patterns, one of which consists of two sub-patterns for the pressure and temperature (Figure 14.1, top and middle), have a meaningful physical interpretation. Below normal temperatures in Bern are associated with high pressure over the British Isles and below normal temperatures in the rest of Europe since the correlation between the local climate pattern (bottom panel, Figure 14.1) and the tropospheric pattern (top two panels, Figure 14.1) is negative. Weakened westerly flow is associated with reduced precipitation; the seasonal mean, standard deviation, and number of ‘wet’ days all tend to be below normal. The large-scale patterns have little effect on wind speed and relative humidity.

The link between the two patterns in Figure 14.1 is strong. The correlation between the coefficient time series (not shown) is -0.89 , and the CCA pattern represents a large proportion of the variance of the local climate (Figure 14.2). More than 50% of interannual variance of the seasonal means of daily mean, minimum and maximum temperature are represented by the first CCA pair. They also represent almost 80% of the interannual variance of DJF precipitation and about 75% of the interannual variance of the number of ‘wet’ days.

14.1 Definition of Canonical Correlation Patterns

14.1.1 One Pair of Patterns. Let us consider an m_X -dimensional random vector $\vec{X}$ and an

¹One could also choose $\vec{f}_X^1$ and $\vec{f}_Y^1$ to maximize the *covariance* between $\vec{X}^T \vec{f}_X^1$ and $\vec{Y}^T \vec{f}_Y^1$. Climatologists sometimes call this *SVD analysis* since the patterns are found by obtaining a singular value decomposition of the cross-covariance matrix. See [14.1.7], Bretherton, Smith, and Wallace [64] and Cherry [83].

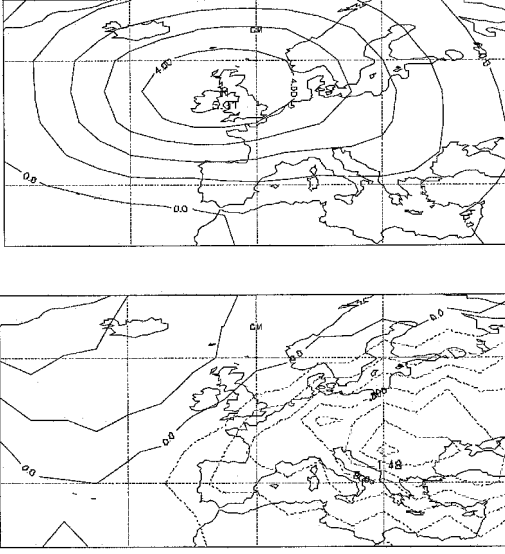


Figure 14.1: First pair of canonical correlation patterns of $\vec{Y}$ = (DJF mean SLP, DJF mean temperature) and a vector $\vec{X}$ of DJF statistics of local weather elements at Bern (Switzerland). Top: The SLP part of the first canonical correlation pattern for $\vec{Y}$.

Middle: The near-surface temperature part of the first canonical correlation pattern for $\vec{Y}$.

Bottom: The canonical correlation pattern for the local variable $\vec{X}$.

Note that the correlation between the corresponding pattern coefficients is negative.

From Gyalistras et al. [152].

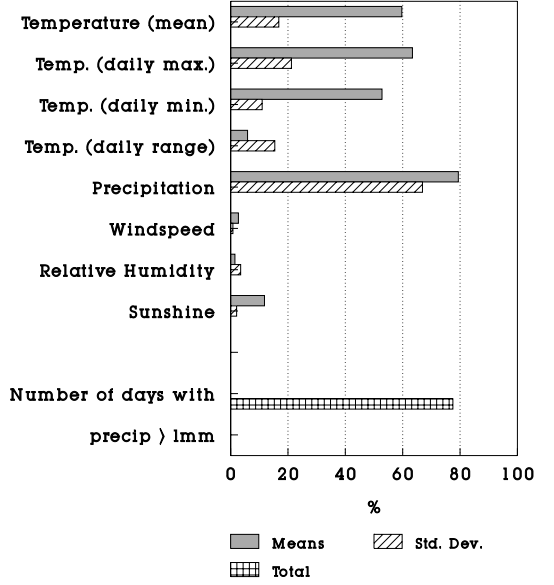


Figure 14.2: Percentage of year-to-year variance of the local climate variables for Bern represented by the first CCA pair.

m_Y -dimensional random vector $\vec{Y}$. We require an m_X -dimensional vector $\vec{f}_X$ and an m_Y -dimensional vector $\vec{f}_Y$ such that the inner products $\beta^X = \langle \vec{X}, \vec{f}_X \rangle$ and $\beta^Y = \langle \vec{Y}, \vec{f}_Y \rangle$ have maximum correlation. That is, we want to maximize

$$\rho = \frac{\text{Cov}(\beta^X, \beta^Y)}{\sqrt{\text{Var}(\beta^X) \text{Var}(\beta^Y)}} \quad (14.1)$$

$$= \frac{\vec{f}_X^T \text{Cov}(\vec{X}, \vec{Y}) \vec{f}_Y}{\sqrt{\text{Var}(\langle \vec{X}, \vec{f}_X \rangle) \text{Var}(\langle \vec{Y}, \vec{f}_Y \rangle)}}.$$

Note that if a pair of vectors $\vec{f}_X$ and $\vec{f}_Y$ maximizes (14.1), then all vectors $\alpha_X \vec{f}_X$ and $\alpha_Y \vec{f}_Y$ do the same for any nonzero α_X and α_Y . Thus the patterns $\vec{f}_X$ and $\vec{f}_Y$ are subject to arbitrary normalization. In particular, we can choose patterns such that

$$\text{Var}(\langle \vec{X}, \vec{f}_X \rangle) = \vec{f}_X^T \Sigma_{XX} \vec{f}_X = 1 \quad (14.2)$$

$$\text{Var}(\langle \vec{Y}, \vec{f}_Y \rangle) = \vec{f}_Y^T \Sigma_{YY} \vec{f}_Y = 1, \quad (14.3)$$

where Σ_{XX} and Σ_{YY} are the covariance matrices of $\vec{X}$ and $\vec{Y}$. Then equation (14.1) can be rewritten as

$$\rho = \vec{f}_X^T \Sigma_{XY} \vec{f}_Y, \quad (14.4)$$

where Σ_{XY} is the cross-covariance matrix

$$\Sigma_{XY} = \mathcal{E}((\vec{X} - \bar{\mu}_X)(\vec{Y} - \bar{\mu}_Y)^T).$$

Vectors $\vec{f}_X$ and $\vec{f}_Y$ are found by maximizing

$$\epsilon = \vec{f}_X^T \Sigma_{XY} \vec{f}_Y + \zeta (\vec{f}_X^T \Sigma_{XX} \vec{f}_X - 1) + \eta (\vec{f}_Y^T \Sigma_{YY} \vec{f}_Y - 1), \quad (14.5)$$

where ζ and η are Lagrange multipliers that are used to account for constraints (14.2) and (14.3). Setting the partial derivatives of ϵ to zero, we obtain

$$\frac{\partial \epsilon}{\partial \vec{f}_X} = \Sigma_{XY} \vec{f}_Y + 2\zeta \Sigma_{XX} \vec{f}_X = 0 \quad (14.6)$$

so that

$$\Sigma_{XX}^{-1} \Sigma_{XY} \vec{f}_Y = -2\zeta \vec{f}_X, \quad (14.7)$$

and

$$\frac{\partial \epsilon}{\partial \vec{f}_Y} = \Sigma_{XY}^T \vec{f}_X + 2\eta \Sigma_{YY} \vec{f}_Y = 0, \quad (14.8)$$

which is equivalent to

$$\Sigma_{YY}^{-1} \Sigma_{XY}^T \vec{f}_X = -2\eta \vec{f}_Y. \quad (14.9)$$

Then (14.9) is substituted into (14.7) and vice versa to obtain a pair of eigen-equations for $\vec{f}_X$ and $\vec{f}_Y$:

$$\Sigma_{XX}^{-1} \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{XY}^T \vec{f}_X = 4\zeta \eta \vec{f}_X \quad (14.10)$$

$$\Sigma_{YY}^{-1} \Sigma_{XY}^T \Sigma_{XX}^{-1} \Sigma_{XY} \vec{f}_Y = 4\zeta \eta \vec{f}_Y. \quad (14.11)$$

An argument similar to that used to establish Theorem [13.2.4] proves that the two matrices share the same non-negative eigenvalues.² The eigenvectors of the two matrices are related to each other through a simple equation: if $\vec{f}_X$ is a solution of equation (14.10), then $\Sigma_{YY}^{-1} \Sigma_{XY}^T \vec{f}_X$ is a solution of equation (14.11), provided that their joint eigenvalue is nonzero. Finally, equation (14.4) is maximized by letting $\vec{f}_X$ and $\vec{f}_Y$ be the solutions of equations (14.10) and (14.11) that correspond to the largest eigenvalue $\lambda = 4\zeta \eta$.

Now that we have found the canonical random variables $\beta^X = \langle \vec{X}, \vec{f}_X \rangle$ and $\beta^Y = \langle \vec{Y}, \vec{f}_Y \rangle$ that are most strongly correlated, the natural next step is to find the value of ρ . Using equations (14.4), (14.6), (14.8), and (14.2), (14.3) in sequence, we find:

$$\begin{aligned} \rho^2 &= \vec{f}_X^T \Sigma_{XY} \vec{f}_Y \vec{f}_Y^T \Sigma_{XY}^T \vec{f}_X \\ &= 4\eta \zeta \vec{f}_X^T \Sigma_{XX} \vec{f}_X \vec{f}_Y^T \Sigma_{YY} \vec{f}_Y \\ &= \lambda. \end{aligned}$$

²Note that if $\vec{f}_X$ is a solution of equation (14.10), then $\Sigma_{XX}^{1/2} \vec{f}_X$ is an eigenvector of $(\Sigma_{XX}^{-1/2})^T \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{XY}^T \Sigma_{XX}^{-1/2}$. Similarly, $\Sigma_{YY}^{1/2} \vec{f}_Y$ is an eigenvector of $(\Sigma_{YY}^{-1/2})^T \Sigma_{XY}^T \Sigma_{XX}^{-1} \Sigma_{XY} \Sigma_{YY}^{-1/2}$. Since these are non-negative definite matrices, their eigenvalues are real and non-negative.

Thus the correlation is the square root of the eigenvalue that corresponds to eigenvectors $\vec{f}_X$ and $\vec{f}_Y$.³

14.1.2 More Pairs. The derivation detailed above can now be repeated to obtain $m = \min(m_X, m_Y)$ pairs of patterns $(\vec{f}_X^i, \vec{f}_Y^i)$ and m corresponding pairs of canonical variates⁴

$$\beta_i^X = \langle \vec{X}, \vec{f}_X^i \rangle \quad (14.12)$$

$$\beta_i^Y = \langle \vec{Y}, \vec{f}_Y^i \rangle \quad (14.13)$$

with correlation

$$\rho_i = \text{Cov}(\beta_i^X, \beta_i^Y) = \sqrt{\lambda_i}.$$

The patterns and canonical variates are indexed in order of decreasing eigenvalue λ_i . Pairs of canonical variates are uncorrelated. That is, for $i \neq j$,

$$\begin{aligned} \text{Cov}(\beta_i^X, \beta_j^X) &= \text{Cov}(\beta_i^Y, \beta_j^Y) \\ &= \text{Cov}(\beta_i^X, \beta_j^Y) = 0. \end{aligned}$$

14.1.3 The Canonical Correlation Patterns. For simplicity, we assume in this subsection that $\vec{X}$ and $\vec{Y}$ are of the same dimension m . Then the canonical variates $\vec{\beta}^X = (\beta_1^X, \dots, \beta_m^X)^T$ and $\vec{\beta}^Y = (\beta_1^Y, \dots, \beta_m^Y)^T$ can be viewed as the result of coordinate transforms that have been applied to $\vec{X}$ and $\vec{Y}$.⁵ The transformations relate $\vec{\beta}^X$ and $\vec{\beta}^Y$ to $\vec{X}$ and $\vec{Y}$ through unknown matrices $\mathcal{F}_X$ and $\mathcal{F}_Y$:

$$\begin{aligned} \vec{X} &= \mathcal{F}_X \vec{\beta}^X \\ \vec{Y} &= \mathcal{F}_Y \vec{\beta}^Y. \end{aligned} \quad (14.14)$$

To find $\mathcal{F}_X$, note that

$$\begin{aligned} \vec{\beta}^X &= ((\vec{X}, \vec{f}_X^1), \dots, (\vec{X}, \vec{f}_X^m))^T \\ &= \mathbf{f}_X^T \vec{X} \end{aligned}$$

³Note that the sign of the correlation is arbitrary since $\vec{f}_X$ and $\vec{f}_Y$ are determined uniquely only up to their signs.

⁴We assume that $(\Sigma_{XX}^{-1/2})^T \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{XY}^T \Sigma_{XX}^{-1/2}$ (or, equivalently, $(\Sigma_{YY}^{-1/2})^T \Sigma_{XY}^T \Sigma_{XX}^{-1} \Sigma_{XY} \Sigma_{YY}^{-1/2}$) has $m = \min(m_X, m_Y)$ distinct, nonzero eigenvalues. Eigenvalues of multiplicity greater than one lead to degeneracy just as in EOF analysis. Uncorrelated canonical variates can still be constructed, but their interpretation is clouded by their non-unique determination. Tools comparable to North's Rule-of-Thumb [13.3.5] are not yet developed for CCA. Note that a pair of degenerate eigenvalues may be an indication of a propagating pattern. See Chapter 15.

⁵The discussion in this subsection is easily generalized to the case in which $\vec{X}$ and $\vec{Y}$ are not of the same dimension.

where $\mathbf{f}_X$ is the $m \times m$ matrix with eigenvector $\tilde{f}_X^i$ in its i th column. Thus

$$\begin{aligned}\text{Cov}(\tilde{\mathbf{X}}, \tilde{\beta}^X) &= \text{Cov}(\tilde{\mathbf{X}}, \mathbf{f}_X^T \tilde{\mathbf{X}}) \\ &= \text{Cov}(\tilde{\mathbf{X}}, \tilde{\mathbf{X}}) \mathbf{f}_X = \Sigma_{XX} \mathbf{f}_X.\end{aligned}$$

However, substituting equation (14.14) for $\tilde{\mathbf{X}}$, we also have

$$\begin{aligned}\text{Cov}(\tilde{\mathbf{X}}, \tilde{\beta}^X) &= \text{Cov}(\mathcal{F}_X \tilde{\beta}^X, \tilde{\beta}^X) \\ &= \mathcal{F}_X \text{Cov}(\tilde{\beta}^X, \tilde{\beta}^X) = \mathcal{F}_X\end{aligned}$$

since $\text{Cov}(\tilde{\beta}^X, \tilde{\beta}^X) = \mathcal{I}$. Thus

$$\mathcal{F}_X = \Sigma_{XX} \mathbf{f}_X \quad (14.15)$$

and similarly

$$\mathcal{F}_Y = \Sigma_{YY} \mathbf{f}_Y. \quad (14.16)$$

The columns of $\mathcal{F}_X$ and $\mathcal{F}_Y$, $\tilde{F}_X^i$ and $\tilde{F}_Y^i$, are called the *canonical correlation patterns*.⁶ The canonical variates β_i^X and β_i^Y are also often called *canonical correlation coordinates*. Since the canonical correlation coordinates are normalized to unit variance, the canonical correlation patterns are expressed in the units of the field they represent, and they indicate the ‘typical’ strength of the mode of covariation described by the patterns.

While the matrix-vector representations of $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{Y}}$ in (14.14) are convenient for the derivation of $\mathcal{F}_X$ and $\mathcal{F}_Y$, they are not very evocative. Therefore, note that (14.14) can also be written as

$$\begin{aligned}\tilde{\mathbf{X}} &= \sum_i \beta_i^X \tilde{F}_X^i \\ \tilde{\mathbf{Y}} &= \sum_i \beta_i^Y \tilde{F}_Y^i.\end{aligned} \quad (14.17)$$

This allows us to see more clearly that (14.14) describes an expansion of $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{Y}}$ with respect to their corresponding canonical correlation patterns. It also suggests that it may be possible to approximate $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{Y}}$ by truncating the summation in (14.17).

14.1.4 Computational Aspects. Once we know one set of vectors, say $\tilde{f}_X^i$, all other vectors are easily obtained through simple matrix operations. Let us assume that we have the vectors $\tilde{f}_X^i$. Then (14.15) yields $\tilde{F}_X^i$. In [14.1.1] we noted that $\Sigma_{YY}^{-1} \Sigma_{XY}^T \tilde{f}_X^i$ is equal to $\tilde{f}_Y^i$ after suitable normalization. Application of (14.16) gives the

⁶Note that neither the eigenvectors $\tilde{f}_X^i$ and $\tilde{f}_Y^i$ nor the canonical correlation patterns $\tilde{F}_X^i$ and $\tilde{F}_Y^i$ are generally orthogonal. However, the columns of $\Sigma_{XX}^{1/2} \mathbf{f}_X = \Sigma_{XX}^{-1/2} \mathcal{F}_X$ and $\Sigma_{YY}^{1/2} \mathbf{f}_Y = \Sigma_{YY}^{-1/2} \mathcal{F}_Y$ are orthonormal.

last set of vectors, the $\tilde{\mathbf{Y}}$ -canonical correlation patterns $\tilde{F}_Y^i$. It is therefore necessary to solve only the smaller of the two eigenproblems (14.10) and (14.11).

14.1.5 Coordinate Transformations. What happens to the canonical correlation patterns and correlations when coordinates are transformed by an invertible matrix $\mathcal{L}$ through $\mathcal{L}\tilde{\mathbf{X}} = \tilde{\mathbf{Z}}$? For simplicity we assume random vector $\tilde{\mathbf{Y}}$ is unchanged.

To get the same maximum correlation (14.1), we have to transform the patterns $\tilde{f}_X^i$ with $\mathcal{L}^{-1}$,

$$\tilde{f}_Z^i = (\mathcal{L}^{-1})^T \tilde{f}_X^i. \quad (14.18)$$

Thus the canonical correlation coordinates $\beta_i^X = \langle \tilde{f}_Z^i, \tilde{\mathbf{Z}} \rangle = \langle \tilde{f}_X^i, \tilde{\mathbf{X}} \rangle$ are unaffected by the transformation. Note that relation (14.18) can also be obtained by verifying that $\tilde{f}_Z^i$ and $\tilde{f}_X^i$ are eigenvectors of the CCA matrices $\Sigma_{ZZ}^{-1} \Sigma_{ZY} \Sigma_{YY}^{-1} \Sigma_{ZY}^T$ and $\Sigma_{XX}^{-1} \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{XY}^T$ with the same eigenvalues.

The canonical correlation patterns $\tilde{F}_X^i$ are determined by the covariance matrix of $\tilde{\mathbf{X}}$ and the $\tilde{f}_X^i$ -pattern (14.15). Therefore,

$$\begin{aligned}\tilde{F}_Z^i &= \Sigma_{ZZ} \tilde{f}_Z^i \\ &= \mathcal{L} \Sigma_{XX} \mathcal{L}^T (\mathcal{L}^{-1})^T \tilde{f}_X^i \\ &= \mathcal{L} \Sigma_{XX} \tilde{f}_X^i \\ &= \mathcal{L} \tilde{F}_X^i.\end{aligned} \quad (14.19)$$

Thus the canonical correlation patterns are transformed in the same way as the random vector $\tilde{\mathbf{X}}$. We may conclude that *the CCA is invariant under coordinate transformations*.

14.1.6 CCA after a Transformation to EOF Coordinates. The CCA algebra becomes considerably simpler if the data are transformed into EOF space before the analysis (Barnett and Preisendorfer [21]). Suppose that only the first k_X and k_Y EOFs are retained, so that

$$\begin{aligned}\tilde{\mathbf{X}} &\approx \sum_i^{k_X} \alpha_i^{X+} \tilde{\mathbf{e}}_X^{i+} \\ \tilde{\mathbf{Y}} &\approx \sum_i^{k_Y} \alpha_i^{Y+} \tilde{\mathbf{e}}_Y^{i+},\end{aligned} \quad (14.20)$$

where we have used the renormalized versions (13.20, 13.21) of the EOFs and their coefficients $\alpha_i^{+} = (\lambda_i)^{-1/2} \alpha_i$ and $\tilde{\mathbf{e}}^{i+} = (\lambda_i)^{1/2} \tilde{\mathbf{e}}^i$. The CCA is then applied to the random vectors $\tilde{\mathbf{X}}' = (\alpha_1^{X+}, \dots, \alpha_{k_X}^{X+})^T$ and $\tilde{\mathbf{Y}}' = (\alpha_1^{Y+}, \dots, \alpha_{k_Y}^{Y+})^T$.

An advantage of this approach is that it is often possible to use only the first few EOFs.

Discarding the high-index EOFs can reduce the amount of noise in the problem by eliminating poorly organized, small-scale features of the fields involved.

Another advantage is that the algebra of the problem is simplified since $\Sigma_{X'X'}$ and $\Sigma_{Y'Y'}$ are both identity matrices. Thus, according to equations (14.10, 14.11), $\vec{f}_X^i$ and $\vec{f}_Y^i$ are eigenvectors of $\Sigma_{X'Y'}\Sigma_{X'Y'}^T$ and $\Sigma_{X'Y'}^T\Sigma_{X'Y'}$ respectively. Since these are non-negative definite symmetric matrices, the eigenvectors are orthogonal. Moreover, the canonical correlation patterns $\vec{F}_{X'}^i = \vec{f}_X^i$ and $\vec{F}_{Y'}^i = \vec{f}_Y^i$.

A minor disadvantage is that the patterns are given in the coordinates of the re-normalized EOF space (14.20). To express the pattern in the original coordinate space it is necessary to reverse transformation (14.20) with (14.18) and (14.19):

$$\begin{aligned}\vec{f}_X^i &= \sum_{j=1}^{k_X} (\lambda_j^X)^{1/2} (\vec{f}_X^i)_j \vec{e}_X^j \\ \vec{f}_Y^i &= \sum_{j=1}^{k_Y} (\lambda_j^Y)^{1/2} (\vec{f}_Y^i)_j \vec{e}_Y^j \\ \vec{F}_X^i &= \sum_{j=1}^{k_X} (\lambda_j^X)^{-1/2} (\vec{f}_X^i)_j \vec{e}_X^j \\ \vec{F}_Y^i &= \sum_{j=1}^{k_Y} (\lambda_j^Y)^{-1/2} (\vec{f}_Y^i)_j \vec{e}_Y^j\end{aligned}\quad (14.21)$$

where $(\cdot)_j$ denotes the j th element of the vector contained within the brackets. The canonical correlation patterns are no longer orthogonal after this backtransformation, and vectors $\vec{f}^i$ and $\vec{F}^i$ are no longer identical.⁷

14.1.7 Maximizing Covariance—the ‘SVD Approach.’ Another way to identify pairs of coupled patterns $\vec{p}_X^i$ and $\vec{p}_Y^i$ in random fields $\vec{X}$ and $\vec{Y}$ is to search for orthonormal sets of vectors such that the covariance between the expansion coefficients $\alpha_i^X = \langle \vec{X}, \vec{p}_X^i \rangle$ and $\alpha_i^Y = \langle \vec{Y}, \vec{p}_Y^i \rangle$,

$$\text{Cov}(\alpha_i^X, \alpha_i^Y) = (\vec{p}_X^i)^T \Sigma_{XY} \vec{p}_Y^i, \quad (14.22)$$

is maximized. Note that we explicitly require orthonormal vectors so that $\vec{X}$ and $\vec{Y}$ can be expanded as $\vec{X} = \sum \alpha_i^X \vec{p}_X^i$ and $\vec{Y} = \sum \alpha_i^Y \vec{p}_Y^i$. The solution of (14.22) is obtained as in [14.1.1] by using Lagrange multipliers to enforce the constraints $(\vec{p}_X^i)^T \vec{p}_X^i = 1$ and $(\vec{p}_Y^i)^T \vec{p}_Y^i = 1$. The result is a system of equations,

$$\begin{aligned}\Sigma_{XY} \vec{p}_Y^i &= s_X \vec{p}_X^i \\ \Sigma_{XY}^T \vec{p}_X^i &= s_Y \vec{p}_Y^i,\end{aligned}\quad (14.23)$$

that can be solved by a singular value decomposition (Appendix B). The same solution is obtained

by substituting the two equations into each other to obtain

$$\begin{aligned}\Sigma_{XY} \Sigma_{XY}^T \vec{p}_X^i &= \lambda_i \vec{p}_X^i \\ \Sigma_{XY}^T \Sigma_{XY} \vec{p}_Y^i &= \lambda_i \vec{p}_Y^i,\end{aligned}$$

where $\lambda_i = s_X s_Y$. These equations share the same eigenvalues $\lambda_i > 0$, and their normalized eigenvectors are related by

$$\begin{aligned}\vec{p}_Y^i &= \frac{\Sigma_{XY}^T \vec{p}_X^i}{\|\Sigma_{XY}^T \vec{p}_X^i\|} \\ \vec{p}_X^i &= \frac{\Sigma_{XY} \vec{p}_Y^i}{\|\Sigma_{XY} \vec{p}_Y^i\|}.\end{aligned}$$

It is easily shown that $\text{Cov}(\alpha_i^X, \alpha_i^Y) = \lambda_i^{1/2}$. Thus the pair of patterns associated with the largest eigenvalue maximizes the covariance. The pair of patterns associated with the second largest eigenvalue and orthogonal to the first pair maximize the covariability that remains in $\vec{X} - \alpha_1^X \vec{p}_X^1$ and $\vec{Y} - \alpha_1^Y \vec{p}_Y^1$, and so on.

This method is often called ‘SVD’ analysis. This wording is misleading because it mixes the definition of a statistical parameter with the algorithm used to calculate the parameter. These patterns *can* be calculated by SVD but there are other ways, such as conventional eigen-analysis, to get the same information. Patterns $\vec{p}_X^i$ and $\vec{p}_Y^i$ are often called *left* and *right singular vectors*. The nomenclature is again misleading because the relevant property of these vectors is that they maximize covariance. We therefore call this method *Maximum Covariance Analysis* (MCA) and call the vectors *Maximum Covariance Patterns*.

Two properties of MCA are worth mentioning.

- MCA is invariant under coordinate transformation only if the transformation is orthogonal. The eigenvalues, and thus the degree of covariability, change when the transformation is non-orthonormal.
- MCA coefficients α_i^X and α_j^X , $i \neq j$, are generally correlated. They are uncorrelated when $\Sigma_{XX} = \sigma_X^2 \mathcal{I}$. This also applies to $\vec{Y}$ -coefficients.

See Wallace, Smith, and Bretherton [411] for examples.

14.1.8 Principal Prediction Patterns. Suppose $\{\vec{Z}_t\}$ is a multivariate time series and define

$$\vec{X}_t = \vec{Z}_t \quad \text{and} \quad \vec{Y}_t = \vec{Z}_{t+\tau} \quad (14.24)$$

⁷Note the similarity between this discussion and that in [14.1.4].

for some positive lag τ . Application of the CCA algorithm, with prior EOF truncation if the dimension of $\vec{Z}_t$ is large, identifies patterns $\vec{F}_0^i = \vec{F}_X^i$ and $\vec{F}_\tau^i = \vec{F}_Y^i$ that tend to appear together, that is, patterns with a fixed time lag in the same variable. Thus the presence of $\vec{F}_0^i$ at a given time indicates that it is likely that pattern $\vec{F}_\tau^i$ will emerge τ time units later. Because of the properties of CCA, patterns $\vec{F}_0^i$ and $\vec{F}_\tau^i$ depict the present and future parts of $\vec{Z}_t$ that are most strongly related. In other words, they are the best linearly auto-predictable components in $\vec{Z}_t$.⁸ An example is given in [14.3.7].

14.2 Estimating Canonical Correlation Patterns

14.2.1 Estimation. Estimates of canonical correlation patterns and coefficients are obtained in the obvious way by replacing Σ_{XX} , Σ_{YY} , and Σ_{XY} with corresponding estimates. We recommend that the problem be kept small by approximating the data with truncated EOF expansions (see [14.1.5] and also Bretherton et al. [64]). This has the added benefit of eliminating small-scale spatial noise.

14.2.2 Making Inferences. As noted previously, very little is known about the sampling variability of the eigenvectors or canonical correlation patterns. However, there are some useful asymptotic results for making inferences about the canonical correlations themselves.

Bartlett [32] proposed a test of the null hypothesis $H_0: \rho_{l+1} = \dots = \rho_m = 0$ that the last $m - l$ canonical correlations are zero when it is known that the first l are nonzero. Here $m = \min(m_X, m_Y)$. Bartlett's test can be used when the canonical correlations have been estimated from a sample $\{(\vec{x}_1, \vec{y}_1), \dots, (\vec{x}_n, \vec{y}_n)\}$ of independent realizations of random vectors $\vec{X}$ and $\vec{Y}$ that are jointly multivariate normal. The test statistic (Bartlett [32])

$$\chi^2 = -(n-1-l) - \frac{1}{2}(m_X + m_Y + 1) + \sum_{i=1}^l \hat{\rho}_i^{-2} \ln \left(\prod_{i=l+1}^m (1 - \hat{\rho}_i^2) \right), \quad (14.25)$$

where $\hat{\rho}_i = \sqrt{\hat{\lambda}_i}$, is approximately distributed as $\chi^2((m_X - l)(m_Y - l))$ under H_0 . The test is

⁸It seems that the idea was first suggested by Hasselmann in an unpublished paper in 1983 but it was not pursued until 1996 [103].

performed at the $(1 - \tilde{p}) \times 100\%$ significance level by comparing χ^2 (14.25) against the $\tilde{p}$ -quantile of the approximating χ^2 distribution (see Appendix E).

Glynn and Muirhead [142] give a bias correction for $\hat{\rho}_i$ and also give an expression for the asymptotic variance of the corrected estimator that is useful for constructing confidence intervals. Using the Fisher z -transform (recall [8.2.3]), Glynn and Muirhead show that if

$$\theta_i - \frac{1}{2} \ln \left(\frac{1 + \rho_i}{1 - \rho_i} \right) \quad \text{and} \quad z_i = \frac{1}{2} \ln \left(\frac{1 + \hat{\rho}_i}{1 - \hat{\rho}_i} \right),$$

then the bias of

$$\hat{\theta}_i = z_i - \frac{1}{2n\hat{\rho}_i} (m_X + m_Y - 2 + \hat{\rho}_i^2 + 2(1 - \hat{\rho}_i^2) \sum_{\substack{j=1; \\ j \neq i}}^m \frac{\hat{\rho}_j^2}{\hat{\rho}_i^2 - \hat{\rho}_j^2})$$

is approximately $\mathcal{O}(n^{-2})$ and

$$\text{Var}(\hat{\theta}_i) = \frac{1}{n} + \mathcal{O}(n^{-2}).$$

Thus the bounds for an approximate $\tilde{p} \times 100\%$ confidence interval for ρ_i are given by

$$\tanh(\hat{\theta}_i \pm z_{(1+\tilde{p})/2}/\sqrt{n}), \quad (14.26)$$

where $z_{(1+\tilde{p})/2}$ is the $(1 + \tilde{p})/2$ -quantile of the standard normal distribution (Appendix D). Muirhead and Waternaux [282] show that asymptotic statistics like equations (14.25, 14.26) are not particularly robust against departures from the multivariate normal assumption. Use of the bootstrap (see Section 5.5) is probably the best practical alternative when this is a concern.

One question rarely mentioned in the context of CCA is the size of sample needed to make good estimates and inferences. Thorndike [365, pp. 183–184] suggests that $n > 10(m_X + m_Y) + 50$ is a reasonable rule of thumb, and argues that $n > (m_X + m_Y)^2 + 50$ may be needed for some purposes. Our experience, however, is that much smaller samples can provide meaningful information about the first few patterns and correlations. However, be aware that the asymptotic results discussed above are not likely to hold under these circumstances. The Monte Carlo experiments discussed in the next subsection give some further insight into what can be accomplished with small samples.

mode		$k = 20$					$n = 250$			
i	ρ_{xy}^i	$n =$	50	100	500	1000	$k =$	10	30	50
1	0.69		0.96	0.83	0.70	0.69		0.68	0.71	0.74
2	0.60		0.92	0.76	0.59	0.58		0.58	0.61	0.65
3	0.37		0.79	0.51	0.33	0.31		0.30	0.36	0.43
4	0.11		0.54	0.28	0.10	0.09		0.06	0.16	0.27
5	0.07		0.46	0.23	0.08	0.06		0.03	0.13	0.25

Table 14.1: The means of 100 canonical correlation estimates computed from simulated samples of n pairs of 251-dimensional random fields (see text). For brevity, only five of the 10 canonical correlations are listed. The true correlations ρ_{xy}^i are given in the second column; the results obtained for variable time series lengths n , with an EOF truncation of $k = 20$, are given in columns three to six. The effect of including different numbers of EOFs k , using a fixed time series length of $n = 250$, is listed in columns seven to nine. From Borgert [55].

14.2.3 Monte Carlo Experiments. Borgert [55] conducted a Monte Carlo study of the performance of CCA on EOF truncated [14.1.6] data. He simulated a pair of 251-point random fields $\vec{X}$ and $\vec{Y}$ that consisted of a random linear combination of 10 pairs of patterns. Each pair of patterns was multiplied by a pair of random coefficients that were independent of all other pairs of coefficients. Thus the random coefficients are the true canonical variables. Each pair of random coefficients was generated from a different bivariate auto-regressive process. In this way the cross-correlations between the pairs of canonical variates, the true canonical correlations, were known. Thus Borgert was able to simulate a pair of random fields with known canonical correlations and patterns.

Borgert used this tool to generate 100 independent samples for a number of combinations of sample size n and EOF truncation point $k = k_X = k_Y$. A canonical correlation analysis was performed on each sample, and statistics assessing the average quality of the CCA were gathered for each combination of n and k . He found that the CCA was really able to identify the correct pairs of patterns: the estimated patterns were close to the prescribed patterns. However, as exemplified in Table 14.1, there were considerable biases in the estimated correlations if too many EOFs were retained or if the time series were too short.

Bretherton et al. [64] reviewed a number of techniques for diagnosing coupled patterns and intercomparing them in a series of small Monte Carlo experiments. They found that CCA with *a priori* EOF truncation and Maximum

Covariance Analysis were more robust than the other techniques considered.

14.2.4 Irregularly Distributed Gaps in the Data. One way to cope with missing data is to fill the gaps by spatial or temporal interpolation. However, this is unsatisfactory if more than just a small amount of data is missing because we end up trying to diagnose connections between real data on the one hand and imputed data with much lower information content on the other. A better procedure is to use only the data that are actually available. This can be achieved by the procedure already outlined in [13.2.7]. The various matrices, such as Σ_{XX} , are estimated by forming sums over only the available pairs of observations (13.31):

$$\hat{\sigma}_{ij} = \frac{1}{|K_i \cap K_j|} \sum_{k \in K_i \cap K_j} (\mathbf{x}_{ki} - \hat{\mu}_i)(\mathbf{x}_{kj} - \hat{\mu}_j)^*$$

where $K_i = \{k: \text{component } i \text{ of } \vec{\mathbf{x}}_k \text{ is not missing}\}$, the notation $|\cdot|$ indicates the number of elements in a set, and $\hat{\mu}_i = \frac{1}{|K_i|} \sum_{k \in K_i} \mathbf{x}_{ki}$. As with EOFs, the calculation of the time coefficients can no longer be done by means of the dot products (14.12) and (14.13). Instead coefficients are determined by least squares, as in equation (13.32).

14.3 Examples

14.3.0 Overview. We will present three examples in this section. The joint variability of a pair of large-scale fields is examined for evidence of a cause-and-effect relationship between the occurrence of large-scale sea-level air pressure and sea-surface temperatures anomalies in the North

Atlantic [14.3.1,2]. In the second example, one of the vector times series is again North Atlantic sea-level pressure but the second ‘partner’ in the CCA is a regional scale variable, namely, precipitation on the Iberian Peninsula [14.3.3,4]. This example is used to demonstrate *statistical downscaling* of GCM output. The last example [14.3.5] illustrates the *Principal Prediction Patterns* introduced in [14.1.7].

The literature also contains many other examples of applications of CCA. Bretherton et al. [64] cite several studies, including classic papers by Barnett and Preisendorfer [21], Nicholls [293] and Barnston and colleagues [26, 28, 30, 346].

14.3.1 North Atlantic SLP and SST: Data and Results.

CCA is used to analyse the relationship between $\bar{\mathbf{X}} = \text{monthly mean sea-level pressure (SLP)}$ and $\bar{\mathbf{Y}} = \text{sea-surface temperature (SST)}$ over the North Atlantic in northern winter (DJF) (see Zorita et al. [438] for details). The data are time series of monthly means of SLP and SST on a grid over the North Atlantic north of about 20°N , for DJF of 1950 to 1986. Anomalies were obtained at each grid point by subtracting the long-term monthly mean from the original values.

The coefficients of the first five EOFs of both fields were retained for the subsequent CCA. They represent 87% and 62% of the total variance of SLP and SST respectively. To check the sensitivity of the results to EOF truncation, the same calculations were performed using five SLP EOFs and either 10 or 15 SST EOFs (77% and 84%, respectively) and essentially the same results were obtained.

The CCA yields two pairs of patterns that describe the coherent variations of the SST and SLP fields. The two patterns are dominant in describing SLP and SST variance.

The first pair of patterns, $\bar{\mathbf{F}}_{SLP}^1$ and $\bar{\mathbf{F}}_{SST}^1$, which corresponds to a canonical correlation of 0.56, represents 21% of the variance of monthly mean SLP and 19% of the variance of monthly mean SST (Figure 1.13).⁹ The two patterns are consistent with the hypothesis first suggested by Bjerknes that atmospheric anomalies cause SST anomalies. The main features of the SLP pattern are a decrease of the westerly wind at about 50°N , and an anomalous cyclonic circulation centred at 40°W and 30°N .¹⁰ North of the cyclone, the

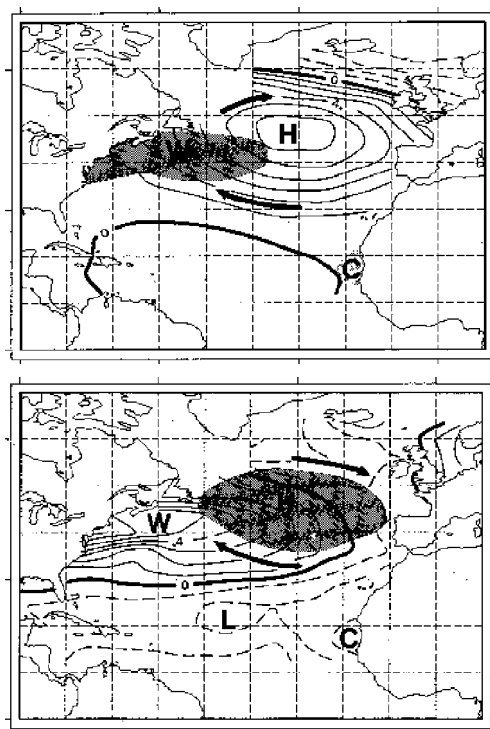


Figure 14.3: The second pair of canonical patterns for monthly mean SLP and SST over the North Atlantic in DJF. The dark shading on each pattern identifies the main positive feature of the opposing pattern.

Top: SLP, contour interval: 1 hPa,

Bottom: SST, contour interval: 0.1 K.

From Zorita et al. [438].

ocean surface is warmer than normal when the westerly wind is reduced. West of the cyclone, just downstream from the cold American continent, the ocean is substantially cooled. The SST anomalies off the African coast are a local response to anomalous winds; coastal upwelling is reduced when there are weaker than normal northerly winds. In contrast, when the circulation produces enhanced westerlies and anomalous anticyclonic flow in the southern part of the area, opposite SST anomalies are expected. The canonical correlation coefficient time series also support the Bjerknes hypothesis: the one month lag correlation is 0.65 when SLP leads SST but it is only 0.09 if SLP lags.

The coefficients of the second pair of patterns, $\bar{\mathbf{F}}_{SLP}^2$ and $\bar{\mathbf{F}}_{SST}^2$, have correlation 0.47 (Figure 14.3). The SLP pattern represents 31% of the total variance and is similar to the first SLP EOF (Figure 13.8), which is related to the North Atlantic Oscillation (see also [13.5.5] and

⁹We have dropped the ‘ $\hat{\cdot}$ ’ notation for now, but be aware that the patterns are parameter estimates. The same applies to canonical coordinate time series when they are discussed.

¹⁰We use the geostrophic wind relationship for the derivation of approximate wind anomalies from pressure anomalies.

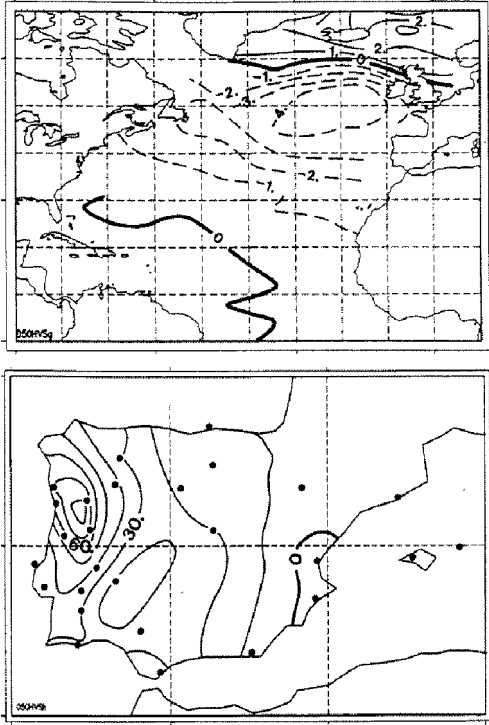


Figure 14.4: First pair of canonical correlation patterns of the North Atlantic winter mean sea-level pressure $\bar{\mathbf{Y}}$ and a vector $\bar{\mathbf{X}}$ of seasonal means of precipitation at a number of Iberian locations [403].

Figure 13.6). The structure of this pair of patterns is also consistent with the Bjerknes hypothesis. The one month lag correlation is 0.48 when SLP leads and 0.03 when SLP lags.

14.3.2 North Atlantic SLP and SST: Discussion.

We described conventional and rotated EOF analysis of the same data in [13.5.6,7]. The CCA of SLP and SST suggests why rotation had a marked effect on the SST EOFs but not on the SLP EOFs. The coherent variations in the atmosphere (SLP) are caused by large-scale internal atmospheric processes so that the EOFs have a simple large-scale structure. In case of the ocean (SST), however, the coherent variations (EOFs) are the oceanic response to the large-scale atmospheric variations. This response really does not have simple structure (recall our description in [14.3.3] of the ocean's response to NAO variations).

14.3.3 North Atlantic SLP and Iberian Rainfall: Analysis and Historic Reconstruction.

In this example, winter (DJF) mean precipitation

from a number of rain gauges on the Iberian Peninsula is related to the air-pressure field over the North Atlantic (see [403] for details). CCA was used to obtain a pair of canonical correlation pattern estimates $\bar{F}_{SLP}^1$ and $\bar{F}_{pre}^1$ (Figure 14.4), and corresponding time series $\beta_1^{SLP}(t)$ and $\beta_1^{pre}(t)$ of canonical variate estimates. These strongly correlated modes of variation (the estimated canonical correlation is 0.75) represent about 65% and 40% of the total variability of seasonal mean SLP and Iberian Peninsula precipitation respectively. The two patterns represent a simple physical mechanism: when $\bar{F}_{SLP}^1$ has a strong positive coefficient, enhanced cyclonic circulation advects more maritime air onto the Iberian Peninsula so that precipitation in the mountainous northwest region ($\bar{F}_{pre}^1$) is increased.

Since the canonical correlation is large, the results of the CCA can be used to forecast or specify winter mean precipitation on the Iberian peninsula from North Atlantic SLP. The first step is to connect $\beta_1^{pre}(t)$ and $\beta_1^{SLP}(t)$ with a simple linear model $\beta_1^{pre}(t) = a\beta_1^{SLP}(t) + \epsilon$. Since $\beta_1^{pre}(t)$ and $\beta_1^{SLP}(t)$ are normalized to unit variance, the least squares estimate of coefficient a is the canonical correlation ρ_1 . Given a realization of $\beta_1^{SLP}(t)$, the canonical variate for precipitation can be forecast as $\hat{\beta}_1^{pre}(t) = \rho_1\beta_1^{SLP}(t)$, and thus the precipitation field is forecast as

$$\hat{\mathbf{R}} = \hat{\beta}_1^{pre}(t)\bar{F}_{pre}^1 = \rho_1\beta_1^{SLP}(t)\bar{F}_{pre}^1. \quad (14.27)$$

Similarly, if several useful canonical correlation patterns had been found, Iberian winter mean precipitation could be forecast or specified as

$$\hat{\mathbf{R}} = \sum_{i=1}^k \rho_i \beta_i^{SLP}(t) \bar{F}_{pre}^i.$$

The analysis described above was performed with the 1950–80 segment of a data set that extends back to 1901. Since the 1901–49 segment is independent of that used to ‘train’ the model (14.27), it can be used to validate the model. Figure 14.5 shows both the specified and observed winter mean rainfall averaged over all Iberian stations for this period. The overall upward trend and the low-frequency variations in observed precipitation are well reproduced by the indirect method indicating the usefulness of the technique (14.27) as well as the reality of both the trend and the variations in the Iberian winter precipitation.

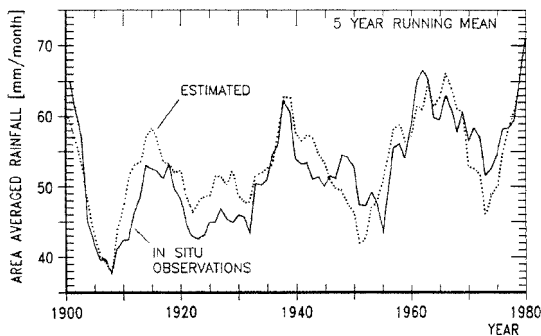


Figure 14.5: Five-year running mean of winter mean rainfall averaged across Iberian rain gauges. The solid curve is obtained from station data, and the dotted curve is imputed from North Atlantic SLP variations [403].

14.3.4 North Atlantic SLP and Iberian Rainfall: Downscaling of GCM output. The regression approach described above has an interesting application in climate change studies. GCMs are widely used to assess the impact that increasing concentrations of greenhouse gases might have on the climate system. But, because of their resolution, GCMs do not represent the details of regional climate change well. The *minimum scale* that a GCM is able to resolve is the distance between two neighbouring grid points whereas the *skilful scale* is generally accepted to be four or more grid lengths. The minimum scale in most climate models in the mid 1990s is of the order of 250–500 km so that the skilful scale is at least 1000–2000 km.

Thus the scales at which GCMs produce useful information does not match the scale at which many users, such as hydrologists, require information. *Statistical downscaling* [403] is a possible solution to this dilemma. The idea is to build a statistical model from historical observations that relates large-scale information that can be well simulated by GCMs to the desired regional scale information that can not be simulated. These models are then applied to the large-scale model output.

The following steps must be taken.

- 1 Identify a regional climate variable $\vec{R}$ of interest.
- 2 Find a climate variable $\vec{L}$ that
 - controls $\vec{R}$ in the sense that there is a

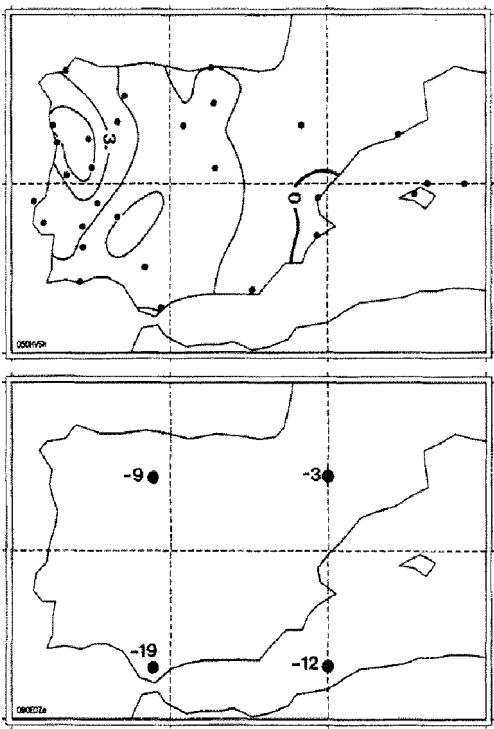


Figure 14.6: Downscaled and grid point response of Iberian precipitation in a ‘2×CO₂ experiment’ [403].

statistical relationship between $\vec{R}$ and $\vec{L}$ of the form

$$\vec{R} = \mathcal{G}(\vec{L}, \vec{\alpha}) + \epsilon \tag{14.28}$$

in which $\mathcal{G}(\vec{L}, \vec{\alpha})$ represents a substantial fraction of the total variance of $\vec{R}$. Vector $\vec{\alpha}$ contains parameters that can be used to adjust the fit of (14.28).

- is reliably simulated in a climate model.
- 3 Use historical realizations $(\vec{r}_t, \vec{l}_t)$ of $(\vec{R}, \vec{L})$ to estimate $\vec{\alpha}$.
 - 4 Validate the fitted model on independent historical data or by cross-validation (see [18.5.2]).
 - 5 Apply the validated model to GCM simulated realizations of $\vec{L}$.

This is exactly the process that was followed in the previous subsection. A model (14.27) was constructed that related Iberian rainfall $\vec{R}$ to North Atlantic SLP $\vec{L}$ through a simple linear functional. The adjustable parameters $\vec{\alpha}$ consisted of the canonical correlation patterns $\vec{F}_{pre}^1$ and

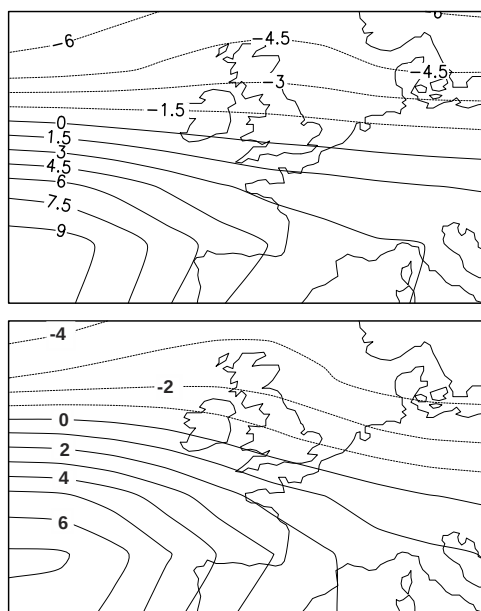


Figure 14.7: *Principal Prediction Patterns $\bar{F}_0^1$ (top) and $\bar{F}_3^1$ with $\Delta = 3$ days (bottom) for the North Atlantic / European Sector daily winter SLP. From Dorn and von Storch [103].*

$\bar{F}_{SLP}^1$ and the canonical correlation ρ_1 . These parameters were estimated from 1950 to 1980 data. Observations before 1950 have been used to validate the model.

Downscaling model (14.27) was applied to the output of a '2×CO₂' experiment performed with a GCM. Figure 14.6 compares the 'downscaled' response to doubled CO₂ with the model's grid point response. The latter suggests that there will be a marked decrease in precipitation over most of the Peninsula whereas the downscaled response is weakly positive. The downscaled response is physically more reasonable than the direct response of the model.

14.3.5 Principal Prediction Pattern of North Atlantic / European SLP. Dorn and von Storch [103] used the Principal Prediction Pattern (PPP) analysis technique to study the synoptic predictability of sea-level pressure (SLP) over the eastern North Atlantic and Western Europe. This particular field was used because a rich data set, consisting of daily analysis since approximately 1900, was available for determining the skill of the PPP model.

The PPP analysis was performed with daily winter SLP maps for 1958–88. The dimensionality of the problem was reduced by projecting the maps

onto the first eight EOFs of daily winter SLP. Analyses were performed for lags $\tau = 1, \dots, 5$ days, but we discuss only the $\tau = 3$ days results below.

The first pair of PPPs is shown in Figure 14.7. The patterns are normalized such that the variance of the coefficient of $\bar{F}_0^1$ is 1, and that of the coefficient of $\bar{F}_3^1$ is $1/\sqrt{\rho_1}$. With this normalization, the coefficient for the regression of the $\bar{F}_0^1$ -coefficient on the $\bar{F}_3^1$ -coefficient is the identity. Also, the patterns are scaled so that if the initial state is a multiple of $\bar{F}_0^1$, then the best predictor is the same multiple of $\bar{F}_3^1$. Patterns $\bar{F}_0^1$ and $\bar{F}_3^1$ are rather similar indicating that the analysis has selected the regional SLP mode that is most persistent on synoptic time scales. The reduction of the magnitude by about 1/3 indicates that this persistence goes with some damping. Thus, the forecast incorporated in this pair of patterns implies constancy in the pattern, but a reduction of the intensity, i.e., 'damped persistence'.

This statement also holds for the other patterns and is further supported by comparing the forecast skill, as given by the anomaly correlation coefficient¹¹ between the true SLP field and the field predicted by either PPP or persistence (Figure 14.8). The skill of the two forecast schemes is practically identical and exhibits the characteristic decay with increasing lag. Thus, the PPP forecast is no more skilful than the simpler 'competitor' persistence.

However, the PPP forecast scheme should not be dismissed out of hand. By conditioning on the proportion of spatial variance represented by $\bar{F}_0^1$, the PPP forecast was found to be more skilful when the proportion is large (Figure 14.8, bottom). Thus the PPP scheme also gives a forecast of forecast skill.

The utility of the PPP technique needs further exploration and the user is advised to examine all results obtained with this technique critically. In particular, surprisingly good results may be generated by using short time series or by failing to adequately reduce the degrees of freedom of the problem.

14.4 Redundancy Analysis

14.4.1 Introduction. So far, we have identified pairs of patterns by maximizing the correlation

¹¹This measure of skill is explained in detail in [18.2.9]. Roughly speaking, it is the mean spatial correlation between the forecast and the verifying field.

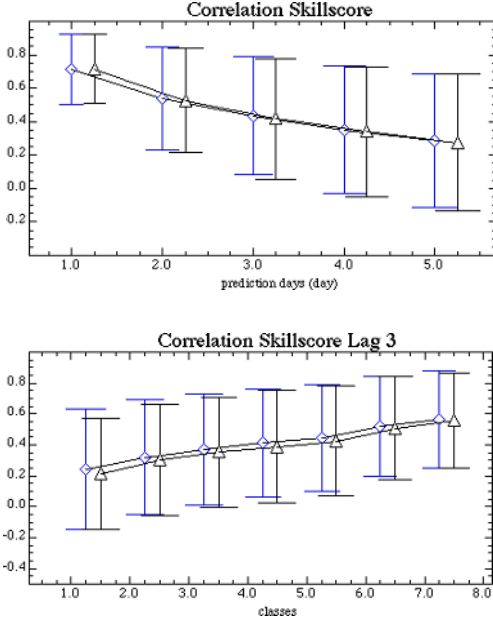


Figure 14.8: Anomaly correlation coefficient of the PPP forecast (diamonds) and of persistence (triangles). The vertical bars indicate $\pm\sigma$ bands, as estimated from all forecast prepared for the winter days from 1900 until 1990. For better readability, the numbers for the two forecast schemes, persistence and PPP, are shifted horizontally.

Top: For lags $\tau = 1, \dots, 5$ days.

Bottom: For lag $\tau = 3$ days. The anomaly correlation coefficients were classified according to the proportion of variance of the initial SLP field described by the PPP (bottom). Class 1 contains cases with proportions in the range $[0.0, 0.4]$, class 2 contains cases with proportions in $(0.4, 0.5]$, and so on up to class 7, which contains cases with proportions in $(0.9, 1]$.

between the corresponding pattern coefficients. We then demonstrated how regression techniques can be used to specify or forecast the value of the pattern coefficients of one of the fields from those of the other field. This regression problem is generically non-symmetric because the objective is to maximize the variance of the predictand that can be represented. Properties of the predictor patterns, such as the amount of variance they represent, are irrelevant to the regression problem. Hence, there is a mismatch between CCA, which treats variables equally, and regression analysis, which focuses primarily on the predictand.

The ‘redundancy analysis’ technique directly addresses this problem by identifying patterns that

are strongly linked through a regression model. Patterns are selected by maximizing predictand variance. This technique was developed in the late 1970s but apparently has not been introduced in climate research literature.

Here we present the *redundancy analysis* as suggested by Tyler [376]. *Note that very little experience has been collected with this technique in the field of climate research. Therefore, the technique should be applied with great care, and results should be appraised critically.*

14.4.2 Redundancy Index. Let us consider a pair of random vectors $(\vec{X}, \vec{Y})$ with dimensions m_X and m_Y . Let us assume further that there is a linear operator represented by a $m_X \times k$ matrix Q_k . How much variance in $\vec{Y}$ can be accounted for by a regression of $Q_k^T \vec{X}$ on $\vec{Y}$?¹² We assume, without loss of generality, that the expected value of both $\vec{X}$ and $\vec{Y}$ is zero.

The regression model that relates $Q_k^T \vec{X}$ is given by

$$\vec{Y} = \mathcal{R}(Q_k^T \vec{X}) + \vec{\epsilon}, \quad (14.29)$$

where $\mathcal{R}$ is an $m_Y \times k$ matrix of regression coefficients. The variance represented by $(Q_k^T \vec{X})$ is maximized when

$$\mathcal{R} = \Sigma_{Y, QX} (\Sigma_{QX, QX})^{-1}, \quad (14.30)$$

where

$$\Sigma_{Y, QX} = \text{Cov}(\vec{Y}, Q_k^T \vec{X}) = \Sigma_{YX} Q_k \quad (14.31)$$

$$\Sigma_{QX, QX} = Q_k^T \Sigma_{XX} Q_k. \quad (14.32)$$

Tyler [376] called the proportion of variance represented by the regression (14.29) the *redundancy index* and labelled it

$$R^2(\vec{Y} : Q_k^T \vec{X}) = \frac{\text{tr}(\text{Cov}(\vec{Y}, \vec{Y}) - \text{Cov}(\vec{Y} - \hat{\vec{Y}}, \vec{Y} - \hat{\vec{Y}}))}{\text{tr}(\text{Cov}(\vec{Y}, \vec{Y}))} \quad (14.33)$$

where $\hat{\vec{Y}} = \mathcal{R}(Q_k^T \vec{X})$ is the estimated value of $\vec{Y}$. The motivation of this wording is that it is a measure of how *redundant* the information in $\vec{Y}$ is if one already has the information provided by $\vec{X}$.

¹²The number of columns (patterns) in Q_k is smaller than the dimension of $\vec{X}$ in most practical situations, so that $k < m_X$ or even $k \ll m_X$. Thus, the operation $\vec{X} \rightarrow Q_k^T \vec{X}$ represents a reduction of the phase space of $\vec{X}$, as in all the other cases we have discussed in this and the previous chapter.

The numerator is the trace (sum of main diagonal elements) of the matrix

$$\begin{aligned} \Sigma_{YY} - (\Sigma_{Y\hat{Y}} + \Sigma_{\hat{Y}Y} - 2\Sigma_{Y\hat{Y}}) \\ = -\mathcal{R}\Sigma_{QX, QX}\mathcal{R}^T + 2\Sigma_{Y, QX}\mathcal{R}^T. \end{aligned}$$

Using (14.31)–(14.33), and simplifying, we find that

$$\begin{aligned} R^2(\vec{Y} : Q_k^T \vec{X}) = \\ \frac{\text{tr}(\Sigma_{YX} Q_k (Q_k^T \Sigma_{XX} Q_k)^{-1} Q_k^T \Sigma_{XY})}{\text{tr}(\Sigma_{YY})}. \end{aligned} \quad (14.34)$$

14.4.3 Invariance of the Redundancy Index to Linear Transformations. The redundancy index has a number of interesting properties. One of these is its invariance to orthonormal transformations of $\vec{Y}$: if $\mathcal{A}$ is orthonormal, then

$$R^2(\mathcal{A}\vec{Y} : Q_k^T \vec{X}) = R^2(\vec{Y} : Q_k^T \vec{X}). \quad (14.35)$$

The significance of this property comes from the fact that we may identify any orthonormal transformation with a linear transformation that conserves variance. Relationship (14.35) does not hold for general non-singular matrices, in particular not for transformations that change the variance since the proportion of captured variance changes when the variance of $\vec{Y}$ is changed.

On the other hand, any square non-singular matrix Q_{m_X} used to transform the specifying variable $\vec{X}$ has also no effect on the redundancy index. In that case, $(Q_{m_X})^{-1}$ exists and $(Q_{m_X}^T \Sigma_{XX} Q_{m_X})^{-1} = Q_{m_X}^{-1} \Sigma_{XX}^{-1} (Q_{m_X}^T)^{-1}$ in the numerator of (14.34), so that

$$R^2(\vec{Y} : Q_{m_X}^T \vec{X}) = R^2(\vec{Y} : \vec{X}). \quad (14.36)$$

The implication of (14.36) is that the coordinate system in which the random vector $\vec{X}$ is given does not matter, so long as it describes the same linear space. This is a favourable property since the information contained in $\vec{X}$ about $\vec{Y}$ should not depend on the specifics of the presentation of $\vec{X}$, such as the metric used to measure the components of $\vec{X}$, or the order of its components.

However, if the linear transformation Q_k maps the m_X -dimensional variable $\vec{X}$ onto a k -dimensional variable $\vec{X}'_k = Q_k^T \vec{X}$, the new variable contains less information about $\vec{Y}$, so that

$$\begin{aligned} R^2(\vec{Y} : \vec{X}'_k) &\leq R^2(\vec{Y} : \vec{X}'_{k+1}) \\ &\leq R^2(\vec{Y} : \vec{X}'_{m_X}) = R^2(\vec{Y} : \vec{X}) \end{aligned} \quad (14.37)$$

provided that ‘column spaces’¹³ of Q_k , Q_{k+1} , and Q_{m_X} are nested and Q_{m_X} is invertible. If, for all k , Q_{k+1} is constructed by adding a column to Q_k , then inequality (14.37) simply reflects the fact that the regression on $\vec{Y}$ has k predictors in the case of $\vec{X}'_k$, and the same k predictors plus one more in the case of $\vec{X}'_{k+1}$.

For a given transformation Q_k , again only the subspace spanned by the columns of Q_k matters. That is, for any invertible $k \times k$ matrix $\mathcal{L}$, we find

$$R^2(\vec{Y} : \mathcal{L}^T (Q_k^T \vec{X})) = R^2(\vec{Y} : Q_k^T \vec{X}). \quad (14.38)$$

Thus, the redundancy index for two variables is a function of the subspace the variable $\vec{X}$ is projected upon, and the way in which $\vec{Y}$ is scaled.

Since R^2 does not depend on the specific coordinates of the variable $\vec{X}$ and $\vec{X}'_k$, we may assume that the columns of Q_k are chosen to be orthogonal with respect to $\vec{X}$; that is,

$$\vec{q}^k{}^T \Sigma_{XX} \vec{q}^j = 0 \quad (14.39)$$

for any $k \neq j$. Then

$$R^2(\vec{Y} : Q_k^T \vec{X}) = \sum_{j=1}^k R^2(\vec{Y} : \vec{q}^j{}^T \vec{X}), \quad (14.40)$$

which may be seen as a special version of (14.37). Note that (14.39) is fulfilled if the vectors $\vec{q}^j$ are the EOFs of $\vec{X}$.

14.4.4 Redundancy Analysis. The theory behind *redundancy analysis*, as put forward by Tyler [376], confirms the existence of a non-singular transformation $\mathcal{B} = (\vec{b}^1 | \vec{b}^2 | \dots | \vec{b}^{m_X})$ so that the index of redundancy (i.e., the amount of $\vec{Y}$ -variance explained through the regression of $\mathcal{B}_k^T \vec{X}$ on $\vec{Y}$) is maximized for any $k = 1, \dots, \min(m_X, m_Y)$. Matrix $\mathcal{B}_k$ contains the first k columns of $\mathcal{B}$.

Thus redundancy analysis determines the k -dimensional subspace that allows for the most efficient regression on $\vec{Y}$. Since we are free to choose the coordinates of this subspace, we may use a linear basis with k orthogonal patterns that satisfies (14.39), so that the redundancy index may be expressed specifically as (14.40).

The following theorem identifies a second set of patterns, $\mathcal{A} = (\vec{a}^1 | \vec{a}^2 | \dots | \vec{a}^k)$, that represent an orthogonal partitioning of the variance of $\vec{Y}$ that is accounted for by the regression of $\vec{X}$ on $\vec{Y}$. More specifically, the regression maps the subspace

¹³The *column space* of a matrix Q is the vector space spanned by the columns of Q .

represented by $\vec{X}_k'$ onto the space spanned by the first k columns of $\mathcal{A}$.

The following subsections describe the mathematics required for the determination of matrices $\mathcal{A}$ and $\mathcal{B}$. The theorems are taken from Tyler's paper [376].

14.4.5 The Redundancy Analysis Transformations. For any random vectors $\vec{Y}$ of dimension m_Y and $\vec{X}$ of dimension m_X , there exists an orthonormal transformation $\mathcal{A}$ and a non-singular transformation $\mathcal{B}$ such that

$$\text{Cov}(\mathcal{B}^T \vec{X}, \mathcal{B}^T \vec{X}) = \mathcal{I} \quad (14.41)$$

$$\text{Cov}(\mathcal{A}^T \vec{Y}, \mathcal{B}^T \vec{X}) = \mathcal{D} \quad (14.42)$$

where $\mathcal{D}$ is an $m_Y \times m_X$ matrix with elements $d_{ij} = 0$ for $i \neq j$ and diagonal elements $d_{jj} = \sqrt{\lambda_j}$ for $j \leq \min(m_X, m_Y)$.

The proof, which is detailed in Appendix M, revolves around two eigen-equations:

$$\Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY} \vec{a}^j = \lambda_j \vec{a}^j \quad (14.43)$$

$$\Sigma_{XX}^{-1} \Sigma_{XY} \Sigma_{YX} \vec{b}^j = \lambda_j \vec{b}^j. \quad (14.44)$$

Both equations have the same positive eigenvalues λ_j , and the eigenvectors $\vec{a}^j$ and $\vec{b}^j$ belonging to the same nonzero eigenvalue λ_j are related through

$$\vec{b}^j = \frac{1}{\sqrt{\lambda_j}} \Sigma_{XX}^{-1} \Sigma_{XY} \vec{a}^j. \quad (14.45)$$

The matrices $\mathcal{A}$ and $\mathcal{B}$, which are composed of eigenvectors $\vec{a}^j$ and $\vec{b}^j$, respectively, are the only matrices that satisfy the requirements of the theorem.

From the computational point of view, it is advisable to solve the eigenproblem with the Hermitian matrix (14.43), then use the identity (14.45). Since (14.43) is a Hermitian problem, all eigenvectors $\vec{a}^j$ are real valued, and since (14.45) involves only real matrices, the 'patterns' $\vec{b}^j$ are also real valued.

14.4.6 Theorem: Optimality of the Redundancy Transformation. The significance of the redundancy transformation originates from the following theorem given by Tyler [376]:

The redundancy index $R^2(\vec{Y} : \mathcal{Q}_k^T \vec{X})$ is maximized by setting $\mathcal{Q}_k = \mathcal{B}_k$, where $\mathcal{B}_k$ is the $m_X \times k$ matrix that contains the k eigenvectors satisfying (14.42) that correspond to the k largest eigenvalues.

Note that the statement holds for all $k \leq m_X$. Thus,

among all possible single patterns $\vec{q}$, the eigenvector $\vec{b}^1$ belonging to the largest eigenvalue of the matrix $\Sigma_{XX}^{-1} \Sigma_{XY} \Sigma_{YX}$ provides the maximum information, in a linear sense, about the variance of $\vec{Y}$:

$$R^2(\vec{Y} : \vec{q}^T \vec{X}) \leq R^2(\vec{Y} : \vec{b}^{1T} \vec{X}) \quad (14.46)$$

for any m_X -dimensional vector $\vec{q}$. Moreover, by equations (14.41) and (14.40), the index of redundancy takes a particularly simple form,

$$R^2(\vec{Y} : \mathcal{B}_k^T) = \sum_{j=1}^k R^2(\vec{Y} : \vec{b}^{jT} \vec{X}). \quad (14.47)$$

Also, note that inequality (14.46) may be generalized to

$$\sum_{j=1}^k R^2(\vec{Y} : \vec{q}^{jT} \vec{X}) \leq \sum_{j=1}^k R^2(\vec{Y} : \vec{b}^{jT} \vec{X}) \quad (14.48)$$

for any set of vectors $\vec{q}^1, \dots, \vec{q}^k$.

14.4.7 The Role of Matrix $\mathcal{A}$. Since $\mathcal{B} = (\vec{b}^1 | \dots | \vec{b}^{m_X})$ is non-singular, random vector $\vec{X}$ can be expanded in the usual manner as

$$\vec{X} = \sum_{j=1}^{m_X} (\vec{X}^T \vec{b}^j) \vec{b}^j, \quad (14.49)$$

where the adjoint patterns $\mathcal{P} = (\vec{p}^1 | \dots | \vec{p}^{m_X})$ are given by $\mathcal{P}^T = \mathcal{B}^{-1}$. When re-expressed in matrix-vector form, equation (14.49) simply reads as

$$\vec{X} = \mathcal{P} \mathcal{B}^T \vec{X}.$$

Similarly, since $\mathcal{A}$ is orthonormal, the part of $\vec{Y}$ that can be represented by $\vec{X}$, that is, $\hat{\vec{Y}}$, can be expanded as

$$\hat{\vec{Y}} = \mathcal{A} \mathcal{A}^T \hat{\vec{Y}} = \sum_j (\hat{\vec{Y}}^T \vec{a}^j) \vec{a}^j. \quad (14.50)$$

When we regress $\vec{Y}$ on $\vec{X}$, we find that $\hat{\vec{Y}} = \Sigma_{YX} \Sigma_{XX}^{-1} \vec{X}$. Thus the expansion coefficients in (14.50), the elements of $\mathcal{A}^T \hat{\vec{Y}}$, are given by

$$\mathcal{A}^T \hat{\vec{Y}} = \mathcal{A}^T \Sigma_{YX} \Sigma_{XX}^{-1} \vec{X}.$$

Now, from equations (14.41) and (14.42) we have that $\Sigma_{XX}^{-1} = \mathcal{B} \mathcal{B}^T$ and $\mathcal{A}^T \Sigma_{YX} \mathcal{B} = \mathcal{D}$. Thus

$$\mathcal{A}^T \hat{\vec{Y}} = \mathcal{A}^T \Sigma_{YX} \mathcal{B} \mathcal{B}^T \vec{X} = \mathcal{D} \mathcal{B}^T \vec{X}.$$

Hence the expansion coefficients in (14.50) are given by

$$\hat{\mathbf{Y}}^T \vec{a}^j = \sqrt{\lambda_j} \vec{\mathbf{X}}^T \vec{b}^j. \quad (14.51)$$

Considering both (14.49) and (14.50), we see that the regression maps variations in the amplitude of $\vec{\mathbf{X}}$ patterns $\vec{p}^j$ onto variations in the amplitude of $\vec{\mathbf{Y}}$ patterns $\vec{a}^j$. On average, $\vec{\mathbf{Y}}_j = \sqrt{\lambda_j} \vec{a}^j$ when $\vec{\mathbf{X}}_j = \vec{p}^j$ (cf. (14.49) and (14.51)). It is easily shown that the patterns themselves are related by¹⁴

$$AD = \Sigma_{YX} \Sigma_{XX}^{-1} P.$$

That is, the $\vec{\mathbf{X}}$ -patterns are transformed into scaled versions of the $\vec{\mathbf{Y}}$ patterns by the regression operator.

Thus, redundancy analysis offers a number of useful insights. First, it helps us to identify an efficient way of specifying a maximum of variance in one random vector from the information provided by another vector. It also guides us in finding those components of the specifying variable that contain the most information about the variable to be specified. Finally, it offers pairs of patterns that are mapped onto each other. If we observe the pattern $\vec{p}^j$ in the specifying field, then the likelihood of observing pattern $\vec{a}^j$ in the field to be specified is increased.

If we consider the full $\vec{\mathbf{X}}$ -space, we find that

$$\Sigma_{\hat{\mathbf{Y}}\hat{\mathbf{Y}}} = \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}. \quad (14.52)$$

When comparing this expression with the eigenproblem (14.43), it becomes obvious that the $\vec{a}$ -vectors are the EOFs of $\hat{\mathbf{Y}}$. Thus the $\vec{a}^1$ coefficient accounts for the largest amount of $\hat{\mathbf{Y}}$ variance (i.e., λ_1), $\vec{a}^2$ accounts for the second largest amount of variance λ_2 , and so on. The total variance of the regressed vector $\hat{\mathbf{Y}}$ is $\sum_j \lambda_j$. Since $\Sigma_{Y\hat{\mathbf{Y}}} = \Sigma_{\hat{\mathbf{Y}}\hat{\mathbf{Y}}}$, we have

$$\begin{aligned} R^2(\vec{\mathbf{Y}} : \hat{\mathbf{Y}}) &= R^2(\vec{\mathbf{Y}} : \vec{\mathbf{X}}) \\ &= \frac{\text{tr}(\Sigma_{\hat{\mathbf{Y}}\hat{\mathbf{Y}}})}{\text{tr}(\Sigma_{YY})} = \frac{\sum_j \lambda_j}{\text{tr}(\Sigma_{YY})}. \end{aligned} \quad (14.53)$$

When we truncate (14.49) to the k components of $\vec{\mathbf{X}}$ that carry the most information about $\vec{\mathbf{Y}}$, we find that

$$R^2(\vec{\mathbf{Y}} : \hat{\mathbf{Y}}) = R^2(\vec{\mathbf{Y}} : B_k^T \vec{\mathbf{X}}) = \frac{\sum_{j=1}^k \lambda_j}{\text{tr}(\Sigma_{YY})}.$$

¹⁴The proof is straightforward:

$$\mathcal{RP} = \Sigma_{YX} \Sigma_{XX}^{-1} (B^T)^{-1} = \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XX} B = \Sigma_{YX} B = AD.$$

14.4.8 Comparison with CCA. Let us now consider the special case in which Σ_{XX} and Σ_{YY} are both identity matrices. Then $\mathcal{B}$ and $\mathcal{P}$ are also identity matrices, and the regressed patterns $\vec{a}$, the EOFs $\hat{\mathbf{Y}}$, are the eigenvectors of $\Sigma_{YX} \Sigma_{XY}$. That is, $\vec{\mathbf{X}}$ provides the most information about the component of $\hat{\mathbf{Y}}$ that lies in the $\vec{a}^1$ direction, where $\vec{a}^1$ is the first eigenvector of $\Sigma_{YX} \Sigma_{XY}$. The best predictor of this component is $\vec{\mathbf{X}}^T \Sigma_{XY} \vec{a}^1$.

When we perform CCA on the same system we must solve the paired eigenvalue problem

$$\begin{aligned} \Sigma_{XY} \Sigma_{YX} \vec{f}_X &= \lambda \vec{f}_X \\ \Sigma_{YX} \Sigma_{XY} \vec{f}_Y &= \lambda \vec{f}_Y. \end{aligned}$$

The first pair of eigenvectors of this system is given by $\vec{f}_X = \Sigma_{XY} \vec{a}^1$, and $\vec{f}_Y = \vec{a}^1$, indicating that $\vec{\mathbf{X}}^T \Sigma_{XY} \vec{a}^1$ is the $\vec{\mathbf{X}}$ -component most strongly correlated with $\vec{\mathbf{Y}}^T \vec{a}^1$.

Thus redundancy analysis and CCA are equivalent in this special case: both identify the same $\vec{\mathbf{X}}$ and $\vec{\mathbf{Y}}$ directions.

In general, however, the methods are not equivalent. Redundancy analysis finds the best predicted (or specified) components of $\vec{\mathbf{Y}}$ by finding the eigenvectors $\vec{a}$ of

$$\Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}$$

and then finding the patterns $\vec{p}$ of $\vec{\mathbf{X}}$ -variations that carry this information. CCA, on the other hand, finds the most strongly correlated components of $\vec{\mathbf{Y}}$ by finding the eigenvectors $\Sigma_{YY}^{-1/2} \vec{f}_Y$ of

$$(\Sigma_{YY}^{-1/2})^T \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY} \Sigma_{YY}^{-1/2}.$$

That is, CCA does redundancy analysis on $\vec{\mathbf{Y}}' = (\Sigma_{YY}^{-1/2})^T \vec{\mathbf{Y}}$, the random vector that is obtained by projecting $\vec{\mathbf{Y}}$ onto its EOFs and scaling each component by its standard deviation. We can therefore anticipate that the two techniques will produce similar results if $\vec{\mathbf{Y}}$ is projected onto a small number of EOFs with similar eigenvalues.

14.4.9 Example: Interdecadal Variability of Intramonthly Percentiles of Significant 'Brent' Wave Height.

We now describe an application in which we use redundancy analysis to specify monthly wave height statistics at the Brent oil field, located northeast of Scotland in the North Atlantic at (61°N, 1.5°E). Wave height (sea state) data are available from visual assessments made on ships of opportunity, at light houses, from wave rider buoys, and shipborne instruments at ocean weather stations. Also, wave height maps

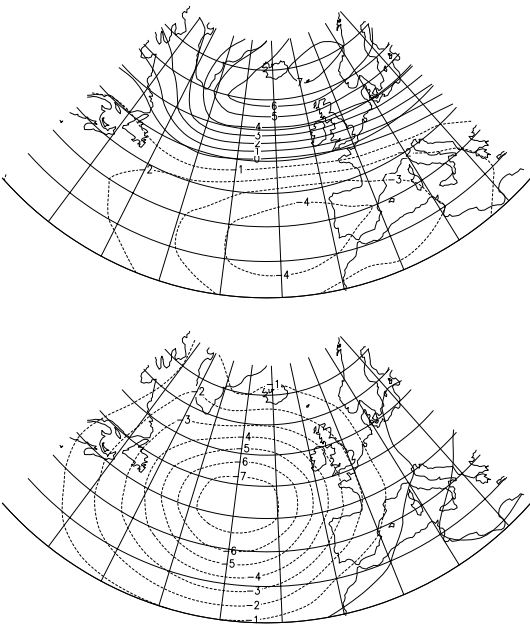


Figure 14.9: First two monthly mean air-pressure anomaly distributions $\hat{p}^k$ identified in a redundancy analysis as being most strongly linked to simultaneous variations of the intramonthly percentiles of significant wave height in the Brent oil field (61° N, 1.5° E; northeast of Scotland).

have been constructed from wind analyses for the purpose of ship routing (Bouws et al. [57]). These data are sparse and suffer from various inhomogeneities. Also, the records are generally too short to allow an assessment of changes during the past century.

Thus, observational data alone do not contain sufficient information about the interdecadal variability of wave statistics. One solution is a combined statistical/dynamical reconstruction of the past that uses a dynamical wave model. The model is forced with recent wind data that are believed to be fairly reliable and not strongly affected by improving analysis techniques.¹⁵ The wave heights derived from the hindcast simulation are treated as observations and are used to build a statistical model linking the wave heights to surface air pressure. Finally, the resulting

¹⁵Note that the homogeneity of weather maps and their surface winds is difficult to assess. Analysis system improvements can introduce artificial signals, such as increasing frequencies of extreme events, into the hindcast. Improved analyses procedures, be it more or better observations or more intelligently designed dynamical and statistical analysis tools, lead to the emergence of more details in weather maps and, therefore, larger extremes.

	Wave height percentile		
	50%	80%	90%
$\hat{a}^1$	−81	−107	−114
$\hat{a}^2$	32	2	−25

Table 14.2: The vectors $\hat{a}^k$ of anomalous intramonthly percentiles of significant wave height are given as rows in the following table. Units: cm.

statistical model is fed with the observed air pressure from the beginning of the century onward, thereby producing a plausible estimate of wave height statistics for the entire century. The statistical model is presented below.

In this case we bring together ‘apples’ and ‘oranges’, that is, two vector quantities that are not directly linked. One vector time series, $\vec{X}_t$, represents the winter (DJF) monthly mean surface air-pressure distributions in the North Atlantic. The other vector time series, $\vec{Y}_t$, is a three-dimensional random vector consisting of the 50th, 80th, and 90th percentiles of the intramonthly distributions of significant wave height¹⁶ in the Brent oil field at (61° N, 1.5° E). Both vector time series are assumed to be centred, so that the air-pressure values and percentiles are deviations from their respective long-term means.

The monthly mean of North Atlantic SLP is indirectly linked to the intramonthly percentiles, since storms affect both the monthly mean air-pressure distribution and the distribution of wave heights within a month at a specific location. Of course, the storm activity may also be seen as being conditioned by the monthly mean state.

The daily wave height data are taken from a 40-year ‘hindcast’ simulation (Günther et al. [153]). The following analysis assumes that the hindcasts and windfield analyses both represent the real world well enough for statistical relationships between the wave and wind fields on the monthly time scale to be reliably diagnosed.

A redundancy analysis of the two vector time series is performed to detect the dominant coupled anomaly patterns in the mean air pressure and in the intramonthly wave height percentiles. The SLP patterns $\hat{p}^1$ and $\hat{p}^2$ are shown in Figure 14.9 and the corresponding intramonthly percentiles $\hat{a}^1$ and $\hat{a}^2$ are listed in Table 14.2. The time coefficients are normalized to unit variance so that

¹⁶Significant wave height is a physical parameter that describes the wave field on the sea surface. The word ‘significant’ does not imply a significance test in this context. See [3.2.4].

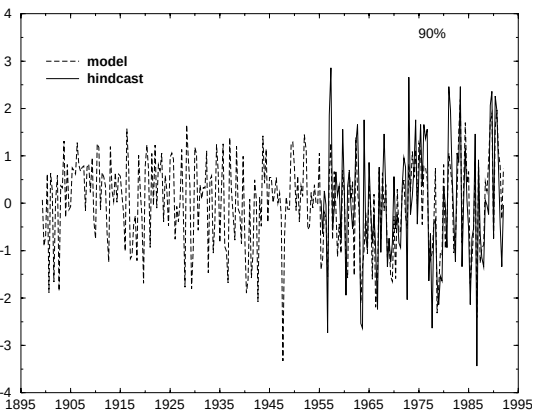


Figure 14.10: Reconstructed (dashed line) and hindcasted (continuous line; 1955–94) anomalies of the 90th percentile of significant wave height at in the Brent oil field. Units: m.

the three components of $\hat{a}^k$ may be interpreted as typical anomalies that occur when the pressure field anomalies are given by $\frac{1}{\sqrt{\lambda_k}} \hat{p}^k$.

The pattern $\vec{a}^1$ accounts for 94% of the variance of $\vec{Y}$, and $\vec{a}^2$ for 5%. The correlation between the coefficient time series of the first pair of vectors is 0.84 while that between the second pair is only 0.08. Thus, the first pair establishes a regression representing $R^2(\vec{Y} : \mathcal{B}_1^T \vec{X}) = 94\% \times 0.84^2 = 66\%$ of the variance of $\vec{Y}$, whereas the second pair represents only $5\% \times 0.08^2 < 0.1\%$ of variance. Thus the redundancy index for $k = 1$ (0.66) can not be usefully increased by adding a second vecor.

The first air-pressure pattern is closely related to the *North Atlantic Oscillation* (see [13.5.5] and Figure 13.6). A weakening of the NAO is associated with a decrease in all three intramonthly percentiles of significant wave height. In effect, this pattern describes a shift of the intramonthly distribution towards smaller waves.

The second pattern describes a mean south-easterly flow across the northern North Sea. The 50th percentile of the significant wave heights is increased by 32 cm, while the 90th percentile is reduced by 25 cm. Thus there is a tendency for the wave height distribution to be widened when pressure anomaly pattern $\hat{p}^2$ prevails. The reversed pattern goes with a narrowed intramonthly distribution of wave heights. This pair of patterns accounts for only 5% of the predictable wave height variance.

The regression model incorporated in the redundancy analysis was used to estimate the time series of the percentiles of significant wave height in the Brent oil field from the observed monthly mean air pressure anomaly fields between 1899 and 1994. The last 40 years may be compared with the hindcast data, whereas the first 50 years represent our best guess and can not be verified at this time. The 90th percentiles of the reconstructed wave height time series for 1899–94 and the corresponding hindcasted time series for 1955–94 are shown in Figure 14.10. The link appears to be strong, as is demonstrated by the correlations and the proportion of described variance, during the overlapping period:

	Wave height percentile		
	50%	80%	90%
Correlation	0.83	0.82	0.77
Described variance	0.70	0.66	0.60

The amount of percentile variance represented by the SLP patterns is consistent with the redundancy index (14.53), which has value 0.66. As with all regression models, the variance of the estimator is smaller than the variance of the original variable. This makes sense, since the details of the wave action in a month are not completely determined by the monthly mean air-pressure field. It is also affected by variations in surface wind that occur on shorter time scales.

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